Introductory Microeconomics: Definitions and Worked Examples

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Scarcity, Choice and the Frontier

1.1 Definitions

Scarcity. Scarcity means that resources are finite relative to what people want. Because resources are finite, there are competing claims on them. Scarcity means constraint. We cannot have more than there is, even if we want more.

Consider an economy with 6,000 hours of labor and no more. An hour that goes into a bicycle is an hour that cannot go into a jacket, and wanting more of both goods does not add a single hour to the supply.

Opportunity cost. An opportunity cost is the value of the next best alternative that is forgone. It is measured in units of the thing that is given up, not necessarily in money. It is the only cost concept that appears in every model in this book.

Suppose a bicycle takes 6 hours of labor to make and a jacket takes 2. The opportunity cost of a bicycle is then 3 jackets, because that is what the 6 hours would otherwise have made.

Rationality. A rational agent is one who makes consistent choices given the objectives they have. They take the best action that serves their interests given all of the information and constraints that they face. Rational agents are in effect goal oriented.

Consider an airline about to depart with one unsold seat. If a standby passenger offers $300 for it, the airline takes the money, because $300 serves its interests better than a seat that flies empty.

Production possibility frontier. Given its technology, a production possibility frontier shows the maximum combinations of two goods that an economy can produce when all of its resources are used fully and efficiently.

Take the economy with 6,000 hours of labor, where a bicycle takes 6 hours and a jacket takes 2. Every hour is used on the frontier, so $6B + 2J = 6000$, which rearranges to $J = 3000 - 3B$. The frontier runs from 3,000 jackets and no bicycles to 1,000 bicycles and no jackets.

One picture, three ideas. The frontier brings together three ideas: scarcity, efficiency and opportunity cost. *Scarcity* shows up as the existence of a boundary, because some combinations are simply not available. *Productive efficiency* means being on the boundary rather than inside it. *Opportunity cost* is the slope of the boundary, leaving aside its minus sign. This is not an analogy. The size of the slope and the opportunity cost are one and the same number.

We can see all three on the frontier $J = 3000 - 3B$. The combination of 400 bicycles and 1,800 jackets is on the boundary, so it is efficient, and the slope of -3 says that one more bicycle costs 3 jackets. No point beyond the line can be reached with the hours available.

Constant opportunity cost gives a straight line. The equation $J = 3000 - 3B$ has a slope of -3, so each extra bicycle costs three jackets. That number is not something read off the graph. It comes straight from the labor data:

$$\frac{6 \text{ hours per bicycle}}{2 \text{ hours per jacket}} = 3 \frac{\text{jackets}}{\text{bicycle}}.$$

That is because the six hours that go into a bicycle could otherwise have made three jackets. Every hour is equally productive at either task, so the cost is the same at every point on the frontier. A constant slope gives a straight line.

Suppose the economy makes 400 bicycles and wants 100 more. At 3 jackets each, the extra bicycles cost 300 jackets, and jackets fall from 1,800 to 1,500.

Increasing opportunity cost. A straight-line frontier assumes that every unit of the resource is equally suited to making both goods. In practice, resources are usually specialized. So as production shifts toward one good, the expanding sector has to draw in resources that are less and less suited to it. Each additional unit therefore costs more than the one before. When the slope gets steeper, the opportunity cost increases and the frontier is bowed.

Suppose the frontier for food and clothing is $C = 50 - 0.02F^2$. The first 10 units of food cost 2 units of clothing, and the last 10 units cost 18.

Three kinds of point. A point *on* the frontier is productively efficient: you cannot get more of one good without giving up some of the other. A point *inside* the frontier is feasible but inefficient, because the same resources could produce more of both goods. A point *outside* the frontier is out of reach with current resources and technology.

On the frontier $C = 50 - 0.02F^2$, an economy that grows 40 units of food can make 18 units of clothing. So 40 food with 18 clothing is efficient, and 40 food with 24 clothing is out of reach. With 40 food and fewer than 18 clothing, the economy is inside its frontier.

Economic growth. Economic growth is an outward shift of the frontier. It enlarges the set of combinations that the economy can attain.

Go back to the point with 40 units of food and 24 of clothing, which the frontier $C = 50 - 0.02F^2$ does not allow. Suppose that after growth the same 40 units of food permit about 32 units of clothing. The point has not moved, but it is now inside the frontier.

Positive and normative statements. A *positive* statement is a claim about how the world is, and in principle evidence can confirm or refute it. A *normative* statement is a claim about how the world ought to be. It rests on values as well as on facts.

The claim that a higher minimum wage reduces teenage employment is positive, because we can look at the data and the data may show it to be wrong. The claim that the government should raise the minimum wage is normative.

1.2 Worked examples

The cost of a year of college. An example illustrates the idea of an opportunity cost. Consider the cost of a year of college. Tuition and fees are the explicit costs. Let's suppose these are $32,000. Forgone earnings are the implicit costs. For example, a year's worth of full-time work might earn $38,000. Therefore the opportunity cost of the year is $70,000.

Notice that room and board don't come into it, because whether you're at college or working full time, you still have to pay for room and board. Only the difference between what they cost at college and what they would cost otherwise is part of the opportunity cost.

The concert ticket. As before, an example illustrates. Suppose you buy a concert ticket that costs $150. It is non-refundable, and you can't sell it on to anybody else. On the night of the concert, it's raining and you would rather stay in. No matter what, the $150 is gone. The reasonable comparison now is between the happiness, or what economists call utility, that you get from going to the concert and the utility you get from spending the evening at home. The price is now irrelevant.

A standard error is to suppose that not going to the concert now wastes $150. However, that is the wrong way to think about things. The $150 was already spent when the ticket was bought. The decision after the cost is sunk depends totally on the utility of the two options, staying at home or going to the concert. There is nothing you can do about the cost. Worrying about it, and thinking it should be part of the decision, is known as the **sunk cost fallacy**.

The empty seat. Once again, a simple example might illustrate this. Suppose an aircraft has 200 seats and is about to depart with one unsold seat. The average cost per passenger is $500. Now suppose there is a standby passenger, and that person offers $300. From the airline's point of view, is the seat worth selling? The answer is yes, sell. Why? The flight is going to go regardless, so the crew, fuel, aircraft and everything else are already committed.

The marginal cost of carrying one more passenger is near zero. It is a drink and a little more fuel, perhaps \$5. So the marginal benefit to the airline is \$300 against a marginal cost of a trivial \$5.

The \$500 average includes costs that do not change when the passenger boards. It answers a different question, and it is the wrong number when deciding what is the rational thing to do.

Tabulating the frontier.

Bicycles B	Substitute	Jackets J
0	$3000 - 3(0)$	3000
200	$3000 - 3(200)$	2400
400	$3000 - 3(400)$	1800
600	$3000 - 3(600)$	1200
800	$3000 - 3(800)$	600
1000	$3000 - 3(1000)$	0

Just as before, we can check one row directly against the constraint. At $B = 400$, $J = 1800$, and $6(400) + 2(1800) = 2400 + 3600 = 6000$. So the bicycles and the jackets between them use up the hours exactly.

Read down the jackets column and each additional 200 bicycles costs exactly 600 jackets. The cost stays the same from row to row, and that constancy is what makes the line straight. Figure 1.1 plots the table and the line.

Tabulating a bowed frontier.

Food F	$C = 50 - 0.02F^2$	Clothing C	Cost of last 10 food
0	$50 - 0.02(0)$	50	—
10	$50 - 0.02(100)$	48	2
20	$50 - 0.02(400)$	42	6
30	$50 - 0.02(900)$	32	10
40	$50 - 0.02(1600)$	18	14
50	$50 - 0.02(2500)$	0	18

The order matters here: square first, then multiply. At $F = 30$, $C = 50 - 0.02(900) = 50 - 18 = 32$.

The last column takes the difference between one clothing entry and the next: 2, 6, 10, 14, 18. Each block of ten units of food costs more clothing than the block before it. The increasing opportunity cost comes from the shape of the equation, and the arithmetic makes it visible. Every entry in the table can be checked on its own. Figure 1.2 plots these six points and the curve through them.

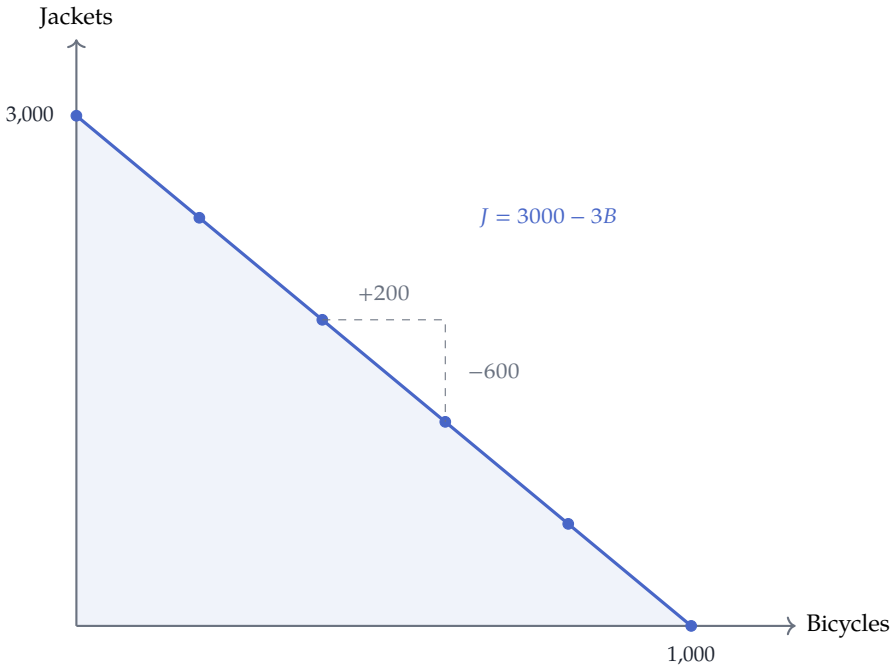


Figure 1.1: The linear frontier. Dots are the table; the line is the equation. The shaded region is everything the economy can produce.

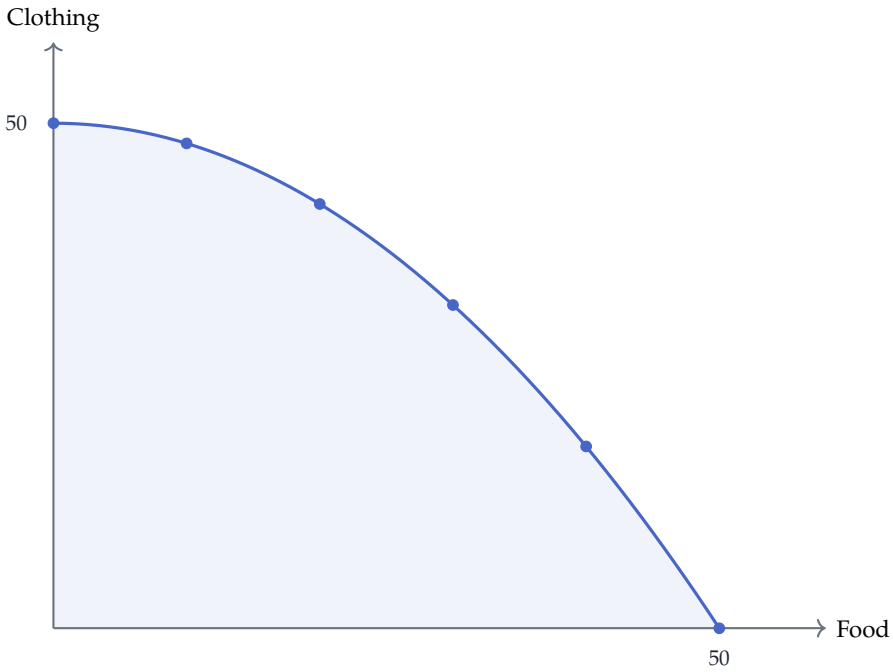


Figure 1.2: The bowed frontier. The slope steepens as food expands: each additional block of food costs more clothing than the last.

Gains from Trade

2.1 Definitions

Specialization and exchange. Agents concentrate their production on a narrow range of goods and get the rest by exchange. Nobody produces everything they consume.

Consider Smith's pin workshop, where ten men who divide the work between them make upwards of forty-eight thousand pins in a day. A man who makes only pins must get his food, clothing and shelter from others, so he specializes and he exchanges.

Absolute advantage. Producing a good with fewer inputs than another producer does. It is a direct comparison of productivities.

Suppose a unit of cloth takes 100 hours of labor in England and 90 in Portugal, and a unit of wine takes 120 hours in England and 80 in Portugal. Portugal needs fewer hours for both goods, so it has the absolute advantage in both.

Comparative advantage. Producing a good at a lower opportunity cost than another producer does.

A unit of wine costs Portugal 0.89 units of cloth and costs England 1.20, so Portugal has the comparative advantage in wine. A unit of cloth costs England 0.83 units of wine and costs Portugal 1.125, so England has the comparative advantage in cloth, even though it needs more hours than Portugal for both goods.

Autarky. The absence of trade. A country consumes precisely what it produces, so its production frontier is also its consumption frontier.

Suppose England puts 1,800 hours into each good. It then makes 18 units of cloth and 15 of wine, and under autarky that is exactly what it consumes.

Specialization raises world output. Now suppose all of each country's labor moves into the good in which that country holds the comparative advantage. England then produces $3600/100 = 36$ units of cloth and no wine. Portugal produces $2880/80 = 36$

units of wine and no cloth. World output is now 36 units of cloth and 36 of wine, against 34 and 33 under autarky.

Here England has 3,600 hours of labor and Portugal 2,880, and the autarky figures suppose that each country splits its hours equally between the two goods. So world output rises by two units of cloth and three of wine, and no hour of labor has been added.

Terms of trade. The rate at which the two goods exchange once trade opens. Let p be the price of one unit of wine in units of cloth. There is no money in the model, so a price is a ratio between goods.
Portugal sells wine, and it will not accept less than its own cost of producing wine. That requires $p > \frac{8}{9} \approx 0.89$. England buys wine, and it will not pay more than its own cost of producing wine, which requires $p < 1.20$. Both conditions hold only within

$$\frac{8}{9} < p < 1.20.$$

Suppose 17 units of wine trade for 17 units of cloth, so that $p = 1$. That price is above Portugal's cost of 0.89 units of cloth and below England's cost of 1.20, so both countries accept it.

Consumption beyond the frontier. With trade, a country's consumption point may lie outside its own production frontier. Production is still bounded by the frontier. Consumption is bounded by what the country produces together with what that production can be exchanged for.

After trade England consumes 19 units of cloth and 17 of wine. Its own frontier allows only 15.6 units of cloth at 17 units of wine, so England is consuming a bundle that it could not have produced.

2.2 Worked examples

Hours of labor required per unit.

	Cloth	Wine
England	100	120
Portugal	90	80

Each entry is a cost, so a larger number means worse performance. Portugal needs fewer hours than England for cloth, 90 against 100. It also needs fewer for wine, 80 against 120. So Portugal is the more productive country in both goods.

Ricardo states these requirements as the labor of a given number of men for one year.

Here they are read as hours. That changes nothing, because only the ratios between the four figures do any work.

Ricardo gave labor requirements but did not say how much labor each country had. Labor requirements fix the slope of a frontier but not its position, so the two endowments must be supplied here: England 3,600 hours and Portugal 2,880. These may be read as ten workers against eight. The figures are chosen so that the outputs derived below come out as whole numbers. The pattern of comparative advantage does not depend on them, although the output figures do.

The two frontiers. Labor put into cloth plus labor put into wine uses up the whole available supply. For England,

$$100C + 120W = 3600 \quad \Longrightarrow \quad C = 36 - 1.2\,W.$$

And for Portugal,

$$90C + 80W = 2880 \quad \Longrightarrow \quad C = 32 - \tfrac{8}{9}\,W.$$

So England can produce at most 36 units of cloth or 30 of wine. For Portugal the limits are 32 units of cloth or 36 of wine.

Both frontiers are straight lines. The slope of each one is that country's opportunity cost of wine, measured in cloth.

Opportunity costs derived for both countries. Start with England. One unit of wine takes 120 hours. Those 120 hours would otherwise have produced $120/100 = 1.2$ units of cloth. Therefore

$$\text{England: 1 wine costs } \tfrac{120}{100} = 1.20 \text{ cloth.}$$

Now run the same calculation the other way. One unit of cloth takes 100 hours, and those hours would otherwise have produced $100/120$ units of wine:

$$\text{England: 1 cloth costs } \tfrac{100}{120} \approx 0.83 \text{ wine.}$$

For Portugal, one unit of wine takes 80 hours, which would otherwise have produced $80/90$ units of cloth. One unit of cloth takes 90 hours, which would otherwise have produced $90/80$ units of wine:

$$\text{Portugal: 1 wine costs } \tfrac{80}{90} \approx 0.89 \text{ cloth}, \qquad 1 \text{ cloth costs } \tfrac{90}{80} = 1.125 \text{ wine.}$$

	1 wine costs (cloth)	1 cloth costs (wine)
England	$120 \div 100 = 1.20$	$100 \div 120 \approx 0.83$
Portugal	$80 \div 90 \approx 0.89$	$90 \div 80 = 1.125$

The two entries in each row are reciprocals, since $1.20 \times 0.83 \approx 1$ and $0.89 \times 1.125 \approx 1$. That is arithmetic rather than coincidence. With two goods and one resource, the cost of

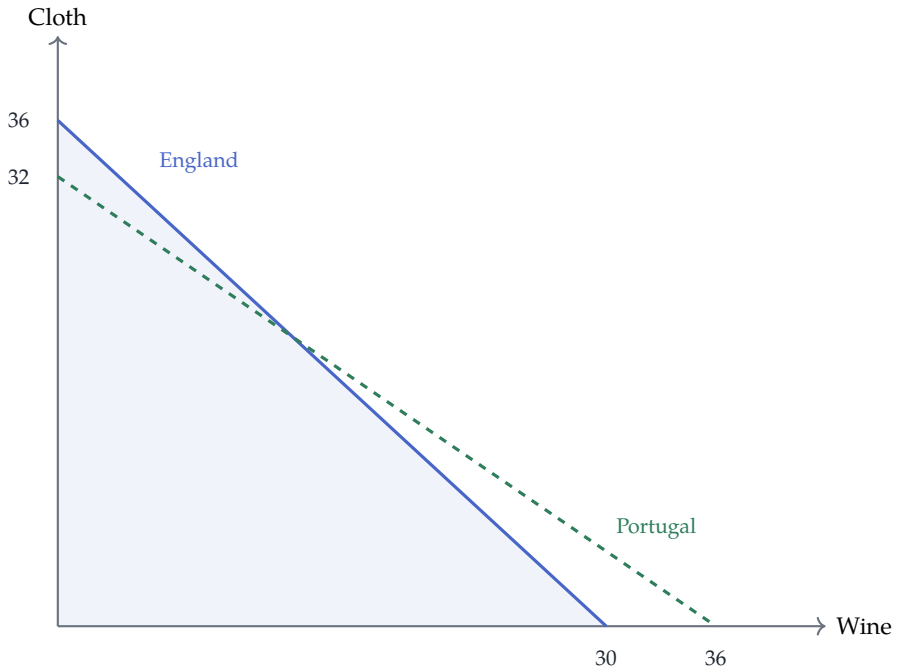


Figure 2.1: The two frontiers. England's is the steeper, so that wine is expensive to England in terms of cloth. Neither frontier lies wholly inside the other.

cloth in terms of wine and the cost of wine in terms of cloth are two readings of a single ratio. So a country that finds one good expensive necessarily finds the other cheap.

The wine column reproduces the slopes of the two frontiers in Figure 2.1, 1.20 and $\frac{8}{9}$. Again this is arithmetic, since the slope of a frontier is an opportunity cost. The cloth column holds their reciprocals.

We need a baseline against which a gain can be measured. Suppose each country splits its labor equally between the two goods: England puts 1,800 hours into each, and Portugal 1,440 into each. This split is assumed rather than derived. The pattern of comparative advantage does not depend on it, although the size of the gains reported below does.

Autarky output.

	Cloth	Wine
England	$1800/100 = 18$	$1800/120 = 15$
Portugal	$1440/90 = 16$	$1440/80 = 18$
World	**34**	**33**

England may be checked against its frontier, $36 - 1.2(15) = 18$, and Portugal against its own, $32 - \frac{8}{9}(18) = 16$. Both countries are on their frontiers, since each uses all of its labor. Gains to the world are measured against 34 units of cloth and 33 of wine. Gains to each country are measured against its own row.

Set $p = 1$, so one unit of cloth trades for one of wine. That price lies inside the band, and it gives whole numbers.

The volume exchanged. Let X be the number of units exchanged. England keeps $36 - X$ units of cloth and receives X units of wine. Portugal keeps $36 - X$ units of wine and receives X units of cloth.

England improves on autarky if $36 - X > 18$ and $X > 15$, which requires $15 < X < 18$. Portugal improves on autarky if $X > 16$ and $36 - X > 18$, which requires $16 < X < 18$. So both conditions hold when $16 < X < 18$. Take $X = 17$.

The outcome.

	Cloth made	Cloth consumed	Wine made	Wine consumed
England	36	$36 - 17 = 19$	0	$0 + 17 = 17$
Portugal	0	$0 + 17 = 17$	36	$36 - 17 = 19$

Before trade, England had 18 units of cloth and 15 of wine. Now it has 19 and 17 (Figure 2.2). Portugal had 16 and 18, and now has 17 and 19. Each country has more of both goods, and neither has worked a single extra hour.

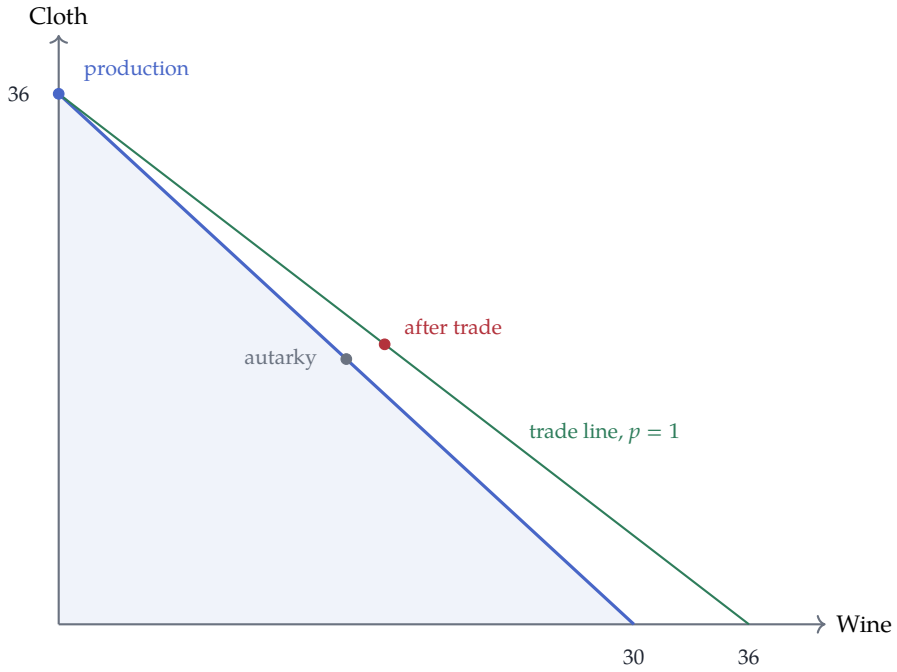


Figure 2.2: England. Production remains at the cloth intercept, while consumption moves beyond the frontier along the trade line.

Supply and Demand

3.1 Definitions

The law of demand. Other things equal, when the price of a good goes up, the quantity demanded of that good goes down.

In the coffee market that runs through this chapter, quantity is measured in thousands of cups per week, price in dollars per cup, and demand is

$$Q^d = 70 - 10P. \tag{3.1}$$

Raise the price from \$3 to \$4 with nothing else changed, and the quantity demanded falls from 40 thousand cups to 30 thousand.

Market demand. The horizontal sum of the individual demand curves. At each price, add up the quantities that each buyer demands.

Suppose the market has two buyers, and at a price of \$3 Ama demands 24 thousand cups while Ben demands 16 thousand. The market quantity demanded at \$3 is then $24 + 16 = 40$ thousand cups, and the two buyers face the same price, so no prices are added.

The law of supply. Other things equal, when the price of a good rises, the quantity supplied of that good rises.

Supply in the coffee market is $Q^s = 10P - 10$. At \$3 sellers offer 20 thousand cups a week and at \$4 they offer 30 thousand, because the higher price makes the extra output worth producing.

Market equilibrium. The price and quantity at which quantity demanded equals quantity supplied. At that price the market clears. Every buyer who is willing to pay it finds a seller, and every seller who is willing to accept it finds a buyer.

In the coffee market, quantity demanded and quantity supplied are both 30 thousand cups a week at a price of \$4, so \$4 is the equilibrium price. At \$3 buyers want 40 thousand cups and sellers offer only 20 thousand, which means that \$3 does not clear the market.

Comparative statics. Comparing the equilibrium before an event with the equilibrium after it. The method compares two resting points and says nothing about the path between them or the time it takes.

Suppose demand for coffee goes up by 20 at every price while supply stays where it was. The equilibrium moves from \$4 and 30 thousand cups to \$5 and 40 thousand cups, and the method reports those two points without saying how the market got from the first to the second.

3.2 Worked examples

The demand schedule.

P (\$/cup)	Substitute into $Q^d = 70 - 10P$	Q^d (000s/week)
1	$70 - 10(1)$	60
2	$70 - 10(2)$	50
3	$70 - 10(3)$	40
4	$70 - 10(4)$	30
5	$70 - 10(5)$	20
6	$70 - 10(6)$	10
7	$70 - 10(7)$	0

The table is not some second object that just happens to agree with Equation (3.1). It is that same equation, evaluated seven times over, and any row can be rebuilt in one line.

Two buyers. Suppose Ama and Ben have the demands

$$Q_A^d = 42 - 6P, \qquad Q_B^d = 28 - 4P.$$

Ama buys more than Ben at every price below \$7, and she responds more strongly to a change in price. Her coefficient is -6, against Ben's -4. Both of them stop buying at \$7, since $42/6 = 7$ and $28/4 = 7$.

P	Ama	Ben	Market Q^d
1	36	24	$36 + 24 = 60$
2	30	20	$30 + 20 = 50$
3	24	16	$24 + 16 = 40$
4	18	12	$18 + 12 = 30$
5	12	8	$12 + 8 = 20$
6	6	4	$6 + 4 = 10$
7	0	0	$0 + 0 = 0$

The final column is just the demand schedule of Section 3.2. In algebra, the sum takes one step:

$$Q^d = (42 - 6P) + (28 - 4P) = 70 - 10P.$$

The intercepts add, $42 + 28 = 70$, and the coefficients add, $6 + 4 = 10$. A single line of algebra does the job of the whole table.

Figure 3.1 draws the three curves.

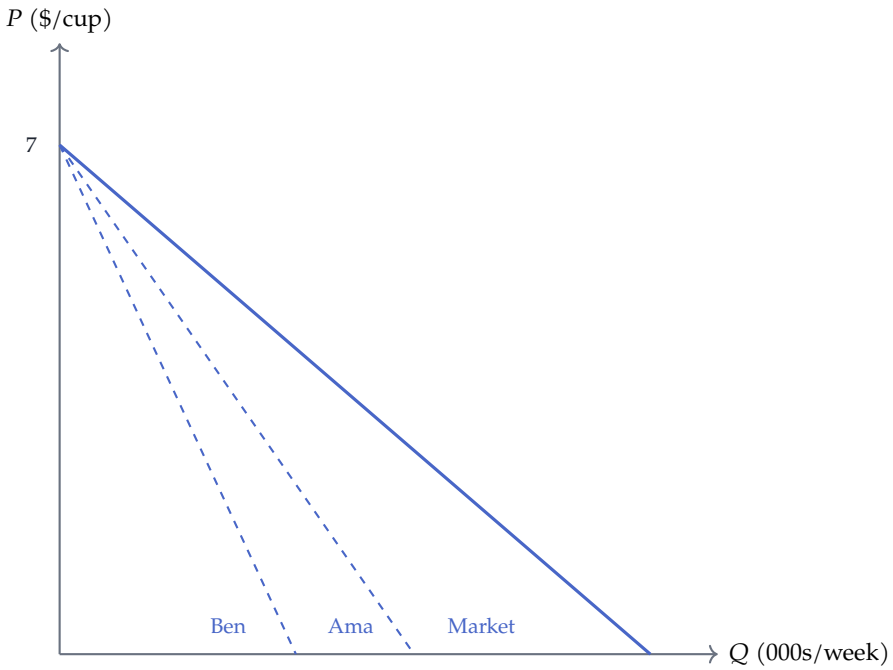


Figure 3.1: Horizontal summation. At any price the market quantity is Ama's plus Ben's, so the market curve lies to the right of Ama's curve by Ben's quantity and to the right of Ben's by Ama's. All three meet the price axis at \$7 because the two buyers share a choke price.

Solving for equilibrium. First, set $Q^d = Q^s$:

$$\begin{aligned} 70 - 10P &= 10P - 10 \\ 70 + 10 &= 10P + 10P \\ 80 &= 20P \\ P^* &= 4. \end{aligned}$$

Then substitute the price back through *both* equations:

$$Q^d = 70 - 10(4) = 30, \qquad Q^s = 10(4) - 10 = 30.$$

The answers agree, and the agreement is what makes it a check. So equilibrium is \$4 per cup and 30 thousand cups per week.

Figure 3.2 shows the equilibrium.

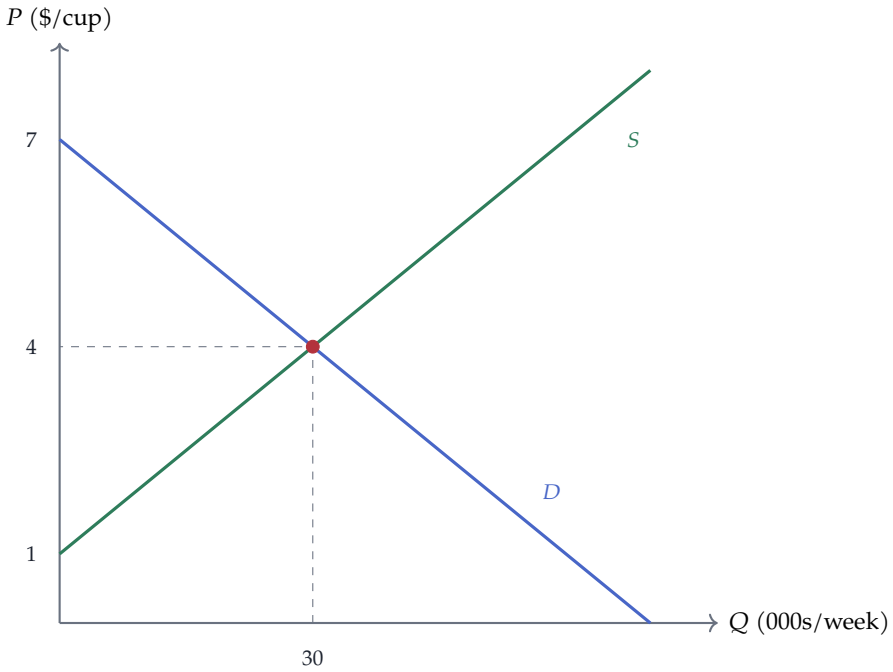


Figure 3.2: Equilibrium at \$4 and 30 thousand cups. The demand curve meets the price axis at the choke price, \$7; the supply curve meets it at the minimum supply price, \$1.

A shortage at \$3.

$$Q^d = 70 - 10(3) = 40, \qquad Q^s = 10(3) - 10 = 20, \qquad \text{shortage} = 20.$$

Queues form, stock runs out, and sellers notice that they can charge more without losing customers. As the price goes up, Q^d falls and Q^s rises simultaneously. So the gap closes from both sides, and it disappears at \$4.

A surplus at \$5.

$$Q^d = 70 - 10(5) = 20, \qquad Q^s = 10(5) - 10 = 40, \qquad \text{surplus} = 20.$$

Unsold stock piles up, and sellers cut their prices to clear it. Q^s falls, Q^d rises, and once again the market ends up at \$4.

Figure 3.3 shows both gaps.

Case 1, demand increases. Suppose a study reports that coffee is good for your health. Demand goes up by 20 at every price, so the intercept moves from 70 to 90. Supply stays where it was.

$$90 - 10P = 10P - 10 \implies 100 = 20P \implies P^* = 5,$$

$$Q^* = 10(5) - 10 = 40.$$

Check it through demand: $90 - 50 = 40$. The price moves \$4 → \$5 and the quantity moves 30 → 40, so both rise.

Figure 3.4 shows the shift.

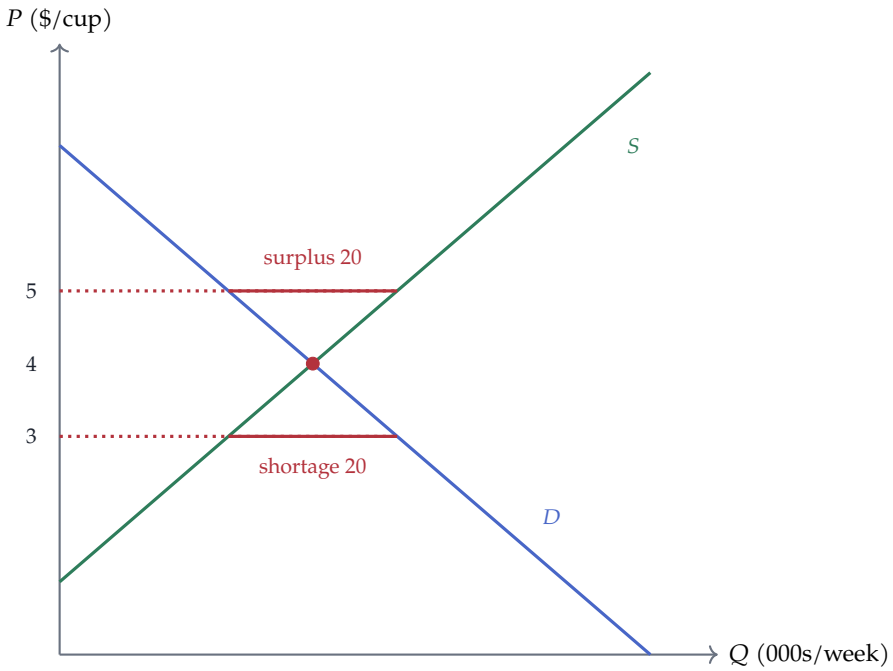


Figure 3.3: Disequilibrium from either side. Above \$4 the horizontal gap is a surplus; below \$4 it is a shortage. Both close as the price moves toward \$4.

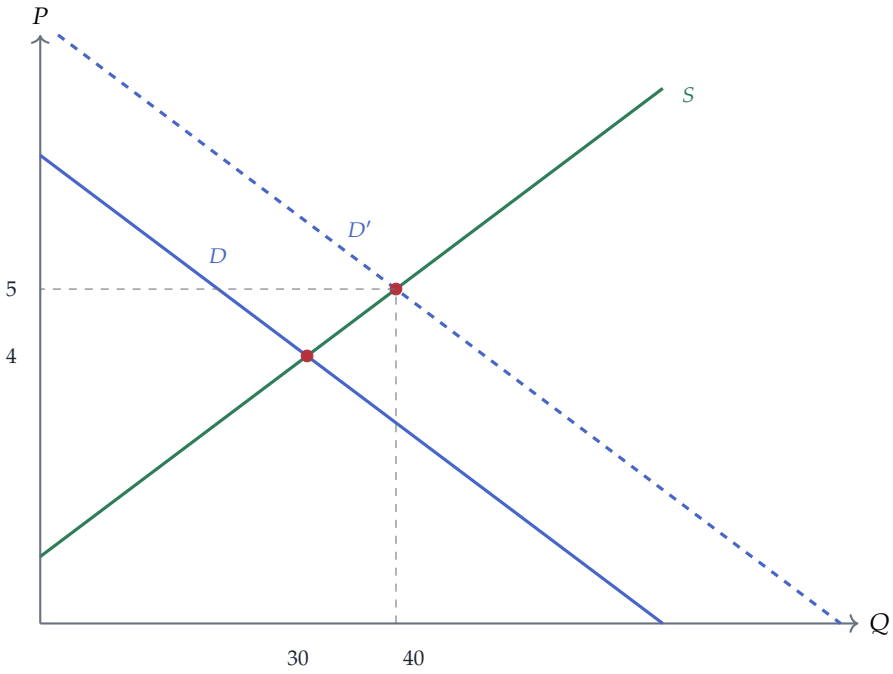


Figure 3.4: Case 1. Demand shifts right and the equilibrium moves up along a fixed supply curve. Price and quantity both rise.

Case 2, supply decreases. Now suppose a frost destroys part of the crop. Supply falls by 20 at every price, so its intercept moves from -10 to -30. This time demand stays put.

$$70 - 10P = 10P - 30 \implies 100 = 20P \implies P^* = 5,$$

$$Q^* = 10(5) - 30 = 20.$$

Again, check through demand: $70 - 50 = 20$. The price moves \$4 → \$5, but the quantity moves 30 → 20. This time they go in opposite directions.

Figure 3.5 shows the shift.

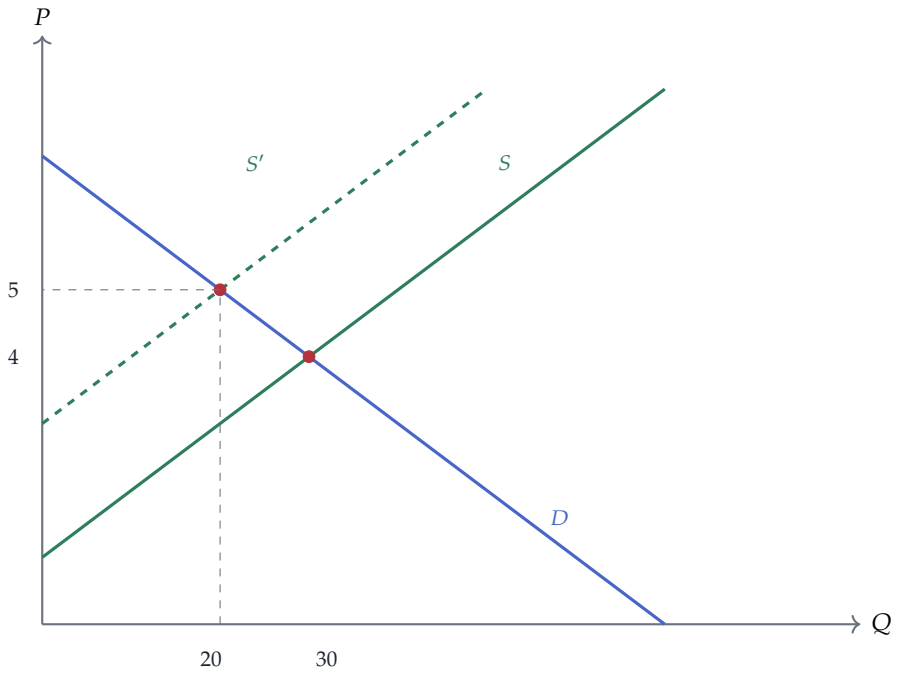


Figure 3.5: Case 2. Supply shifts left and the equilibrium moves up along a fixed demand curve. The price rises and the quantity falls.

Elasticity

4.1 Definitions

Elasticity. The percentage change in one variable divided by the percentage change in another variable. Because the numerator and the denominator are both percentages, the units cancel and the ratio is a pure number. It has no cups in it, no dollars and no bushels.

For example, a slope of 10 in the coffee market means thousands of cups per dollar, so it cannot be put beside a slope from the housing market. An elasticity of 1.80 can be compared across markets, because the units have gone.

Price elasticity of demand.

$$E_d = \frac{\%\,\Delta Q^d}{\%\,\Delta P}.$$

Quantity goes in the numerator and price goes in the denominator. If you swap them, you are answering a different question.

On the coffee curve, a price change of 22.2 percent goes with a quantity change of 40 percent, so $|E_d| = 40.0/22.2 = 1.80$. The law of demand makes the sign negative every time, and for that reason we report the absolute value.

The vocabulary.

Condition	Term	Buyers ...
$\lvert E_d\rvert > 1$	elastic	respond more than proportionately
$\lvert E_d\rvert = 1$	unit elastic	respond exactly in proportion
$\lvert E_d\rvert < 1$	inelastic	respond less than proportionately
$\lvert E_d\rvert = 0$	perfectly inelastic	do not respond at all
$\lvert E_d\rvert \to \infty$	perfectly elastic	flee any price rise entirely

For example, when coffee is \$4 the elasticity is 1.33, so demand at that price is elastic. At \$2 it is 0.40, and demand is inelastic.

Point elasticity of demand. The price elasticity of demand, evaluated at a single point on a straight-line demand curve $Q^d = a - bP$,

$$|E_d| = \left|\frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}\right| = b\,\frac{P}{Q}.$$

Along a straight line, $\Delta Q/\Delta P = -b$ at every point, and for a price change of any size. That means the measure is exact.

The coffee curve is $Q^d = 70 - 10P$, so $b = 10$. At a price of \$4.50 the quantity is 25, and the point elasticity is $10 \times 4.50/25 = 1.80$.

Arc elasticity of demand. The price elasticity of demand, evaluated at the averages of two points on a straight-line demand curve $Q^d = a - bP$,

$$|E_d| = \left|\frac{\Delta Q}{\Delta P} \cdot \frac{\bar{P}}{\bar{Q}}\right| = b\,\frac{\bar{P}}{\bar{Q}}.$$

Here $\bar{P}$ and $\bar{Q}$ are the averages of the two prices and of the two quantities. Take the coffee prices of \$5 and \$4, where the quantities are 20 and 30. The averages are 4.50 and 25, so the arc elasticity is $10 \times 4.50/25 = 1.80$.

The two extremes. *Perfectly inelastic*, the vertical case. For example, a transplant patient will buy the prescribed dose of anti-rejection medication at very nearly any price. There is no substitute for the medication, and there is no option of taking less. So a change in price moves the quantity bought very little, and demand comes close to the vertical case.
Perfectly elastic, the horizontal case. Consider one wheat farmer among thousands, selling wheat identical to everyone else's. If the farmer charges a cent above the going price, every buyer goes elsewhere. If the farmer charges a cent below it, the farmer is simply giving money away.

On the diagram, the first case is a vertical demand curve, with $|E_d| = 0$. The second is a horizontal one, where $|E_d| \to \infty$.

The five determinants.

1. **Close substitutes.** Butter's price rises and buyers switch to margarine; eggs have no stand-in, so egg buyers stay. More substitutes mean more elasticity.
2. **How the market is drawn.** *Food* has no substitute and its demand is inelastic. *One bakery's sourdough loaf* has a substitute in every other bakery and its demand is highly elastic. The narrower the market, the more elastic the demand.

3. **Necessity or luxury.** A doctor's visit happens at almost any price; a sailboat does not.
4. **Share of income.** Rent doubling rearranges a life; salt doubling goes unnoticed. A bigger budget share makes demand more elastic.
5. **Time to adjust.** Gasoline jumps and this month the driving to work continues. Over years come a smaller car, a carpool, a move closer in.

The estimated elasticities show the determinants at work. Eggs, at 0.1, have no close substitute and take a small share of the budget, while restaurant meals, at 2.3, have substitutes everywhere.

Total revenue. The total amount that buyers pay and that sellers receive, $TR = P \times Q$. On the diagram, total revenue is a rectangle whose height is the price and whose width is the quantity. So revenue is not an extra concept attached to supply and demand. It is an area that has been inside the picture all along.

For example, at the coffee equilibrium the price is \$4 and the quantity is 30 thousand cups, so $TR = 4 \times 30 = \$120$ thousand per week.

Elasticity and total revenue. Suppose the price rises. The *price effect* is that each unit now sells for more, and that pushes TR up. The *quantity effect* is that fewer units are sold, and that pushes TR down.
Which of the effects wins? The elasticity settles that, because it measures exactly what is in dispute: whether quantity moves proportionately more or less than price.

	Price rises	Price falls
Elastic ($\lvert E_d \rvert > 1$)	TR falls	TR rises
Unit elastic ($\lvert E_d \rvert = 1$)	TR unchanged	TR unchanged
Inelastic ($\lvert E_d \rvert < 1$)	TR rises	TR falls

When demand is elastic, it is the quantity effect that wins. When demand is inelastic, it is the price effect.

Suppose 100 units sell at \$10, and the price rises by 20 percent, to \$12. If $\lvert E_d \rvert = 2$, quantity falls to 60 and revenue falls from \$1,000 to \$720. If $\lvert E_d \rvert = 0.5$, quantity falls only to 90, and revenue rises to \$1,080.

Price elasticity of supply.

$$E_s = \frac{\%\,\Delta Q^s}{\%\,\Delta P}.$$

Because supply slopes upward, both changes have the same sign, and so E_s comes out positive on its own, with no need for an absolute value. The vocabulary carries over unchanged. Supply is elastic above one, unit elastic at one and inelastic below one, and the same two extremes sit at the ends.

Suppose a bakery supplies 3,000 loaves a week at \$3 a loaf, and 5,000 loaves at \$4. By the midpoint method, quantity changes by 50 percent and price by 28.6 percent, so $E_s = 1.75$ and supply is elastic.

The hard limit. Some quantities are fixed by nature rather than by cost. No price, however high, will create more coastline. So the physical supply of beachfront land is perfectly inelastic, $E_s = 0$. This is the one constraint that no amount of time will relax. At the other extreme is $E_s \to \infty$, which is a horizontal supply curve. At the going price, the quantity supplied responds without limit.

On the diagram, the supply of beachfront land is a vertical line, with the same quantity at every price. The other extreme is a horizontal line at the going price.

Tax incidence. Who actually bears the burden of a tax, as opposed to who is legally required to hand the money over. The person who writes the check is not necessarily the person who pays.

For example, take a \$1 tax in a market where $|E_d| = 0.43$ and $E_s = 1.71$. Buyers pay \$6.80, up from \$6.00, and sellers keep \$5.80, so buyers bear 80 cents of every dollar of tax and sellers bear 20 cents.

4.2 Worked examples

One line, seven elasticities. $Q^d = 70 - 10P$, which has a slope of 10 everywhere.

P (\$)	Q (000s)	$10 \times P/Q$	$\lvert E_d \rvert$	Verdict
6	10	$10 \times 6/10$	6.00	elastic
5	20	$10 \times 5/20$	2.50	elastic
4	30	$10 \times 4/30$	1.33	elastic
3.50	35	$10 \times 3.5/35$	1.00	unit elastic
3	40	$10 \times 3/40$	0.75	inelastic
2	50	$10 \times 2/50$	0.40	inelastic
1	60	$10 \times 1/60$	0.17	inelastic

The slope is 10 in every row, and that is what it means for a line to be straight. The only

thing that changes is the fraction P/Q, but it changes a great deal. As you move up the curve, P rises while Q falls, so the ratio is pushed from both ends at once. Going from 1/60 to 6/10 is a factor of thirty-six, and yet the slope of the line never moved.

Figure 4.1 puts three rows of the table on the demand curve.

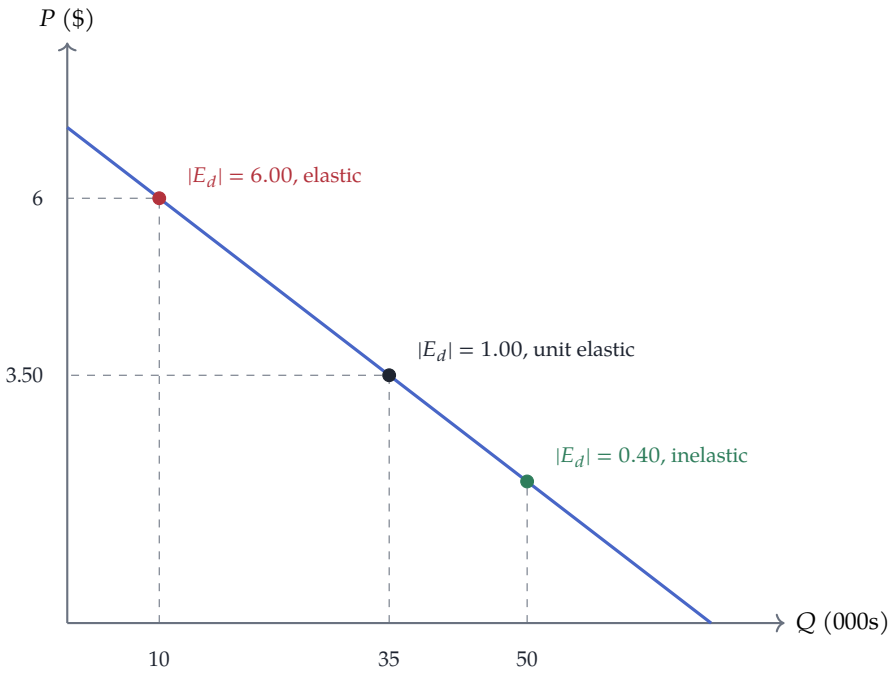


Figure 4.1: Three rows of the table, on one straight line. Elastic at the top, unit elastic at the midpoint, inelastic at the bottom.

Revenue along the coffee curve.

P ($)	Q (000s)	$TR = P \times Q$	$\lvert E_d \rvert$	Region
1	60	$1 \times 60 = 60.0$	0.17	inelastic
2	50	$2 \times 50 = 100.0$	0.40	inelastic
3	40	$3 \times 40 = 120.0$	0.75	inelastic
3.50	35	$3.5 \times 35 = 122.5$	1.00	unit elastic
4	30	$4 \times 30 = 120.0$	1.33	elastic
5	20	$5 \times 20 = 100.0$	2.50	elastic
6	10	$6 \times 10 = 60.0$	6.00	elastic

Now look down the revenue column: 60, 100, 120, 122.5, then 120, 100, 60. Revenue climbs, turns, and then falls. And the turn does not come roughly at the unit-elastic price. It comes exactly at that price.

Figure 4.2 plots the revenue column against quantity.

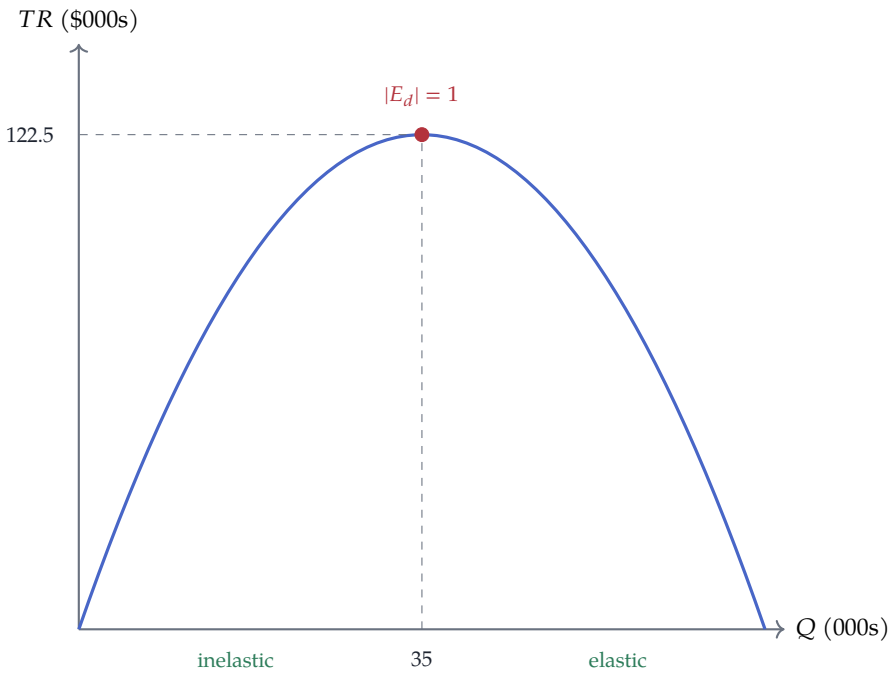


Figure 4.2: The revenue hill. Climbing the demand curve from the right, revenue rises through the inelastic region, peaks where $|E_d| = 1$, and falls through the elastic region.

Price Controls and Taxes

5.1 Definitions

Price ceiling. A legal *maximum* on the price at which a good can be sold. The standard example is rent control, and there the motive is usually to protect buyers.

Take the coffee market, where $Q^d = 70 - 10P$ and $Q^s = 10P - 10$, which clears at a price of \$4 with 30 thousand cups sold a week. A ceiling of \$5 sits above that price and forbids nothing that was going to happen anyway, whereas a ceiling of \$3 sits below it, and so \$3 becomes the price that holds.

Price floor. A legal *minimum* on the price at which a good can be sold. The standard example here is the minimum wage, and the motive is usually to protect sellers.

In the same coffee market, a floor of \$5 sits above the \$4 equilibrium, so it forbids the price the market was heading toward and it binds. A minimum wage works in the same way: where the labor market would clear at \$8 an hour, a floor of \$10 is binding.

5.2 Worked examples

A binding ceiling at \$3. In this case the ceiling binds, so the price is \$3. The demand and supply schedules both stay as they were. The only thing that has been fixed is the price.

$$Q^d = 70 - 10(3) = 40, \qquad Q^s = 10(3) - 10 = 20.$$

So the shortage is 20 thousand cups per week. The quantity actually traded is also 20 thousand, because a cup that was never produced cannot be bought by anybody.

Ordinarily, a shortage like this would push the price back up to \$4, and the shortage would close itself. But the ceiling forbids that price. So the mechanism that would have closed the gap no longer operates, and the shortage persists for as long as the ceiling is in place.

Figure 5.1 shows this market.

A binding floor at \$5.

$$Q^d = 70 - 10(5) = 20, \qquad Q^s = 10(5) - 10 = 40.$$

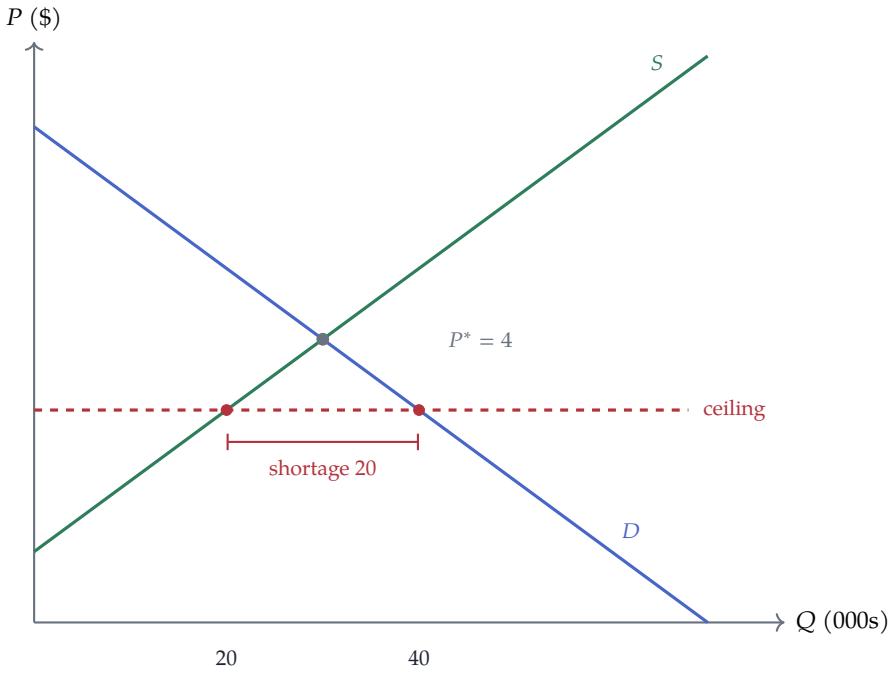


Figure 5.1: A binding ceiling at \$3. The shortage is a horizontal distance: a gap between two quantities at one price, never a gap between two prices.

The surplus is 20 thousand cups that sellers would supply but buyers will not take, whereas the ceiling produced unmet demand. The price cannot fall back to \$4 and clear it away, so the surplus persists.

Figure 5.2 shows this market.

Finding the equilibrium wage.

W	$L^d = 150 - 10W$	$L^s = 10W - 10$
\$6	$150 - 10(6) = 90$	$10(6) - 10 = 50$
\$7	$150 - 10(7) = 80$	$10(7) - 10 = 60$
\$8	$150 - 10(8) = 70$	$10(8) - 10 = 70$
\$9	$150 - 10(9) = 60$	$10(9) - 10 = 80$
\$10	$150 - 10(10) = 50$	$10(10) - 10 = 90$

The algebra gives the same answer. Setting demand equal to supply, $150 - 10W = 10W - 10$, gives $160 = 20W$, so $W^* = \$8$ and $L^* = 70$.

At \$8, every teenager who is willing to work at \$8 has a job, and every firm that is willing to pay \$8 has the workers it wants. So before the floor is introduced, there is no involuntary unemployment in this market.

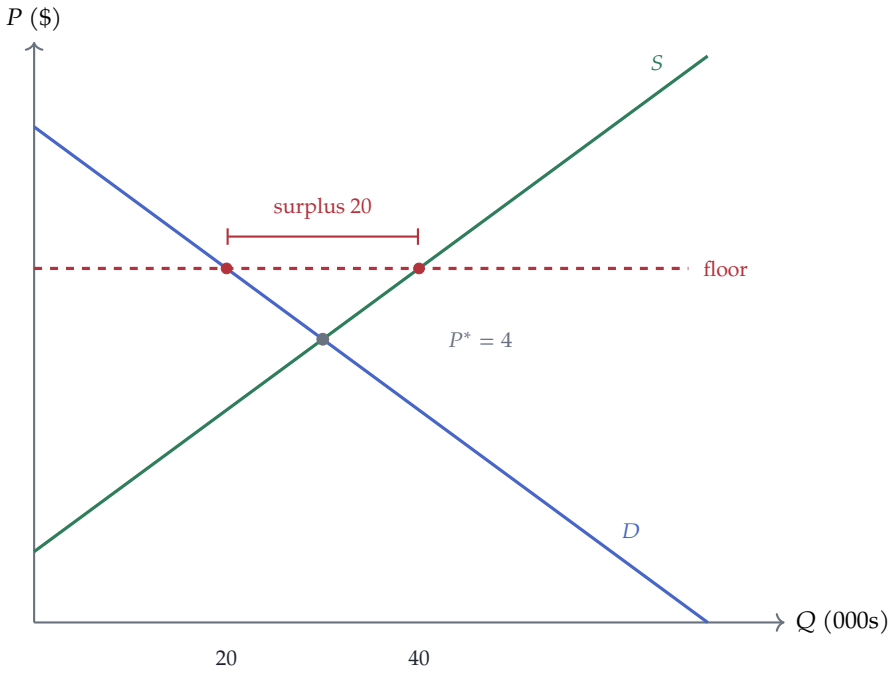


Figure 5.2: A binding floor at \$5. A control line above equilibrium always leaves supply exceeding demand; one below it always leaves demand exceeding supply.

A floor at \$10. Ten is above eight, so this floor is binding.

$$L^d = 150 - 10(10) = 50, \qquad L^s = 10(10) - 10 = 90.$$

The surplus is 40 thousand workers. A surplus of labor is what we call unemployment.

That 40 thousand is really made up of two things, which are equal here by construction. Employment fell from 70 to 50, so **20 thousand jobs were destroyed**. At the same time, labor supply rose from 70 to 90, so **20 thousand more were drawn into the search**. They came because \$10 an hour is worth working for, whereas \$8 was not. That second group were not unemployed before the floor, because they were not looking for work.

Figure 5.3 shows this market.

Sellers decide how much to produce on the basis of what they *keep*. With a \$1 tax, a till price of P leaves them with $P - 1$. So wherever the supply equation used to have P, it now has $P - 1$:

$$Q^s = 10(P - 1) - 10 = 10P - 20.$$

Solving, and checking both ways.

$$70 - 10P = 10P - 20 \implies 90 = 20P \implies P = \$4.50.$$

Buyers pay the till price, $P_B = \$4.50$. Sellers keep $P_S = 4.50 - 1.00 = \$3.50$, which is the till price less the tax. The quantity traded is 25 thousand cups.

There are two ways to check it. Demand at the buyers' price gives $70 - 10(4.50) = 25$. The original supply curve at the sellers' price gives $10(3.50) - 10 = 25$. So the buyers'

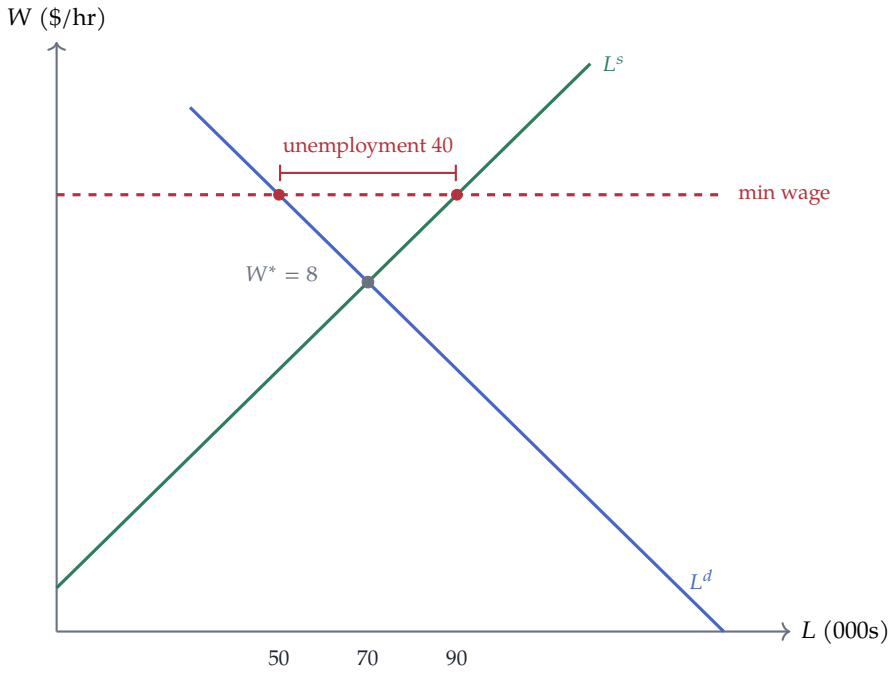


Figure 5.3: A minimum wage of \$10. The width of the red bracket depends on how far the floor sits above equilibrium and on how elastic the two curves are.

price goes into the demand equation, and the sellers' price goes into the original supply equation.

The burden, and why it is even.

Buyers: $4.50 - 4.00$	$+\$0.50$
Sellers: $3.50 - 4.00$	$-\$0.50$
Check: $0.50 + 0.50$	$= \$1.00$

The burden splits evenly because, at this equilibrium, the elasticities of demand and supply are equal:

$$|E_d| = 10 \times \tfrac{4}{30} = 1.33, \qquad E_s = 10 \times \tfrac{4}{30} = 1.33.$$

The equal split is a choice we made in designing the example. The general rule is that the more inelastic side bears more of the tax. This market has deliberately been made symmetric, so that attention falls on the machinery rather than on how the burden is split.

Now change exactly one thing. The same \$1 tax is collected from the buyer at the register instead. Buyers then respond to their all-in cost, $P + 1$:

$$Q^d = 70 - 10(P + 1) = 60 - 10P.$$

The intercept of the demand curve falls from 70 to 60, so the whole curve shifts down by exactly the tax. Supply returns to its original form, $Q^s = 10P - 10$, because sellers now remit nothing to the government.

Solving the buyer-side tax.

$$60 - 10P = 10P - 10 \implies 70 = 20P \implies P = \$3.50.$$

The till price is now \$3.50, and that is what sellers keep. Buyers pay 3.50 + 1.00 = \$4.50 all in. The quantity is 25 thousand cups.

Figures 5.4 and 5.5 show the seller-side and the buyer-side tax in turn.

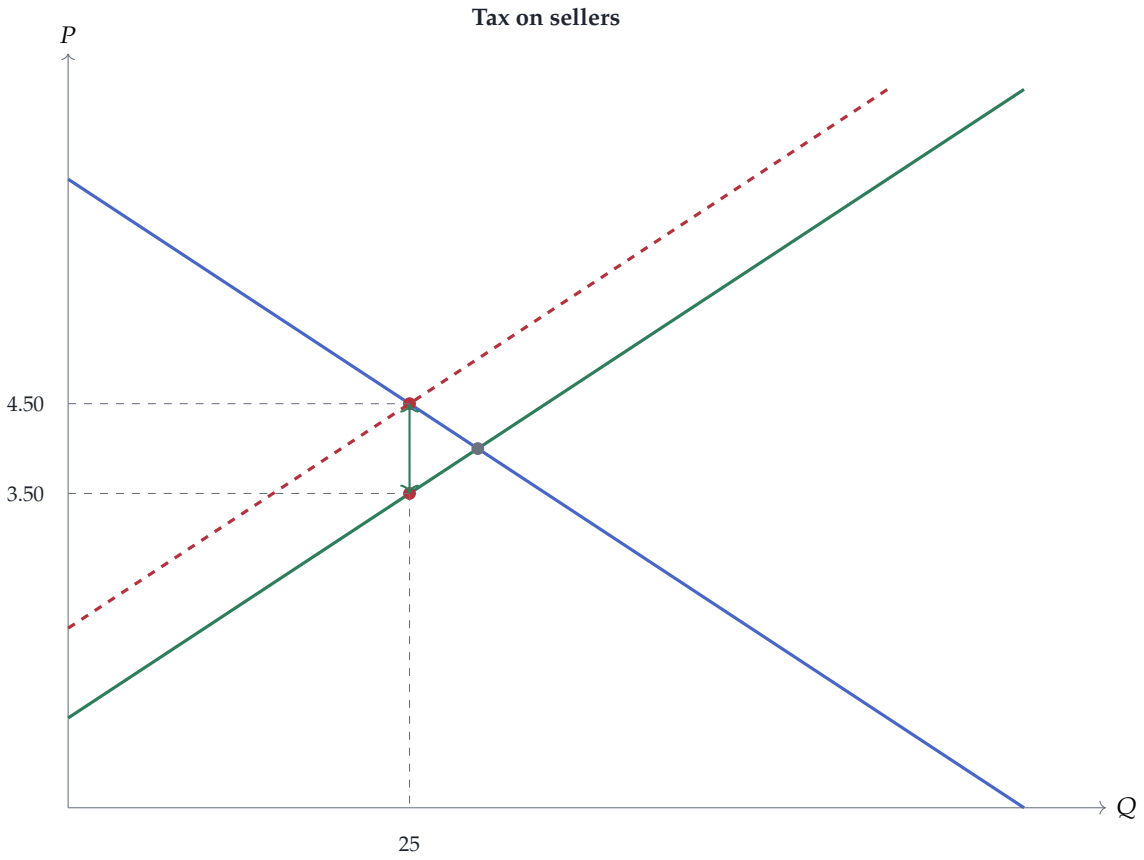


Figure 5.4: The \$1 tax collected from sellers. The dashed curve is supply, shifted.

The trades the tax prevents. If we invert the two curves, then for the Qth cup a buyer's value is $\frac{70-Q}{10}$ and a seller's cost is $\frac{Q+10}{10}$.

Cup	Buyer would pay	Seller's cost	Gain	Survives \$1 wedge?
24	$\frac{70-24}{10} = 4.60$	$\frac{24+10}{10} = 3.40$	1.20	yes
26	$\frac{70-26}{10} = 4.40$	$\frac{26+10}{10} = 3.60$	0.80	no
28	$\frac{70-28}{10} = 4.20$	$\frac{28+10}{10} = 3.80$	0.40	no
30	$\frac{70-30}{10} = 4.00$	$\frac{30+10}{10} = 4.00$	0.00	no

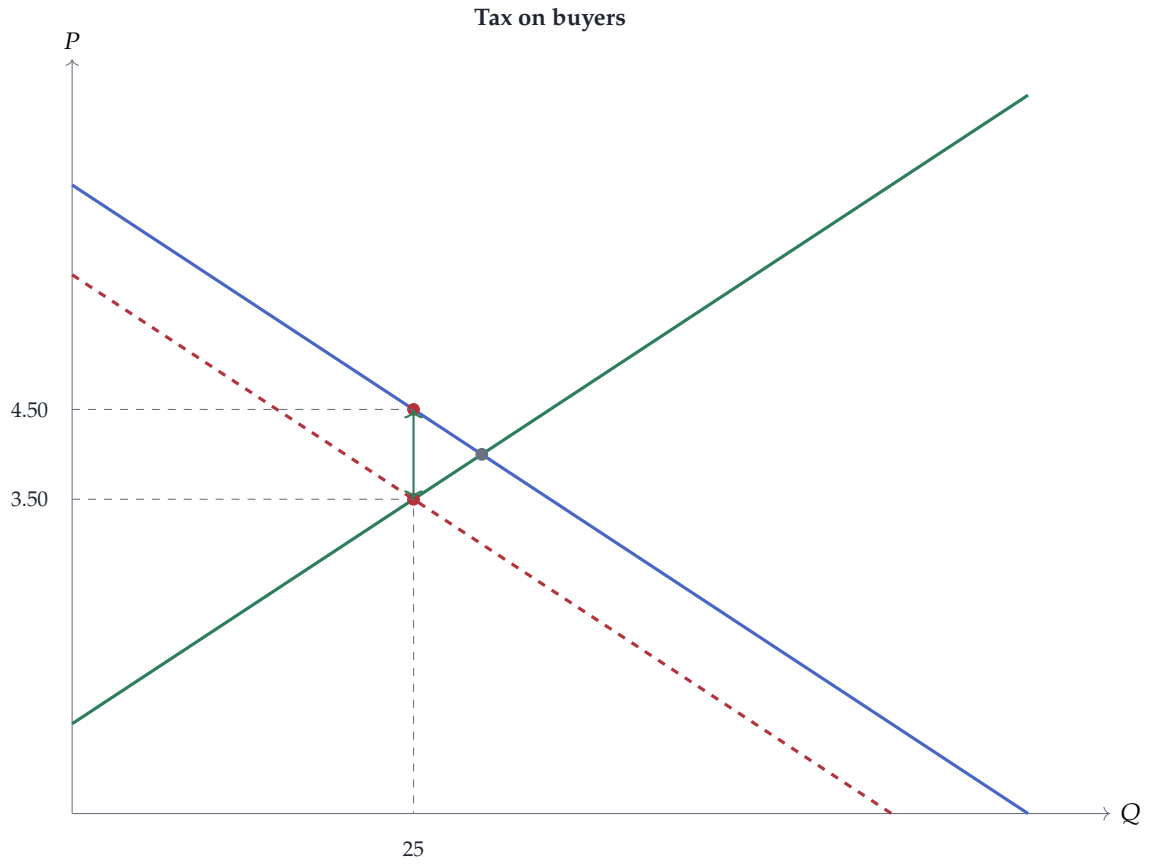


Figure 5.5: The same market, with the tax collected from buyers. The dashed curve is on a different side. Every number is the same as in Figure 5.4.

Take cup 26 thousand as an example. A buyer values it at $4.40, and a seller can make it for $3.60. So the trade would create 80 cents of value, and both sides want it to happen. But for the trade to happen, the seller must receive at least $3.60. With the $1 tax on top, that means the buyer must pay at least $4.60. The buyer, however, will pay at most $4.40. No price satisfies both of them. Eighty cents of gain cannot cover a dollar wedge.

The same is true of every trade whose gain is smaller than the tax. It fails to happen, even though both sides still wanted it.

Market Efficiency

6.1 Definitions

Willingness to pay. Willingness to pay is the most that a buyer would hand over for one unit of a good. If the seller charges even one cent more than that, the buyer does not trade.

For example, Maya would hand over at most \$6.00 for a cup of coffee, and Ben would hand over at most \$2.00. So at a price of \$4.00, Maya buys and Ben does not.

Diminishing marginal utility. Each additional unit that a buyer consumes adds less satisfaction than the unit before it. So if we hold fixed what a dollar is worth to the buyer, the dollar value that the buyer places on each successive unit falls.

Consider Maya's morning, cup by cup. Her first cup is worth \$6.00 to her, her second \$4.50, her third \$3.00 and her fourth \$1.50, so at a price of \$4.00 she buys two cups and then stops.

Consumer surplus. Consumer surplus is what a buyer would have paid minus what the buyer actually pays, CS $=$ WTP $- P$. It measures the buyer's gain from the trade, in dollars.

Suppose the price of a cup is \$3.50. Maya would have paid \$6.00, so her consumer surplus is $6.00 - 3.50 = \$2.50$, while Priya, who would have paid exactly \$3.50, gains \$0.

Producer surplus. Producer surplus is what a seller is paid minus the seller's cost, PS $= P -$ cost. On a diagram, it is the area above the supply curve and below the price, out to the quantity traded.

For example, it costs Rosa \$1.50 to supply a cup from her cart. At a price of \$3.00, her producer surplus on that cup is $3.00 - 1.50 = \$1.50$.

Total surplus. Total surplus is consumer surplus plus producer surplus. It is the whole gain that the market creates for the people who take part in it.

The worked examples below use a coffee market with demand $Q^d = 70 - 10P$ and supply $Q^s = 10P - 10$, which clears at a price of \$4 with 30 thousand cups traded a week. In that market, consumer surplus is \$45,000 and producer surplus is \$45,000, so total surplus is $45 + 45 = \$90{,}000$ per week.

Efficiency. An allocation is efficient when it maximizes total surplus. In an efficient allocation, no change in who buys, who sells or how much is traded can create more value.

In the coffee market, the efficient quantity is 30 thousand cups. At 20 thousand, the marginal cup is worth \$5.00 to a buyer and costs \$3.00 to make, so stopping there gives up value, and at 40 thousand it is worth \$3.00 and costs \$5.00, so the last cups destroy value.

Market failure. Market failure is a situation in which the equilibrium of an unregulated market does not maximize total surplus.

For example, suppose a trade affects people other than its own buyer and seller. Then the value and the cost that the market acts on are not the full value and the full cost, and the quantity it trades is not the one that maximizes total surplus.

6.2 Worked examples

Consumer surplus in the coffee market. At $P^* = \$4$ the market trades $Q^* = 30$ thousand cups. The demand curve starts at \$7. That is the choke price, and it is also the value of the first cup. From there the curve runs down to \$4, which is the value of the marginal cup at 30 thousand. So the region is a triangle. Its base is 30, and its height is $7 - 4 = 3$:

$$\text{CS} = \tfrac{1}{2} \times 30 \times (7 - 4) = \tfrac{1}{2} \times 30 \times 3 = 45.$$

So buyers gain \$45,000 a week in total.

Figure 6.1 shows this triangle.

A price of \$5. Now suppose the price rises to \$5. At that price, quantity demanded falls to 20 thousand cups, and

$$\text{CS} = \tfrac{1}{2} \times 20 \times (7 - 5) = 20,$$

so consumer surplus falls from \$45,000 to \$20,000. That is a loss of \$25,000, and the loss has two parts. First, buyers pay \$1 more on each of the 20 thousand cups that they still buy, and that comes to \$20,000. Second, some buyers drop out and no longer buy. They lose the surplus they were earning on 10 thousand cups, which were worth between \$4 and \$5 to them. That lost surplus is a small triangle, $\tfrac{1}{2} \times 10 \times 1 = \$5{,}000$. Add the two parts together, and we are back at \$25,000.

Figure 6.2 shows the two parts of the loss.

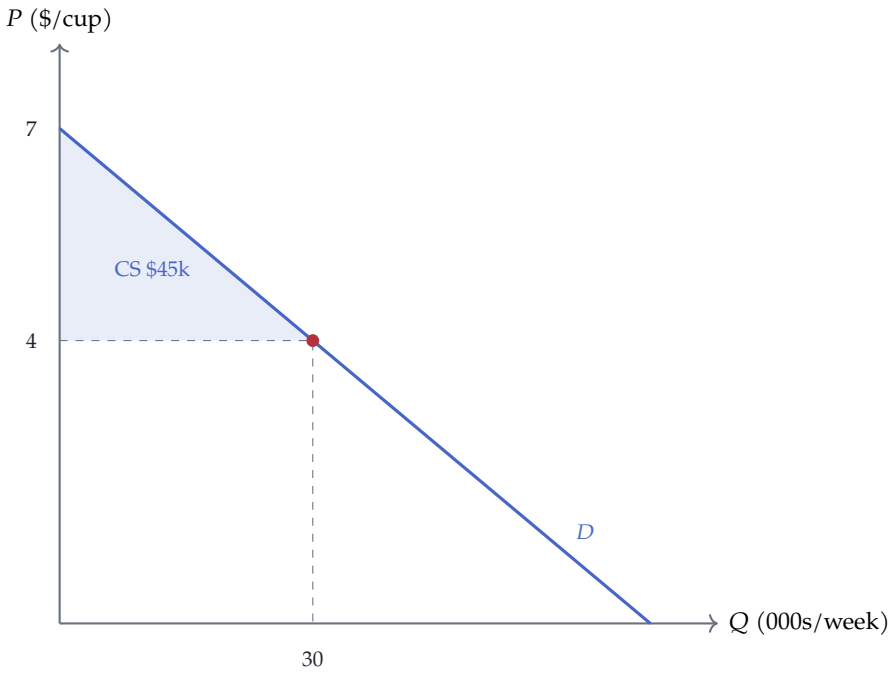


Figure 6.1: Consumer surplus at the equilibrium price. The shaded triangle lies below the demand curve and above $4, out to 30 thousand cups.

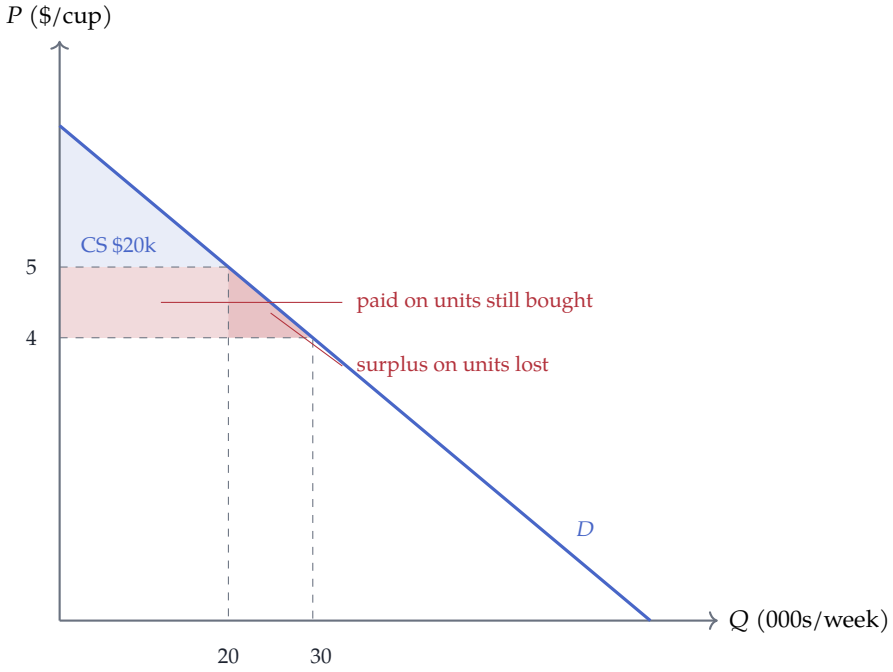


Figure 6.2: A rise in the price from $4 to $5 reduces consumer surplus in two parts: a rectangle paid by buyers who continue to buy, and a triangle of surplus on trades that no longer occur.

Producer surplus in the coffee market. At $P^* = \$4$ the supply curve starts at \$1. That is the minimum supply price, the cost of the cheapest cup to make. The curve then runs up to \$4, which is the cost of the marginal cup at 30 thousand:

$$\text{PS} = \tfrac{1}{2} \times 30 \times (4 - 1) = \tfrac{1}{2} \times 30 \times 3 = 45.$$

So sellers gain \$45,000 a week. As on the buyers' side, the marginal cup earns exactly zero surplus, because at the margin the price equals the cost.

Figure 6.3 shows the sellers' triangle.

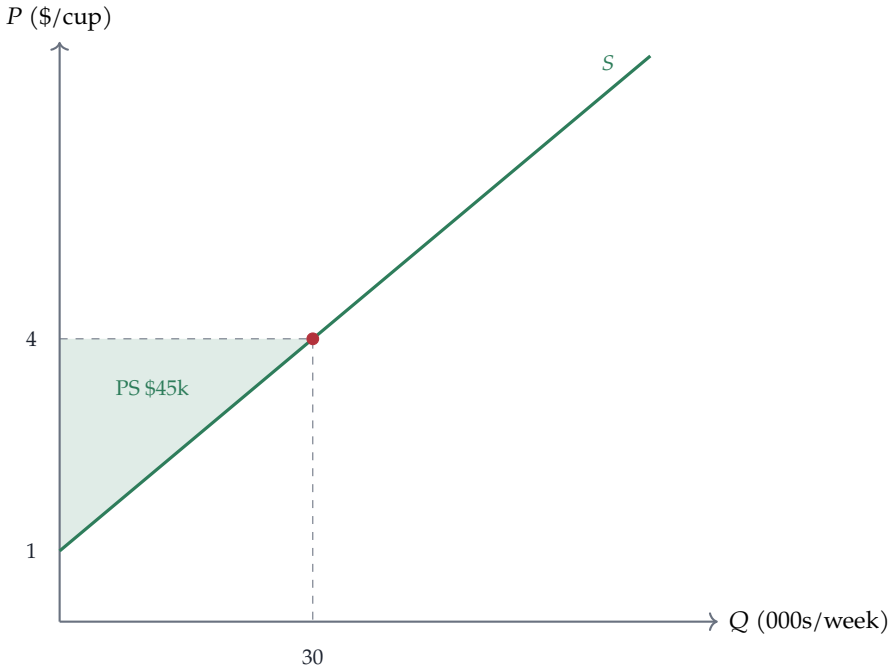


Figure 6.3: Producer surplus at the equilibrium price. The shaded triangle lies above the supply curve and below \$4, out to 30 thousand cups.

Is 30 thousand the right quantity?

Q (000s)	Marginal value	Marginal cost	Gap
20	$\frac{70-20}{10} = 5.00$	$\frac{20+10}{10} = 3.00$	+2.00
25	$\frac{70-25}{10} = 4.50$	$\frac{25+10}{10} = 3.50$	+1.00
30	$\frac{70-30}{10} = 4.00$	$\frac{30+10}{10} = 4.00$	0.00
35	$\frac{70-35}{10} = 3.50$	$\frac{35+10}{10} = 4.50$	−1.00
40	$\frac{70-40}{10} = 3.00$	$\frac{40+10}{10} = 5.00$	−2.00

Below 30 thousand, the gap between value and cost is positive. Each further cup is worth more than it costs, so stopping early gives up gains that were there to be had. Above 30 thousand, the gap is negative. Each further cup costs more than it is worth, so producing it

destroys value. At 30 thousand, the gap is zero. That is the quantity at which total surplus reaches its maximum. It is also the quantity at which the market settles, without anyone giving instructions.

A cap on the quantity. Suppose a licensing rule permits only 20 thousand cups per week, and the price is free to adjust. The trades between 20 and 30 thousand cups were creating value. Over that whole range, marginal value was above marginal cost, so the gap was positive. The licensing rule now prohibits those trades.

The surplus that is lost is a triangle. At 20 thousand cups, the gap between marginal value and marginal cost is \$2.00. The gap then shrinks to zero at 30 thousand. So the loss is $\frac{1}{2} \times 10 \times 2 = \$10{,}000$, and total surplus falls from \$90,000 to \$80,000.

There is a second, independent route to the same figure. On the Qth cup, the gain from trade is the difference between marginal value and marginal cost,

$$\frac{70 - Q}{10} - \frac{Q + 10}{10} = \frac{60 - 2Q}{10} = 6 - 0.2Q,$$

which is \$6.00 on the first cup and \$2.00 on the twenty-thousandth. The gain falls along a straight line, so take the average over that range: $(6.00 + 2.00)/2 = \$4.00$. Then multiply the average by the number of cups, and $4.00 \times 20 = \$80{,}000$.

The two routes give the same answer. Each of them is an exact computation of the same quantity. So if they disagreed, that would point to a slip in the arithmetic, rather than to a disagreement about the economics.

Externalities

7.1 Definitions

Externality. A cost or a benefit, arising from producing or consuming a good, that lands on someone who is not a party to the decision. That someone is called a bystander. The bystander neither pays for the effect nor is paid for it, through any price or contract.

Consider the market for firewood in a valley town, where $Q^d = 100 - 10P$ and $Q^s = 10P - 20$, and Q is measured in hundreds of bundles per week. Every bundle that is burned sends smoke onto households that bought no wood, and we suppose that the smoke from each bundle does \$2.00 of harm to them.

Social cost. The private cost plus the external cost. Put another way, it is what one more unit costs society once the bystanders are included.

For example, at 4,000 bundles a week the marginal bundle of firewood has a private cost of \$6.00. Add the \$2.00 of harm from its smoke, and the social cost of that bundle is \$8.00.

Deadweight loss. The total surplus that is lost to an inefficient outcome, most often because the quantity traded is not the quantity that maximizes total surplus. It is value destroyed either by trading too much or by trading too little, and nobody collects it.

The firewood market trades 4,000 bundles a week, but the quantity that maximizes total surplus is 3,000. Each of the 1,000 bundles in between costs society more than it is worth to its buyer, and the value destroyed adds up to \$1,000 per week.

Internalizing an externality. Changing incentives so that the parties to a trade face the full social cost of that trade, or its full social value.

Suppose the firewood seller has to pay the \$2.00 of harm on every bundle. Then the seller of bundle 4,000 faces the full \$8.00 that the bundle costs society, and not only the \$6.00 that it costs to produce.

Corrective tax. A tax on each unit, set equal to the external cost that the unit imposes. It lifts the curve that sellers respond to by exactly the amount by which that curve fell short of social cost.

A corrective tax is also called a Pigovian tax, and for firewood it is \$2.00 a bundle. With the tax in place, buyers pay \$7 a bundle, sellers keep \$5, and the quantity falls from 4,000 bundles a week to 3,000.

Social value. Private value, meaning the buyer's willingness to pay, plus the external benefit. In other words, it is what one more unit is worth to society when the bystanders are counted.

Consider a bike-share system in a congested city, where $Q^d = 80 - 10P$ and $Q^s = 10P - 20$, and Q is measured in hundreds of rides per day, and suppose each ride is worth \$2.00 to the drivers who are still on the road. At 3,000 rides a day the marginal ride is worth \$5.00 to its rider, so its social value is \$7.00.

Tradable pollution permits. Each permit allows its holder to discharge one unit. The number of permits issued is equal to the cap, and firms may buy and sell them.

Suppose two factories each discharge 80 tons a month into the same river, and the town wants the total brought down to 100 tons. The town issues 100 permits, and a factory that cuts a ton can sell the permit this frees up to the other factory.

Coase theorem. If the affected parties can bargain at no cost, they will reach the efficient outcome by themselves, whichever side the law favors to begin with.

For example, suppose live music is worth \$400 a week to a café, and costs the bookshop next door \$600 a week in lost sales. The bookshop can offer the café \$500 to keep quiet, and that leaves each of them \$100 better off, even though the law gives the café the right to play.

7.2 Worked examples

The firewood market before the smoke is counted.

$$100 - 10P = 10P - 20 \implies 120 = 20P \implies P^* = 6,$$

and $Q^* = 100 - 10(6) = 40$, or 4,000 bundles a week. As a check, substitute the price into supply instead: $10(6) - 20 = 40$. The two routes give the same answer, and that agreement confirms it.

What a bundle is worth and what it costs society.

Q (00s)	Worth to buyer	Private cost	Social cost	Society's gain
20	$\frac{100-20}{10} = 8.00$	$\frac{20+20}{10} = 4.00$	$4.00 + 2 = 6.00$	+2.00
30	$\frac{100-30}{10} = 7.00$	$\frac{30+20}{10} = 5.00$	$5.00 + 2 = 7.00$	0.00
40	$\frac{100-40}{10} = 6.00$	$\frac{40+20}{10} = 6.00$	$6.00 + 2 = 8.00$	−2.00

Start at 2,000 bundles. The marginal bundle is worth \$8.00 and costs society \$6.00, so burning it leaves the valley \$2.00 better off. At 3,000 bundles, the value and the social cost are both \$7.00, so the gain has run out. At 4,000, which is where the market settles, the bundle is worth \$6.00 but costs society \$8.00. So burning that bundle leaves the valley \$2.00 worse off. Figure 7.1 shows the two quantities.

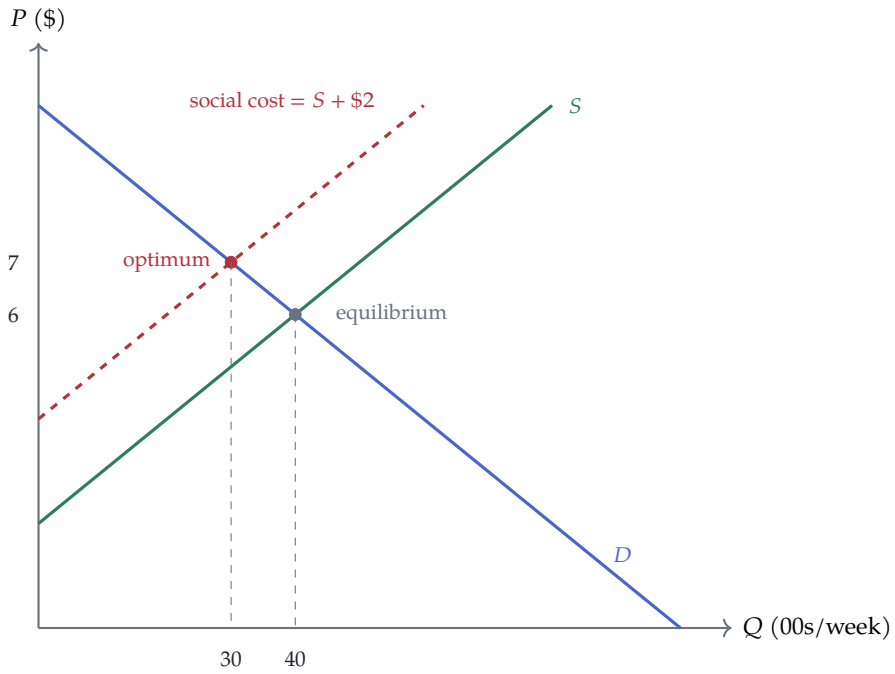


Figure 7.1: The firewood market. The dashed curve is the supply curve lifted by \$2 at every quantity, which is what a bundle costs once the smoke is counted. The market settles where demand meets private supply, at 40, and the surplus-maximizing quantity is 30.

The deadweight loss from the smoke. The loss on the marginal bundle grows steadily. It is \$0.00 at 3,000 bundles, where value still equaled social cost, and it reaches \$2.00 at 4,000. When a quantity rises in a straight line from 0 to 2 across 1,000 bundles, the total adds up to a triangle:

$$\text{DWL} = \tfrac{1}{2} \times 1{,}000 \times \$2.00 = \$1{,}000 \text{ per week.}$$

Figure 7.2 shows the triangle.

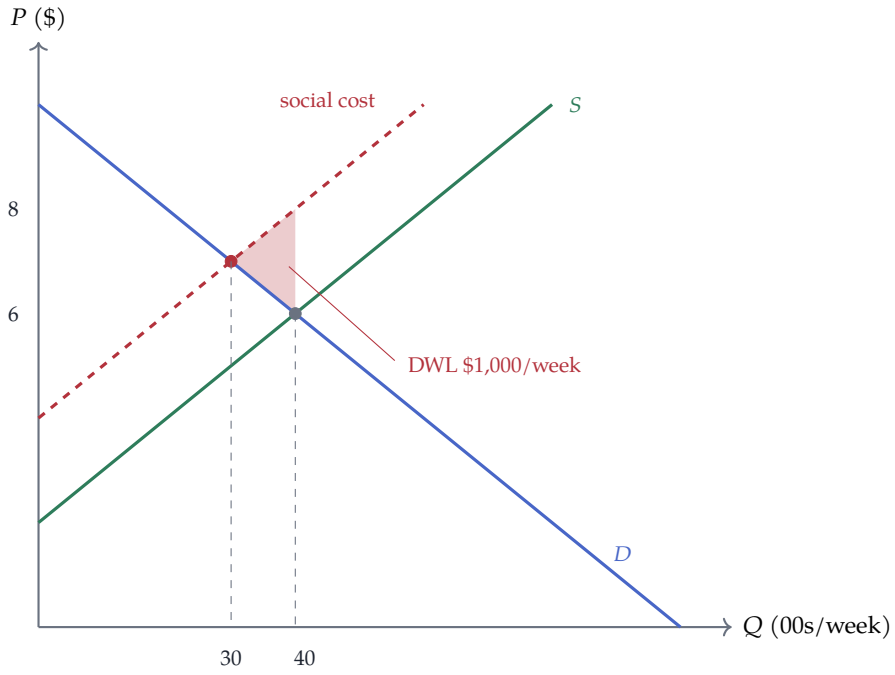


Figure 7.2: The deadweight loss is the shaded triangle between the demand curve and the social cost curve, from the optimum at 30 to the equilibrium at 40. Its vertical side at 40 measures \$2.00, the value destroyed by the last bundle burned.

A \$2.00 tax per bundle. After the tax, sellers keep $P - 2$, and so the supply curve, as the market sees it, becomes

$$Q^s = 10(P - 2) - 20 = 10P - 40.$$

Now set demand equal to this new supply curve,

$$100 - 10P = 10P - 40 \implies 140 = 20P \implies P = 7,$$

and $Q = 100 - 70 = 30$. So buyers pay \$7 a bundle and sellers keep \$5, and the town collects $\$2 \times 3{,}000 = \$6{,}000$ a week.

The bike-share market before the spillover is counted.

$$80 - 10P = 10P - 20 \implies 100 = 20P \implies P^* = 5,$$

and $Q^* = 80 - 10(5) = 30$, or 3,000 rides a day. If we substitute into supply instead, we get $10(5) - 20 = 30$, so the two routes agree.

The rides that do not happen.

Q (00s)	Social value	Cost	Society's gain
30	$\frac{100-30}{10} = 7.00$	$\frac{30+20}{10} = 5.00$	+2.00
35	$\frac{100-35}{10} = 6.50$	$\frac{35+20}{10} = 5.50$	+1.00
40	$\frac{100-40}{10} = 6.00$	$\frac{40+20}{10} = 6.00$	0.00
45	$\frac{100-45}{10} = 5.50$	$\frac{45+20}{10} = 6.50$	−1.00

At 3,000 rides, where the market stops, one more ride is worth $7.00 to society and costs $5.00 to provide. At 3,500 rides the gain is still $1.00, and by 4,000 it has fallen to zero. So the optimum solves

$$\frac{100 - Q}{10} = \frac{Q + 20}{10} \implies 100 - Q = Q + 20 \implies 2Q = 80 \implies Q = 40,$$

at a height of $6.00. So there are a thousand rides a day that would be worth more than they cost, and they never happen.

Two factories on one river. Northside Mill and Harbor Chemical each discharge 80 tons a month into the same river, for a total of 160 tons. The town wants to bring that total down to 100 tons, which means cutting 60 tons between them. Cutting one ton costs the Mill $50, and it costs Harbor $150.

Suppose the rule requires each firm to cut 30 tons:

$$\text{Mill: } 30 \times \$50 = \$1{,}500, \qquad \text{Harbor: } 30 \times \$150 = \$4{,}500,$$

for a total of $6,000 a month, and the target is met exactly.

The same cap through permits. The town issues 100 permits, 50 to each firm. Because there are only two firms, the price has to be negotiated between them, and any price between the Mill's $50 and Harbor's $150 suits both. They agree on $100.

The Mill can cut a ton for $50, and each permit it holds is worth $100. So if it cuts a ton and sells the permit that this frees up, the Mill nets $50. It cuts 60 tons, which is as many as Harbor will buy permits for. That costs $60 \times \$50 = \$3{,}000$. The Mill is then emitting 20 tons and holding 30 spare permits, which it sells for $30 \times \$100 = \$3{,}000$. So its net cost is $0.

Harbor cuts nothing, because cutting costs it $150 a ton while a permit costs $100. So it buys the Mill's 30 permits for $3,000 and goes on emitting 80 tons.

Emissions total $20 + 80 = 100$ tons, the same cap that the rule achieved. The only real resources used up are the Mill's $3,000 of abatement. The $3,000 in permit payments is a transfer between the two firms. It buys no cleanup and destroys nothing. So the true cost is $3,000, compared with $6,000 under the rule.

Figure 7.3 sets the two bills side by side.

Two plants and a 40-ton target. Plant A can cut a ton for $80, and plant B for $30. Each plant emits 50 tons, and the state wants the total brought down to 60, so 40 tons must be cut.

Under a uniform rule, each plant cuts 20 tons, and that costs $20 \times \$80 + 20 \times \$30 = \$1{,}600 + \$600 = \$2{,}200$.

Now suppose the state issues 60 permits instead, 30 to each plant, and the two plants agree on a price of $55. Plant B cuts all 40 tons, which costs $40 \times \$30 = \$1{,}200$, and emits 10. It sells 20 permits for $20 \times \$55 = \$1{,}100$, so its net cost is $100. Plant A cuts nothing and buys those 20 permits for $1,100. Emissions are $10 + 50 = 60$, and society's real abatement bill is B's $1,200.

For both firms to prefer trading to the rule, the permit price must lie between the two firms' own costs. Take plant A first. It buys 20 permits, instead of cutting 20 tons at a cost

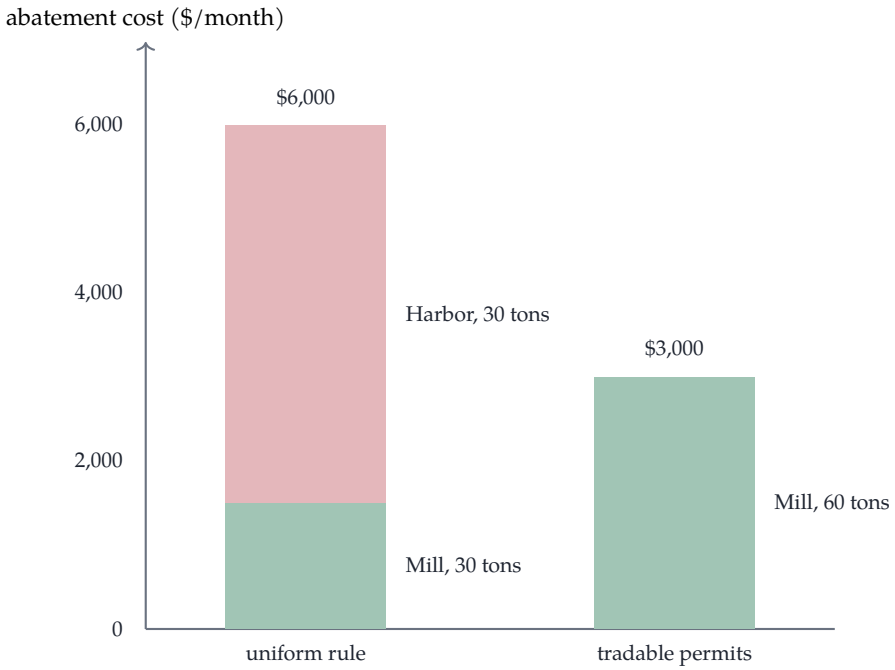


Figure 7.3: What 60 tons of abatement costs under each instrument. The rule buys 30 of the tons at Harbor's $150, and the permit market moves all 60 to the Mill at $50.

of $1,600, when $20p < 1{,}600$, that is, when $p < \$80$. Now take plant B. It cuts 40 and sells 20, at a net cost of $1{,}200 - 20p$. It prefers that to the $600 it would pay under the rule when $1{,}200 - 20p < 600$, that is, when $p > \$30$. Both conditions hold when

$$\$30 < p < \$80.$$

At a price of $80, plant A is indifferent between buying a permit and cutting the ton itself. At $30, it is plant B that is indifferent. So each end of the range is one firm's own cost per ton.

A café, a bookshop and a Friday night. Live music on Fridays at Café Solana is worth $400 a week to the café. The noise carries into Harriman Books next door, and it costs the bookshop $600 a week in lost evening sales. Under the law, the café has the right to play.

Quiet is worth $600 and the music is worth $400. So quiet creates more value, and yet the law allows the outcome that creates less.

Suppose the bookshop offers the café $500 to keep Fridays quiet. The café compares that $500 with the $400 value it places on the music, and it accepts. The bookshop avoids $600 of losses and pays $500, so it is $100 better off. The café gives up $400 of value and receives $500, so it too is $100 better off. No regulator is involved, and yet the efficient outcome is reached, because the bystander stopped being a bystander by making an offer.

Public Goods and Common Resources

8.1 Definitions

Excludable. A good is excludable if a supplier can stop people who have not paid for it from using it.

A burrito is excludable, because if there is no payment there is no burrito. Street lighting is not excludable, since the light that falls on a paying pedestrian falls just as much on the next one.

Rival in use. A good is rival in use if one person's use of a unit reduces what is left over for everybody else.

A fish, once landed, cannot be landed a second time, so fish are rival in use. A streamed show is not rival, because one person's viewing of it takes nothing away from anyone else's.

Public good. A good that is neither excludable nor rival. Nobody can be kept out, and a single supply serves everybody at once.

The mosquito treatment program at Alder Pond is a public good. A licensed crew treats the pond for $2,500 a season, and all 120 households around it get the benefit whether they pay or not.

Free rider. A person who enjoys the benefit of a good without paying for it.

The neighborhood association asks each of the 120 households around Alder Pond for $25. Take a household that values a mosquito-free season at $50. If it keeps its $25 and the program runs, it is a free rider, because the treatment reaches its yard either way.

Cost-benefit analysis. Comparing the total benefit of a project, to everyone it affects, with its total cost, in order to decide whether the project is worth doing.

For the Alder Pond program, the total benefit to the 120 households is $3,300 a season and the total cost is $2,500. The benefit is larger by $800, so the program is worth doing.

Common resource. A good that is rival in use but not excludable. Anyone may take from it, and every taking leaves less for the rest.

Birch Lake is a common resource, because it is open to anyone with a boat. Catch per boat is $1200 - 100\,b$ dollars of fish a day, where b is the number of boats on the water, so every boat already fishing loses $100 of catch when one more boat launches.

The tragedy of the commons. A user weighs the full private gain against the fraction of the true cost that lands on that user. Take any use whose full cost to everyone exceeds its gain, but whose private share of that cost does not. That use gets undertaken, and those are exactly the uses that should not happen.

A boat-day on Birch Lake costs $400. The fifth boat lands $700 of fish, so it launches, but $400 of that catch is fish the other four boats would have landed anyway.

Rent dissipation. The rent of a resource is the value that the resource itself generates over and above the cost of working it. Under open access, entry continues until every user exactly breaks even. A fleet in which every boat breaks even generates no net value at all. The rent is fully dissipated.

On Birch Lake a fleet of four boats earns a net $1,600 a day. Under open access eight boats launch, each one catches $400 against a cost of $400, and the net is $0.

8.2 Worked examples

The two columns a household faces.

	program runs anyway	program fails anyway
contribute $25	$50 - 25 = \$25$	−$25
keep the $25	$50	$0

Keeping the money is better in both columns. Contributing pays only in the one case the table leaves out, where the fund stands at exactly $2,475 without this check, and among 120 households that case is a knife edge. A choice that is best whatever everyone else does is called a dominant choice. Here keeping the money is dominant everywhere except on that knife edge. So the household barely needs to predict what its neighbors will do.

The Alder Pond program, valued. The 120 households do not value the program equally. Forty of them sit on the pond, and each would pay up to $50 for a calm summer. Another 60 are set back from the water and would pay up to $20 each. The last 20 are on the far ridge, where the mosquitoes barely reach, and would pay up to $5.

group	benefit	running total
40 households × \$50	= \$2,000	\$2,000
60 households × \$20	= \$1,200	\$3,200
20 households × \$5	= \$100	\$3,300
total benefit B		\$3,300
total cost C		\$2,500
net gain $B - C$		3,300 − 2,500 = \$800

So the town is \$800 a season richer with the program than without it, and the test says build it.

Figure 8.1 draws the two totals of this example side by side.

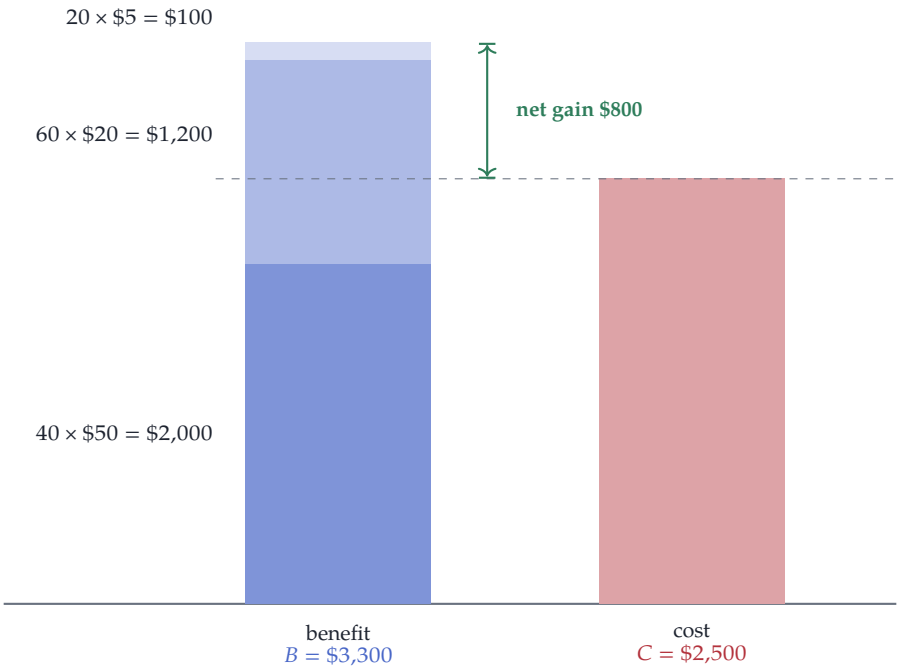


Figure 8.1: The benefit column, stacked by group, against the cost column. The dashed line carries the cost level across, and the benefit clears it by \$800 a season.

Equal shares. Suppose the cost is split equally, at 2,500/120 = \$20.83 per household, and the program is put to a referendum.

group	value	tax	net	vote
40 households	\$50	\$20.83	$50 - 20.83 = +\$29.17$	yes
60 households	\$20	\$20.83	$20 - 20.83 = -\$0.83$	no
20 households	\$5	\$20.83	$5 - 20.83 = -\$15.83$	no
result				no, by 80 to 40

The project makes the town \$800 a season richer, and it loses two to one.

The lake, tabulated.

boats b	catch per boat	total catch	extra from the last boat
1	$1200 - 100 = 1100$	$1 \times 1100 = 1100$	—
2	$1200 - 200 = 1000$	$2 \times 1000 = 2000$	$2000 - 1100 = 900$
3	$1200 - 300 = 900$	$3 \times 900 = 2700$	$2700 - 2000 = 700$
4	$1200 - 400 = 800$	$4 \times 800 = 3200$	$3200 - 2700 = 500$
5	$1200 - 500 = 700$	$5 \times 700 = 3500$	$3500 - 3200 = 300$
6	$1200 - 600 = 600$	$6 \times 600 = 3600$	$3600 - 3500 = 100$
7	$1200 - 700 = 500$	$7 \times 500 = 3500$	$3500 - 3600 = -100$
8	$1200 - 800 = 400$	$8 \times 400 = 3200$	$3200 - 3500 = -300$
9	$1200 - 900 = 300$	$9 \times 300 = 2700$	$2700 - 3200 = -500$

Total catch climbs to \$3,600 at six boats and then falls, because crowding eventually outweighs the extra headcount. From the seventh boat onward the extra catch is negative, so the fleet lands less fish than it would with one boat fewer.

The net column.

boats b	total catch	cost $400b$	net
1	1100	400	$1100 - 400 = 700$
2	2000	800	$2000 - 800 = 1200$
3	2700	1200	$2700 - 1200 = 1500$
4	3200	1600	$3200 - 1600 = 1600$
5	3500	2000	$3500 - 2000 = 1500$
6	3600	2400	$3600 - 2400 = 1200$
7	3500	2800	$3500 - 2800 = 700$
8	3200	3200	$3200 - 3200 = 0$
9	2700	3600	$2700 - 3600 = -900$

The net column peaks at four boats and \$1,600 a day. At six boats, where total catch peaks, net is still \$1,200. So maximizing the catch is not the same as maximizing the value, because catching fish costs money. Net reaches exactly zero at eight boats, and that figure comes back later.

Figure 8.2 plots the net column.

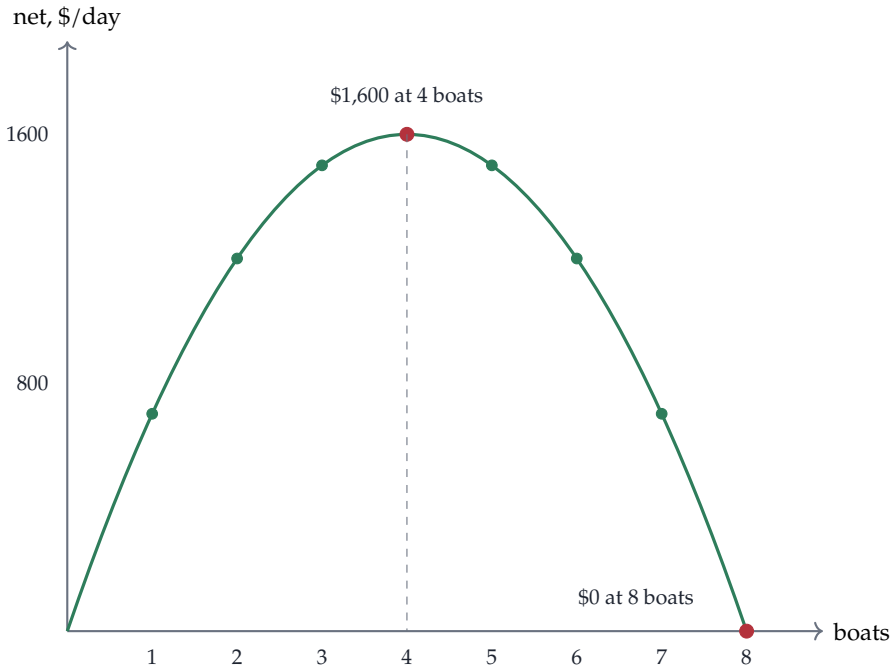


Figure 8.2: Net value of the fishery, total catch less $400 a boat, at each fleet size. The curve peaks at four boats and returns to zero at eight, which is where open access settles.

Boat by boat against cost.

boat	extra catch	against $400	verdict
3rd	$700	$700 > 400$	launch
4th	$500	$500 > 400$	launch
5th	$300	$300 < 400$	stay ashore

Entry stops at four boats. The two routes agree, and they had to. The net column rises exactly as long as each added boat brings in more than it costs, and it falls once it does not. So the peak of net and the last boat worth adding are one fact, read two ways. That is what makes the agreement a useful check: two computations that must agree will differ only if one of them contains an error.

Entry under open access.

boat	catch per boat	against $400	decision
6th	$600	$600 > 400$	launch
7th	$500	$500 > 400$	launch
8th	$400	$400 = 400$	launch, breaking even
9th	$300	$300 < 400$	stay ashore

Entry stops at eight boats, which is twice the four boats that maximize the lake's total profit.

Figure 8.3 shows both entry rules against the cost of a boat-day.

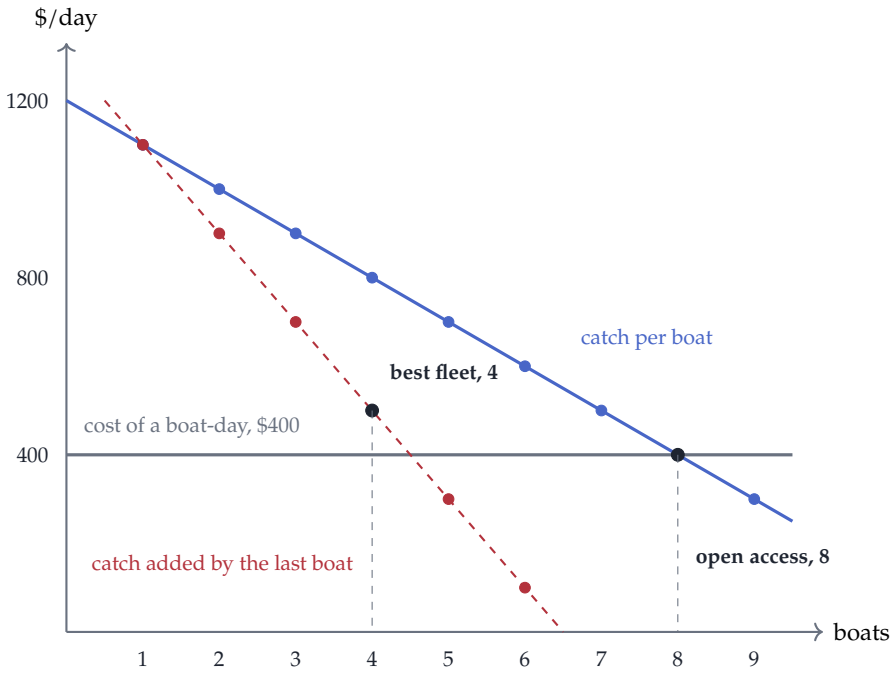


Figure 8.3: Catch per boat and catch added by the last boat, against the $400 cost of a boat-day. The added catch falls twice as fast, because each entrant both catches less itself and removes $100 from every boat already fishing. The added-catch line crosses $400 between the fourth and fifth boats and the catch-per-boat line crosses it at the eighth, and those crossings fix the two fleets.

Two fleets, side by side.

	best fleet, 4 boats	open access, 8 boats
total catch	$3,200	$3,200
cost of boats	$4 \times 400 = \$1{,}600$	$8 \times 400 = \$3{,}200$
net per day	$1,600	$0

Both catch figures come straight from the first table, where $4 \times 800 = 3{,}200$ and $8 \times 400 = 3{,}200$. So the two fleets land the same amount of fish.

Public Choice

9.1 Definitions

Rent created by a restriction. The price minus the marginal cost at the restricted quantity, multiplied by that quantity. The rent is paid to whoever holds the right to sell, and it exists only because the restriction exists.

Take the coffee market of Chapter 6 under a cap of 20 thousand cups a week. Buyers pay \$5 a cup and the twenty-thousandth cup costs \$3 to produce, so the rent is $(5-3)\times 20 = 40$, or \$40,000 a week. It is the rectangle in Figure 9.1.

Collective action problem. The failure of a group of people to act in their common interest.

Suppose 10 sellers hold the licenses, with \$78,000 a year at stake each, and 10,000 buyers purchase the coffee, with \$130 a year at stake each. So a holder can gain by paying for a campaign to keep the cap, and a buyer cannot gain by paying for one to remove it, even though the buyers lose more as a group.

Rational ignorance. Choosing to stay uninformed about a policy because the expected benefit of learning about it is less than the cost of learning.

Consider a buyer with \$130 a year at stake and a 1 in 100,000 chance of casting the decisive vote. The expected benefit of learning about the cap is \$0.0013, and learning takes an hour valued at \$20, so the buyer stays uninformed.

Rent-seeking. Spending real resources to obtain a transfer that a policy has created. The spending determines who gets the transfer, and it adds nothing to output.

Suppose the council will issue 10 licenses, each a right to sell 2,000 cups a week and each worth \$4,000 a week. If applicants spend the full \$40,000 on lawyers and consultants to win them, the cap costs society \$50,000 a week and not \$10,000, as Figure 9.2 shows.

Regulatory capture. The decisions of a regulator come to serve the industry that it regulates.

A licensing board could raise the cap to 30 thousand cups. Each holder would then lose $1,500 a week and each buyer would gain $2.50, so the arguments the board hears about the size of the cap come mainly from the holders.

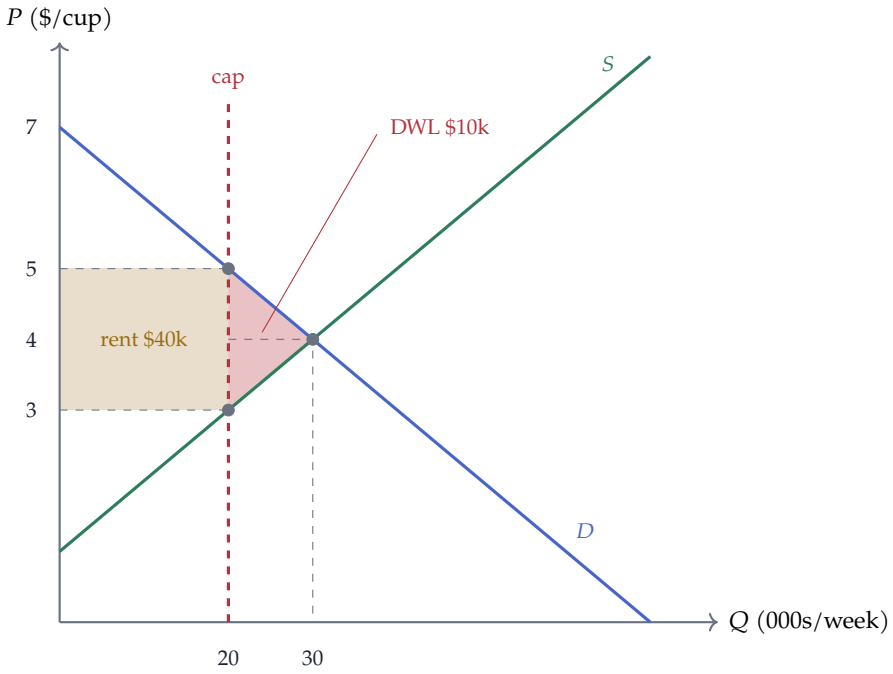


Figure 9.1: A cap of 20 thousand cups a week. The rectangle between $3 and $5 is the rent of $40,000 a week, paid to whoever holds a license. The triangle is the deadweight loss of $10,000 found in Chapter 6.

9.2 Worked examples

Who gains and who loses from the cap. The equilibrium values are the ones found in Chapter 6. Consumer surplus under the cap is the figure that Chapter 6 found at a price of $5.

	No cap	Cap of 20 thousand	Change
Consumer surplus	$45,000	$\frac{1}{2} \times 20 \times 2 = \$20{,}000$	−$25,000
Producer surplus	$45,000	$3 \times 20 = \$60{,}000$	+$15,000
Total surplus	$90,000	$80,000	−$10,000

The fall in total surplus is the deadweight loss of $10,000 from Chapter 6, which we have now reached by a second route.

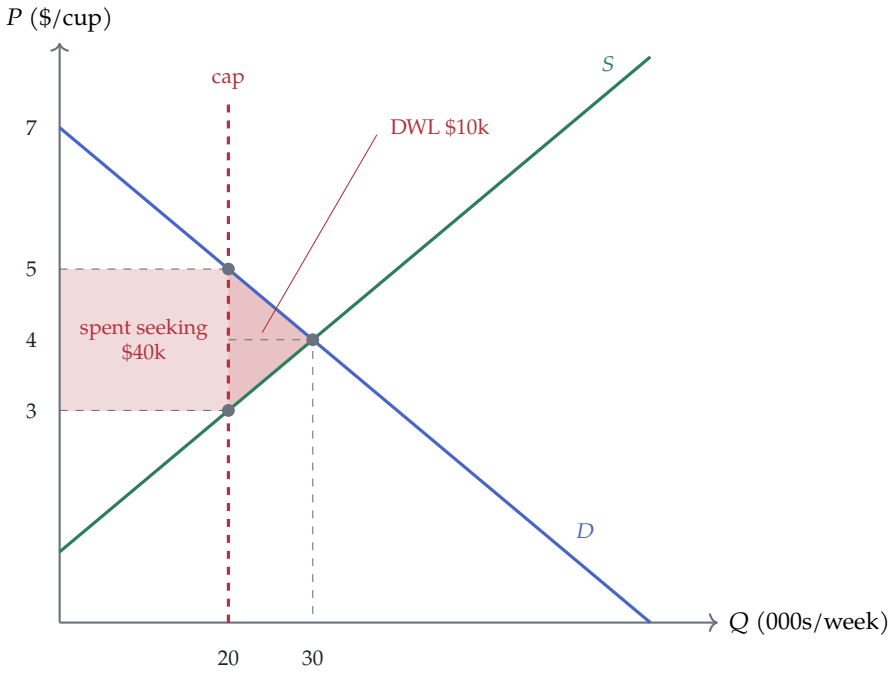


Figure 9.2: The same cap when the licenses are contested. Spending to obtain them can reach the value of the rectangle, so the social cost of the cap rises from $10,000 to as much as $50,000 a week.

The stake of each member.

	Members	Group, weekly	Each, weekly	Each, yearly
License holders	10	+$15,000	+$1,500	+$78,000
Buyers	10,000	−$25,000	−$2.50	−$130

Each row takes the group's change, divides it by the number of members in the group, and multiplies by 52 weeks: $15{,}000/10 = 1{,}500$ and $1{,}500 \times 52 = 78{,}000$ for holders, and $25{,}000/10{,}000 = 2.50$ and $2.50 \times 52 = 130$ for buyers. Taken as a group, buyers lose more than holders gain, since $25,000 is more than $15,000. Per person, the comparison runs the other way. Each holder gains $1{,}500/2.50 = 600$ times what each buyer loses.

India, 1964. The table shows the value of rents from five sources, in millions of rupees, as Krueger (1974, pp. 293–294) estimated them.

Source of rent	Rs million
Public investment	365
Imports	10,271
Controlled commodities	3,000
Credit rationing	407
Railways	602
Total	365 + 10,271 + 3,000 + 407 + 602 = 14,645

National income in 1964 was Rs 201 billion. So rents were 14,645/201,000 = 7.3 percent of national income. The largest source was import licenses. Krueger valued them at 75 percent of the value of 1964 imports. That is a conservative figure, since the government's Santhanam Committee reported that licenses were worth between 100 and 500 percent of their face value. Investment licensing is left out, because there is no basis on which to value it.

Turkey, 1968. Krueger (1974, p. 294) bases the estimate on a sample of imports that covers about 10 percent of the 1968 total, with values in millions of Turkish lira. At the border, the sample cost TL 547 million. Once duties, taxes and landing charges were added, it cost TL 1,443 million. At wholesale it sold for TL 3,568 million. Wholesaling is work too, so allow a markup of 50 percent on landed cost for it. Then the value of the licenses is

$$3{,}568 - 1.5 \times 1{,}443 = 3{,}568 - 2{,}164.5 = 1{,}403.5,$$

which Krueger rounds to TL 1,404 million. That is 1,404/547 = 2.57 times the border value of the imports, a ratio that Krueger describes as almost three times. Now convert this into a share of national income. Imports at border prices were 6 percent of national income. So license rents were about $2.57 \times 6 = 15.4$ percent, which Krueger reports as about 15 percent of GNP.

Production and Costs

10.1 Definitions

Profit. Total revenue minus total cost, written $\pi = TR - TC$, where $TR = P \times Q$.

Consider an engineer who works as an independent consultant and bills $280,000 in a year. Once all of her costs are counted, they come to $210,000, so her profit is $70,000.

Explicit and implicit costs. Explicit costs are input costs that require a direct money payment, such as wages, rent, materials and the interest on a loan. Implicit costs are input costs that involve no money payment. They are the value of what the owner gives up, for example the salary not earned elsewhere and the interest the invested savings could have earned.

The engineer pays $30,000 in office rent and $20,000 for equipment, and both are explicit costs. The $150,000 salary she gave up to work for herself is an implicit cost, and so is the $10,000 of interest that her $200,000 of savings no longer earns at 5 percent.

Accounting profit and economic profit. Accounting profit is total revenue minus explicit costs. Economic profit is total revenue minus all opportunity costs, both explicit and implicit.

On revenue of $280,000, the engineer's accounting profit is $280{,}000 - 50{,}000 = \$230{,}000$, and her economic profit is $280{,}000 - 210{,}000 = \$70{,}000$. The gap between the two is $160,000, which is exactly her implicit costs, $150{,}000 + 10{,}000$.

Short run and long run. In the short run at least one input is fixed. For example, the firm can hire workers at short notice, but it cannot build a bigger factory at short notice. In the long run all inputs are variable, including the size of the factory.

A food truck can change everything about its operation in perhaps six weeks, while a nuclear plant may need a decade. So the long run is not a number of months. It is however long the fixed input takes to become a matter of choice.

Production function. The relationship between the inputs a firm uses and the output it produces, written here as $Q = f(L)$.

Take a firm that assembles electronic components, with its machines and floor space fixed and L workers per shift. Its production function is

$$Q = 24L - L^2, \qquad 0 \le L \le 12, \tag{10.1}$$

with Q measured in components per shift, so 3 workers produce $72 - 9 = 63$ components.

Marginal product. The additional output produced by one more unit of an input, holding the other inputs fixed: $MP = \Delta Q / \Delta L$.

The assembler makes 63 components with 3 workers and 80 with 4. The marginal product of the fourth worker is therefore $80 - 63 = 17$ components.

Diminishing marginal product. As more and more of a variable input is added, with the other inputs held fixed, the marginal product of that input eventually falls.

The assembler's first worker adds 23 components and the twelfth adds only 1. The twelfth worker is no less capable than the first. She adds less because the same machines are now shared among 12 people.

Marginal cost is the wage divided by marginal product. Hiring one more worker costs the wage w and yields MP extra units of output, so each of those units costs

$$MC = \frac{w}{MP}.$$

The assembler pays \$50 for a worker-shift. The first worker adds 23 components, so each of them costs $50/23 = \$2.17$. The ninth worker adds 7, so each of those costs $50/7 = \$7.14$, although the wage has not changed.

Fixed and variable costs. Fixed costs, FC, do not vary with output, and they are owed even at $Q = 0$. For the workshop, $FC = 100$. Variable costs, VC, vary with output. In this case $VC = 60Q - 12Q^2 + Q^3$. Every dollar of total cost is one or the other, so $TC = FC + VC$.

The workshop here builds oak chairs, and its daily total cost in dollars for Q chairs is $TC = 100 + 60Q - 12Q^2 + Q^3$. At 6 chairs a day total cost is \$244, of which \$100 is fixed and $244 - 100 = \$144$ is variable.

The three average costs. Average fixed cost is $AFC = FC/Q$, which is the fixed bill spread over the units produced. Average variable cost is $AVC = VC/Q$, the variable bill per unit. Average total cost is $ATC = TC/Q$, which is what the typical unit costs. Since $TC = FC + VC$, dividing through by Q gives $ATC = AFC + AVC$.

At 6 chairs a day the workshop has $AFC = 100/6 = \$16.67$, $AVC = 144/6 = \$24.00$ and $ATC = 244/6 = \$40.67$. We can check the identity on these numbers, since $16.67 + 24.00 = 40.67$.

Marginal cost. The increase in total cost that comes from producing one more unit, $MC = \Delta TC/\Delta Q$.

The workshop's total cost is \$225 at 5 chairs and \$244 at 6, so the marginal cost of the sixth chair is $244 - 225 = \$19$. None of the fixed \$100 is in that figure, because the \$100 was owed anyway.

The pulling principle. The cost of each new unit is a new mark entering the average. If $MC < ATC$, the new unit is cheap and pulls the average down. If $MC > ATC$, the newcomer is expensive and drags the average up. If $MC = ATC$, the average stands still.
As long as MC lies below ATC, average total cost is falling. Once MC lies above ATC, average total cost is rising. A quantity that falls and then rises has its minimum at the changeover. In a table the changeover falls between two rows, where MC passes from below ATC to above it. On smooth cost curves it is the point where $MC = ATC$ exactly. Marginal cost therefore crosses average total cost at the minimum of average total cost.

The workshop's seventh chair costs \$31 and joins an average of \$40.67, so the average falls to \$39.29. The eighth chair costs \$49, which is more than \$39.29, and the average rises to \$40.50.

Efficient scale. The quantity of output at which average total cost is lowest.

For the chair workshop the efficient scale is 7 chairs a day, where average total cost is \$39.29. This is not necessarily the output the workshop should choose, because profit depends on price as well as on cost.

Long-run average cost. The lowest average total cost attainable at each level of output when the firm is free to choose its factory. Geometrically, it is the lower envelope of the short-run average total cost curves.

Suppose a firm can build a small, a medium or a large plant. The small plant is the

cheapest per unit when orders are thin and the large plant when volume is high, and the long-run curve takes whichever is lowest at each output.

Economies and diseconomies of scale. Economies of scale occur where *LRAC* falls as output rises. Constant returns to scale occur where *LRAC* is flat. Diseconomies of scale occur where *LRAC* rises as output rises.

Adam Smith's pin factory of ten workers split the making of a pin into distinct operations. It produced tens of thousands of pins a day, where a lone artisan performing every step himself could scarcely make 20. That is an economy of scale. Diseconomies arrive later, when management layers multiply and the cost of coordination rises faster than output.

10.2 Worked examples

A baker who leaves salaried work. Here is a second example. Nadia leaves a $70,000 job as a staff baker to open her own bakery. Each year she pays rent of $24,000, ingredients and utilities of $36,000, and $20,000 for a part-time assistant. She also takes $50,000 of savings, which had been earning 4 percent, and puts it into ovens. Her revenue is $160,000.

Add up the explicit costs first: $24{,}000 + 36{,}000 + 20{,}000 = \$80{,}000$. The implicit costs are the salary she gave up plus the interest her savings stopped earning, $70{,}000 + 0.04 \times 50{,}000 = \$72{,}000$. So her total opportunity cost is $152,000.

	Accounting	Economic
Revenue	160,000	160,000
Costs	80,000	152,000
Profit	80,000	8,000

It is economic profit that decides whether she stays, because it is the figure that compares the bakery with the life she gave up. She is beating that life by $8,000. If her revenue were $152,000, she would be exactly indifferent.

The chair workshop's full cost schedule.

Q	TC	$AFC = \frac{100}{Q}$	$AVC = \frac{VC}{Q}$	$ATC = \frac{TC}{Q}$	$MC = \Delta TC$
0	100	—	—	—	—
1	149	100.00	49.00	149.00	49
2	180	50.00	40.00	90.00	31
3	199	33.33	33.00	66.33	19
4	212	25.00	28.00	53.00	13
5	225	20.00	25.00	45.00	13
6	244	16.67	24.00	40.67	19
7	275	14.29	25.00	39.29	31
8	324	12.50	28.00	40.50	49
9	397	11.11	33.00	44.11	73
10	500	10.00	40.00	50.00	103

Q and TC are given, and the other four columns are computed from them (Figure 10.1). As a check, audit the row for $Q = 6$ from scratch. Variable cost is $VC = 244 - 100 = 144$, so $AVC = 144/6 = 24.00$. For the remaining averages, $AFC = 100/6 = 16.67$ and $ATC = 244/6 = 40.67$. The sixth chair costs $MC = 244 - 225 = 19$. The identity holds in that row, since $16.67 + 24.00 = 40.67$. It also holds at $Q = 4$, where $25.00 + 28.00 = 53.00$, and at $Q = 9$, where $11.11 + 33.00 = 44.11$.

Choosing the factory. The per-unit cost of each factory, read off the three curves, is as follows:

Q	Small	Medium	Large
2	\$4.00	\$6.80	\$23.20
6	\$5.60	\$2.00	\$5.60
10	\$23.20	\$6.80	\$4.00

A firm that plans on 2 units a day builds small. One that plans on 6 builds medium, and one that plans on 10 builds large. At two units a day the large plant costs over \$23 per unit, because it is mostly idle and its rent is spread over almost nothing. At ten units the small plant costs over \$23 a unit, for the opposite reason. It has been pushed far past its capacity, and crowding has taken over. So the two mechanisms behind these high costs are quite different. Figure 10.2 draws the three curves.

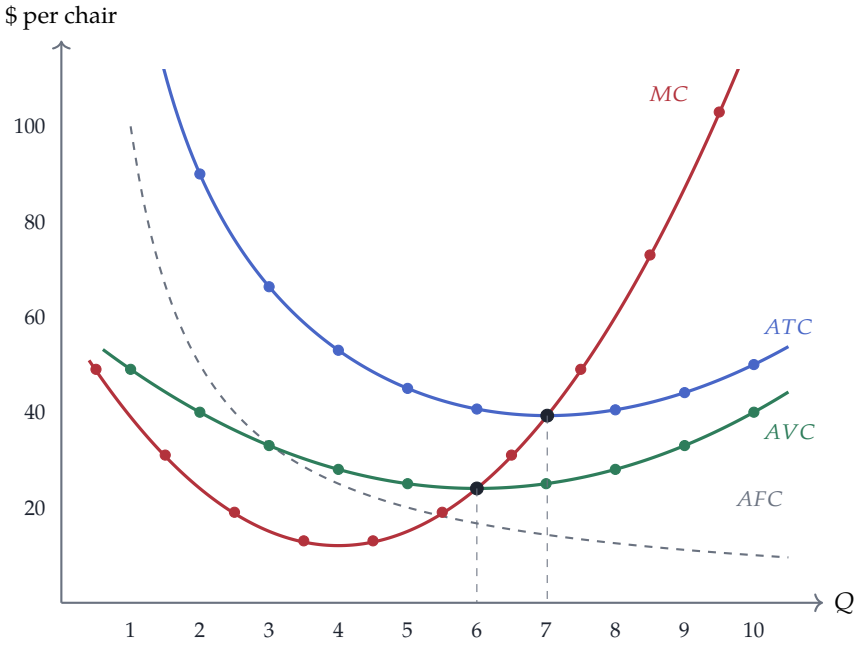


Figure 10.1: The four per-unit cost measures for the chair workshop, all in the short run. The blue and green dots are the ATC and AVC columns of the table. The red dots are the MC column, each plotted at $Q - \frac{1}{2}$ because the cost of the Qth chair belongs between $Q-1$ and Q. Marginal cost meets AVC at its minimum of \$24.00 and ATC at its minimum of \$39.29.

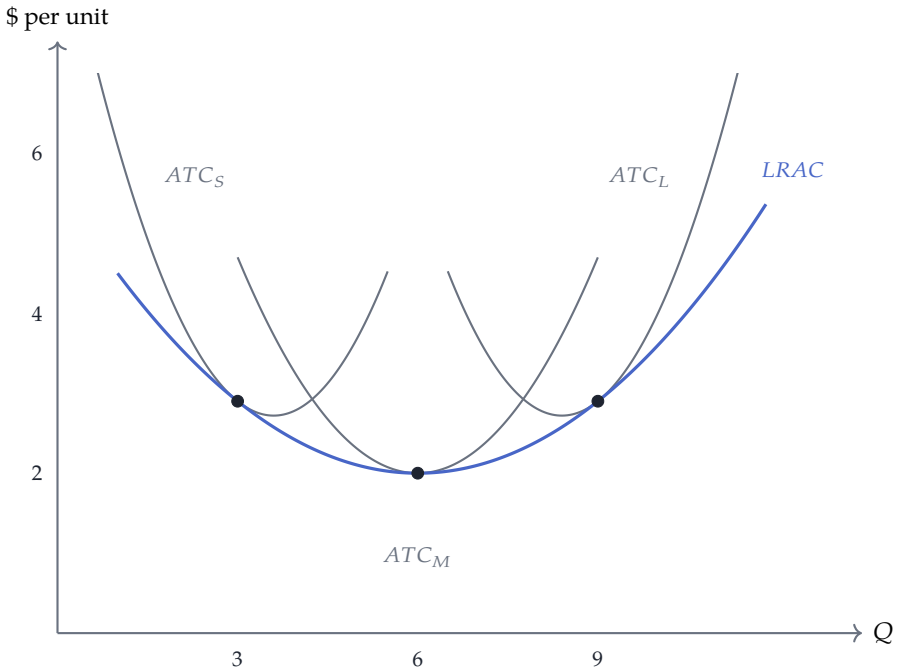


Figure 10.2: Three short-run average total cost curves, one for each of three factory sizes, with the long-run average cost curve, the envelope over every possible factory size, beneath them. Each plant touches the envelope at the output it was built for and lies above it everywhere else.

Firms in Competitive Markets

11.1 Definitions

Four ways a market can be organized.

Structure	Sellers	Product	Power over price
Perfect competition	many	identical	none
Monopolistic competition	many	differentiated	a little
Oligopoly	few	either	often considerable
Monopoly	one	unique	the most

The fewer the sellers and the more distinctive the product, the more each firm can typically say about its own price.

The running example of this chapter sits in the top row. It is a maker of standardized electronic connectors, one of 120 identical firms, with total cost $TC = 144 + 36q - 8q^2 + q^3$ dollars a day, where q is one firm's output and Q is the output of the whole market.

Competitive market. A market with many buyers and many sellers, in which the goods are identical across sellers and there is free entry and exit.

The connector market meets all three conditions. One supplier's output is indistinguishable from another's, no plant is big enough to matter on its own, and anyone may open a plant or close one.

Price taker. A buyer or seller who takes the market price as given and cannot influence it.

Suppose the market price of a connector is \$48. A firm that charges \$49 loses every buyer to a rival, and a firm that charges \$47 wins no extra sales, because it could already sell its whole output at \$48. It simply hands a dollar to each buyer.

Average and marginal revenue. Average revenue is the revenue per unit sold, $AR = TR/q$. Marginal revenue is the change in total revenue when the firm sells one more unit, $MR = \Delta TR$.

At \$48 the connector firm's revenue is $TR = 48q$, so a seventh connector raises revenue from \$288 to \$336, and $MR = 336 - 288 = \$48$. Marginal revenue equals the price because the firm never has to cut its price to sell more.

The output rule. If $MR > MC$, the next unit brings in more than it costs, so making it raises profit, and the firm expands. If $MR < MC$, the last unit cost more than it brought in, so cutting it raises profit, and the firm cuts back. So profit is greatest at the last unit whose MR still covers its MC. When output can be varied in fractions, that point is exactly where $MR = MC$. The rule holds for every firm that maximizes profit. For a price taker, $MR = P$, so the rule becomes $P = MC$.

At \$48 the sixth connector costs \$39 to make and the seventh costs \$59. So the firm makes the sixth, skips the seventh, and produces $q^* = 6$.

Constructing the profit rectangle. First find the output, where the price line meets marginal cost. Then read average total cost at that same output, not at its minimum. The vertical gap between the price line and the ATC curve is profit per unit. The rectangle runs from zero to q^*, with its top at the price line and its bottom at ATC, and its area is profit.

At a price of \$71 the firm produces 7, where ATC is \$49.57. The rectangle is $71 - 49.57 = 21.43$ high and 7 wide, so its area is \$150 (Figure 11.1).

The firm's short-run supply curve. The firm's short-run supply curve is its marginal cost curve above average variable cost, and zero below it. In whole units the curve rises in steps, and the smooth curve is the version in which output is divisible.

The connector firm supplies nothing at \$15, 5 at \$31, 6 at \$48 and 7 at \$71. At exactly \$20, the minimum of its average variable cost, it is indifferent between producing 4 and producing nothing.

Exit. Leaving the market permanently. The plant is sold and the lease ends, so the fixed costs stop too.

A connector firm that exits stops paying its fixed bill of \$144 a day. A firm that merely shuts down keeps paying it and waits for the price to recover.

Two thresholds, two runs. In the short run the fixed cost is sunk, so the test compares the price with the minimum of AVC. In the long run no future cost is fixed, because the firm can avoid every cost it has still to pay by leaving, so the test becomes the price against the minimum of ATC.

Price	Short run	Long run
$P < 20$	shut down	exit
$20 \le P < 48$	operate at a loss	exit
$P = 48$	operate, $\pi = 0$	stay, and no one enters
$P > 48$	operate, $\pi > 0$	stay, and entrants come

At a price of \$31 the connector firm makes 5 and loses \$94 a day, but \$31 is above its average variable cost of \$21 at that output, so it operates today. It also arranges to leave when the lease expires, because \$31 does not cover all of its costs.

Entry and exit. In the long run the number of firms is not fixed, and it is profit that changes it. A profit draws in entrants, which shifts market supply to the right and pushes the price down. A loss drives firms out, which shifts supply to the left and pushes the price up. In the short run a shift in demand moves the price. In the long run it moves the number of firms.

When connector demand rises, the price first spikes to \$71 and then falls back to \$48.20 as 58 new firms enter, so the industry grows from 120 firms to 178 (Stages 1 to 3 below).

Long-run competitive equilibrium. The state in which $P = MC$, so that each firm maximizes profit; P equals the minimum of ATC, so that economic profit is zero and no firm enters or leaves; and each firm produces at its efficient scale.

The three conditions are one point, because marginal cost crosses average total cost at its minimum (Chapter 10). For the connector firm the point is $q = 6$ at \$48.

Rent to a scarce advantage. A surplus earned by a firm whose costs are below those of the marginal entrant, because entrants cannot replicate the advantage that produces the lower costs. It survives in long-run equilibrium.

Suppose entrants to a farming market face a minimum ATC of \$48 at $q = 6$, and a few incumbent farms on unusually good land face \$40. The price settles at \$48, set by the marginal entrant, so a good-land farm earns at least $(48 - 40) \times 6 = \$48$ a day, and entry cannot compete it away because entrants cannot copy the land.

What makes long-run supply flat. The long-run supply curve is horizontal at the minimum of ATC when input prices are unaffected by the size of the industry and all firms have the same costs. Where input prices rise as the industry grows, or where each further entrant has higher costs than the one before it, the curve slopes upward,

and the price settles at whatever covers the marginal entrant's costs.

Suppose connector plants need a specialized alloy in limited supply. A wave of entry bids up the alloy's price, every firm's cost curves shift up, and a bigger market needs a permanently higher price.

11.2 Worked examples

The connector firm's costs.

q	TC	$VC = TC - 144$	$MC = \Delta TC$	$AVC = VC/q$	$ATC = TC/q$
0	144	0	—	—	—
1	173	29	29	29.00	173.00
2	192	48	19	24.00	96.00
3	207	63	15	21.00	69.00
4	224	80	17	20.00	56.00
5	249	105	25	21.00	49.80
6	288	144	39	24.00	48.00
7	347	203	59	29.00	49.57
8	432	288	85	36.00	54.00

Only q is given, and TC is the cost function evaluated at that output. Take the row for $q = 6$ as a check. Total cost is $TC = 144 + 216 - 288 + 216 = 288$. Subtracting the fixed cost gives $VC = 288 - 144 = 144$. Dividing by output gives $AVC = 144/6 = 24.00$ and $ATC = 288/6 = 48.00$. And for the sixth unit, $MC = 288 - 249 = 39$. The marginal cost column needs one comment. The entry recorded against q is the cost of the qth unit, so it belongs between $q - 1$ and q. The table puts it in the row for q, and the figures plot it at $q - \frac{1}{2}$.

Marginal revenue against marginal cost at \$48.

unit	MR	MC	$MR - MC$	verdict
1	48	29	+19	make it
2	48	19	+29	make it
3	48	15	+33	make it
4	48	17	+31	make it
5	48	25	+23	make it
6	48	39	+9	make it
7	48	59	−11	skip it
8	48	85	−37	skip it

The firm produces $q^* = 6$. Why is this optimal? Look at the seventh unit. It would bring

in \$48 of revenue, but it would cost \$59 to make. So the firm would lose \$11 on it. It does not pay to produce the seventh unit. Now look at it the other way. If the firm stopped at five, it would give up the sixth unit, and the sixth unit adds \$9. So the firm stops at six.

Now add up what each of the six units brings in over and above its own cost: $19 + 29 + 33 + 31 + 23 + 9 = \$144$. Notice that \$144 is exactly the fixed cost. In other words, what the firm earns over its running costs pays the fixed bill exactly, with nothing left over. So at a price of \$48, economic profit is exactly zero.

A price of \$71. At \$71 the seventh unit costs \$59 and is worth making, but the eighth costs \$85 and is not, so $q^* = 7$. Using totals,

$$\pi = 71 \times 7 - 347 = 497 - 347 = +\$150.$$

Using the per-unit route instead, ATC at $q = 7$ is $347/7 = \$49.57$, so

$$\pi = (P - ATC) \times q = (71 - 49.57) \times 7 = 21.43 \times 7 = \$150.$$

The two routes give the same answer (Figure 11.1).

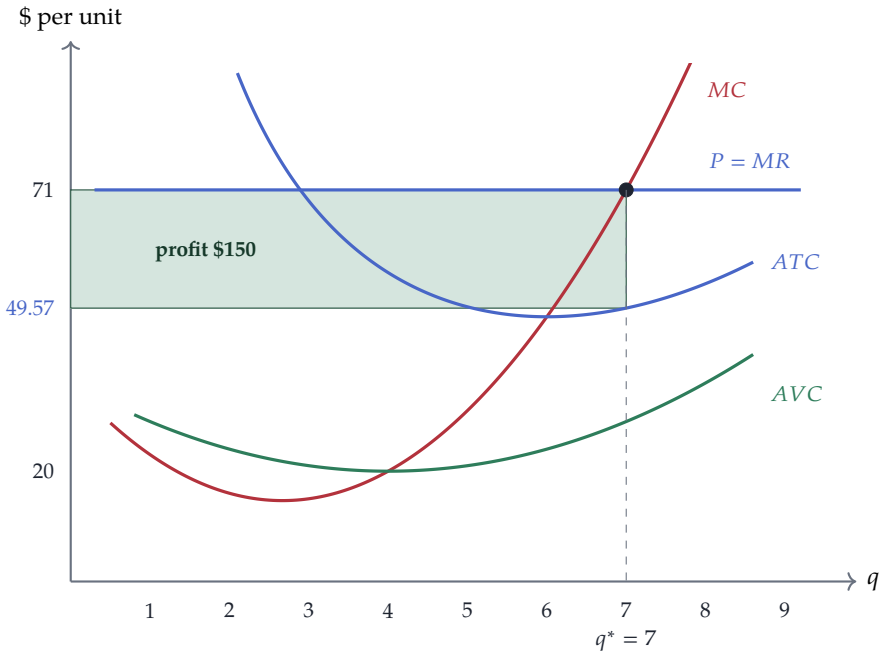


Figure 11.1: The profit rectangle in the short run. The output is fixed where the price line meets marginal cost, at 7, and the rectangle's height is the gap between the price and average total cost read at that output, \$21.43.

Operating at a loss on purpose.

	operate at $q = 5$	shut down
revenue	155	0
variable cost	105	0
fixed cost	144	144
result	−\$94	−\$144

Operating is better by \$50, and that \$50 has a precise meaning. Revenue of \$155 covers all \$105 of variable cost and leaves \$50 to put toward the fixed bill. This is $(31 - 21) \times 5$, the gap between the price and average variable cost on each of the five units. Fifty dollars of the rent paid means fifty dollars less lost.

Stage 1, the market at rest. Demand is $Q^d = 1200 - 10P$. At the zero-profit price of \$48, buyers demand $1200 - 480 = 720$, and 120 firms supply $120 \times 6 = 720$. So the market balances.

Call this point A: $P = 48$, $Q = 720$, 120 firms, each producing $q = 6$ and earning $\pi = 0$. Here nothing moves. There is no profit to attract entrants and no loss to drive incumbents out, and each firm is already at its best output for this price.

Stage 2, demand jumps. Now a new technology creates a need for connectors everywhere, and demand shifts to $Q^d = 1550 - 10P$. Buyers now want 350 more at every price. At the old price of \$48 they demand $1550 - 480 = 1070$ against a supply of 720, so there is a shortage of 350, and a shortage bids the price up.

The cast is still frozen at 120 firms, so the price rises along their existing supply schedule until the two sides meet. At \$71 buyers demand $1550 - 710 = 840$, and the firms supply $120 \times 7 = 840$. Call this point B: $P = 71$, $Q = 840$, still with 120 firms.

Each firm has moved up its marginal cost curve from 6 units to 7, because at \$71 the seventh unit, which costs \$59, is now worth making. The market's extra 120 units come from existing firms working harder, since there are no others yet. Each firm earns $71 \times 7 - 347 = +\$150$ a day.

Stage 3, where entry stops. If the price went all the way back to \$48, buyers would take $1550 - 480 = 1070$. Firms at their efficient scale make 6 each, so the market would hold $1070/6 = 178.33$ firms. A third of a firm cannot enter, so we have to test the whole numbers on either side. In whole units each firm produces 6 at any price between \$39 and \$59, so $Q = 6n$ and $P = (1550 - Q)/10$. Profit per firm is then $6P - 288$, since TC at $q = 6$ is 288.

	$Q = 6n$	$P = (1550 - Q)/10$	$\pi = 6P - 288$
$n = 178$	1068	\$48.20	$289.20 - 288 = +\$1.20$
$n = 179$	1074	\$47.60	$285.60 - 288 = -\$2.40$

The 178th firm still covers all of its costs. A 179th firm would push the price to \$47.60 and lose \$2.40, so it stays out, and entry stops at 178 firms.

Monopoly

12.1 Definitions

Monopoly. A firm that is the sole seller of a product for which there are no close substitutes.

The sensor firm is the sole maker of a specialized industrial sensor, and a patent protects it. It faces market demand $P = 100 - 5Q$, so if it chooses $Q = 4$ the demand curve answers $P = 100 - 20 = 80$.

How a monopoly arises. There are three kinds of barrier. A **key resource** held in one firm's hands, for example control of the essential input. A **government grant**, such as a patent, a copyright or a license, where the law itself keeps rivals out. A **natural monopoly**, where one firm can serve the whole market more cheaply than two firms could. In that case the cost structure itself does the excluding.

The sensor firm's barrier is a government grant. Its patent gives it the exclusive right to sell, typically for 20 years.

Barriers to entry. In a competitive market, profit attracted new entrants, and the entrants competed it away. A monopoly persists only because something blocks the door. If there is no barrier, there is no monopoly. Free entry always wears market power down.

The sensor firm earns $120 thousand a month. In a competitive market a profit of that size would draw entrants, but here the patent holds the door shut.

Natural monopoly. An industry in which a single firm can supply the whole market at a lower average cost than two or more firms could.

A city's electricity grid costs $TC = 160 + 10Q$, so $ATC = 160/Q + 10$. Suppose 8,000 homes are to be served. One firm serving all of them has an average cost of $30 a home, while two firms serving 4,000 each have $50, because each pays the $160 thousand fixed cost for its own poles and wires.

Why marginal revenue falls short of the price. To sell more, the firm must let the price fall. The lower price applies to every unit, including those it was already selling, rather than only to the new one. So $MR < P$ for any firm that faces a downward-sloping demand curve, and the gap between them is the price effect. A competitive firm escapes the price effect because its demand curve is flat, so $MR = P$.

To sell 8 thousand sensors rather than 7 thousand, the firm cuts the price from \$65 to \$60. The output effect is +60 and the price effect is $7 \times (-5) = -35$, so marginal revenue is only 25.

The profit rectangle, and how to build it. Start at the point where marginal revenue meets marginal cost, which fixes Q_m. Climb straight up to the demand curve to get P_m. Then drop down to the ATC curve at the same quantity. The gap is profit per unit. The profit rectangle runs from zero to Q_m, with its top at the price and its bottom at ATC.

For the sensor firm, with $TC = 200 + 20Q$, $\pi = (60 - 45) \times 8 = 120$ (Figure 12.1). If the license fee doubled the fixed cost to 400, ATC at 8 would be 70, the rectangle would sit above the price line, and $\pi = (60 - 70) \times 8 = -80$.

Monopoly compared with perfect competition. The efficient quantity is found by making every unit whose value covers its cost, and stopping where value falls to cost. Competition pushes the price down to marginal cost. The monopolist is a case of trading too little. The deadweight loss triangle sits between the marginal cost line and the demand curve, and spans the quantities from Q_m to Q^*. The profit is a transfer and the triangle is the loss. A monopoly price works like a private tax. It drives a wedge between what buyers pay and what production costs. The difference is that the revenue goes to the firm, rather than to a treasury.

A competitive sensor market would sell 16 thousand at \$20, and the monopolist sells 8 thousand at \$60. Of the \$640 thousand of surplus, \$320 thousand is transferred to the firm and \$160 thousand is destroyed (Figure 12.2).

Price discrimination. Selling the same good to different buyers at different prices, when it costs the same to serve each buyer.

At its single price of \$60, the sensor firm lets the buyers of the second thousand, whose sensors were worth \$90 to them, keep \$30 each. It also excludes units 9 through 16, worth from \$55 down to \$20, that would be profitable to serve at a lower price.

Regulating a natural monopoly. For a natural monopoly of this kind, average total cost is above marginal cost everywhere. So a price equal to marginal cost is necessarily below average cost, and a price that covers average cost is necessarily above marginal cost. With a single price per unit, only a subsidy avoids the choice.

If the city distributor, facing $P = 70 - 5Q$, is capped at its marginal cost of \$10, it connects all 12 thousand homes worth connecting but loses its whole fixed cost of \$160 thousand a month. Capping it at average cost lets the firm break even at \$30 and leaves homes 8,000 to 12,000 unserved (Figure 12.3).

12.2 Worked examples

The sensor firm's revenue schedule.

Q	$P = 100 - 5Q$	$TR = P \times Q$	$MR = \Delta TR$
0	100	$100 \times 0 = 0$	—
1	95	$95 \times 1 = 95$	$95 - 0 = 95$
2	90	$90 \times 2 = 180$	$180 - 95 = 85$
3	85	$85 \times 3 = 255$	$255 - 180 = 75$
4	80	$80 \times 4 = 320$	$320 - 255 = 65$
5	75	$75 \times 5 = 375$	$375 - 320 = 55$
6	70	$70 \times 6 = 420$	$420 - 375 = 45$
7	65	$65 \times 7 = 455$	$455 - 420 = 35$
8	60	$60 \times 8 = 480$	$480 - 455 = 25$
9	55	$55 \times 9 = 495$	$495 - 480 = 15$
10	50	$50 \times 10 = 500$	$500 - 495 = 5$
11	45	$45 \times 11 = 495$	$495 - 500 = -5$

In the table, only Q is given. The price is read off the demand curve. Total revenue is price times quantity. Marginal revenue is the difference between successive total revenues. So the entry against each Q is the revenue added by the Qth thousand sensors, and strictly it belongs between $Q - 1$ and Q.

Marginal revenue against marginal cost.

unit (000s)	MR	MC	$MR - MC$	verdict
1	95	20	+75	make it
2	85	20	+65	make it
7	35	20	+15	make it
8	25	20	+5	make it
9	15	20	−5	skip it
10	5	20	−15	skip it

The margins shrink by \$10 a row, following the marginal revenue column, and the rows left out between 2 and 7 follow the same pattern. The firm produces $Q_m = 8$ thousand sensors, where marginal revenue meets marginal cost (Figure 12.1).

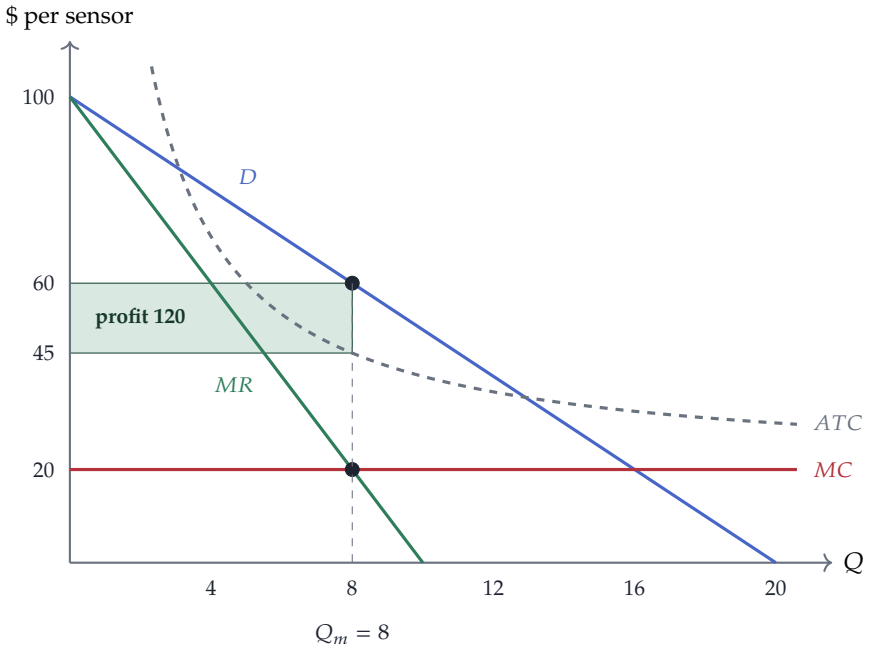


Figure 12.1: The sensor monopolist. The marginal revenue line meets marginal cost at 8 thousand, which fixes the quantity; the demand curve above that quantity fixes the price at \$60; and average total cost read at the same quantity is \$45. The shaded rectangle is profit, \$15 a sensor across 8 thousand sensors.

The units the monopolist refuses to make.

unit (000s)	worth to its buyer	marginal cost	what happens
9	$100 - 45 = 55$	20	worth making, never made
12	$100 - 60 = 40$	20	worth making, never made
16	$100 - 80 = 20$	20	the last one worth making, never made

Every unit from the ninth thousand to the sixteenth is valued at or above its cost, and none of them is traded. Take the ninth thousand. Its buyers value it at \$55 a sensor, and it costs \$20 a sensor to make. So that trade would create \$35 a sensor, or \$35 thousand in all, and it does not happen (Figure 12.2).

Perfect price discrimination. Suppose the firm knew every buyer's exact valuation and charged each buyer precisely that amount. Then every unit worth at least \$20 is sold. Each buyer pays their own valuation, and any valuation above \$20 leaves the firm a margin. So output reaches $Q = 16$, which is the efficient quantity. Revenue is the whole area under the

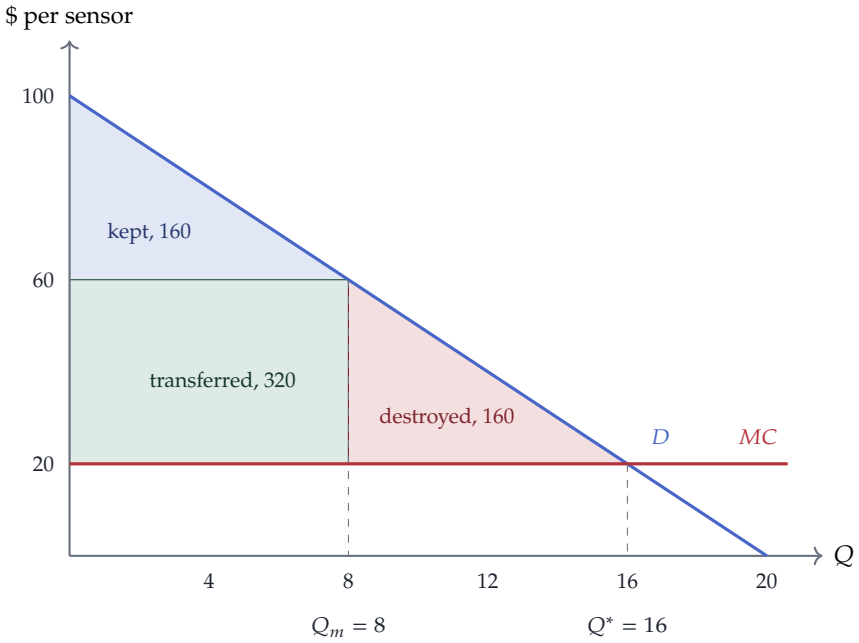


Figure 12.2: The $640 thousand of surplus a competitive sensor market would create, divided three ways under monopoly. The upper triangle is the surplus buyers keep, the rectangle below it is the surplus transferred to the firm by the higher price, and the right-hand triangle is the deadweight loss on the 8 thousand units never made. Each region is labeled with its name and its value.

demand curve out to 16,

$$\tfrac{1}{2}(100 + 20) \times 16 = 960,$$

and profit is $960 - (200 + 320) = 440$, compared with 120 under a single price.

	consumer surplus	deadweight loss	total surplus
one price	160	160	480
perfect discrimination	0	0	640

Consumer surplus is zero. Every buyer pays exactly what the unit was worth to them, so nobody keeps anything. Deadweight loss is zero as well, and total surplus is the full 640, which is the same as under perfect competition.

Capping the price at average cost. Now let the firm break even, and let the equation find the point:

$$70 - 5Q = \frac{160}{Q} + 10.$$

Multiplying through by Q and collecting terms gives $5Q^2 - 60Q + 160 = 0$, or $Q^2 - 12Q + 32 = 0$, which factors as $(Q - 4)(Q - 8) = 0$.

There are two break-even points, at $Q = 4$ and $Q = 8$. The larger one serves more homes at a lower price, so we take $Q = 8$ and $P = 70 - 40 = 30$. To check, $ATC = 160/8 + 10 = 30 = P$, so the firm survives with no subsidy.

The price of \$30 still exceeds the marginal cost of \$10, so homes 8,000 to 12,000 remain unserved (Figure 12.3) and

$$DWL = \tfrac{1}{2}(30 - 10)(12 - 8) = 40.$$

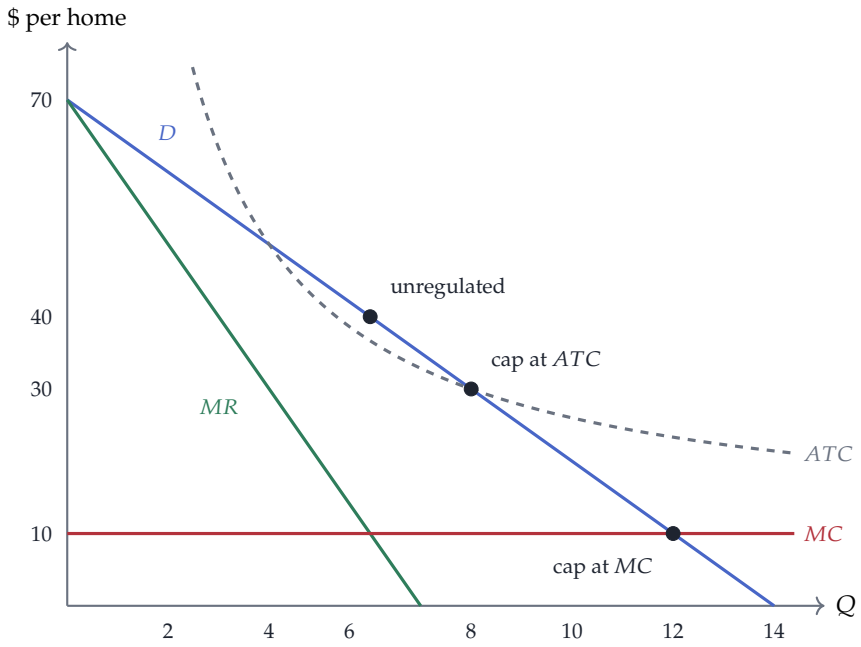


Figure 12.3: The city distributor, with average total cost lying above marginal cost at every quantity. The three dots are the three policies: the unregulated price high on the demand curve, the average-cost price where demand meets ATC, and the marginal-cost price where demand meets MC, which lies below ATC and therefore loses money.

Oligopoly

13.1 Definitions

Oligopoly. A market structure in which only a few sellers offer similar or identical products.

Consider a valley town with exactly two ready-mix concrete plants, Ridge Concrete and Vale Concrete, selling an identical product. Market demand is $P = 60 - Q$, with Q in yards a day, and each plant produces at a marginal cost of \$12 a yard, so each plant's profit is $(P - 12)$ times the yards it sells.

Strategic interdependence. Each firm's best choice depends on the choices the other firms make, and every firm knows it.

Suppose Ridge produces q while Vale produces 12. The price is then $60-(12+q) = 48-q$, so Ridge's profit,

$$(48 - q - 12)\, q = (36 - q)\, q, \tag{13.1}$$

depends on Vale's output as well as its own.

Concentration ratio. The percentage of a market's total output supplied by its four largest firms.

If a town's six grocery stores have market shares of 30, 25, 20, 10, 10 and 5 percent, the four largest supply $30 + 25 + 20 + 10 = 85$ percent. A market of ten firms with 10 percent each has a ratio of only 40 percent.

Collusion and cartels. **Collusion** is an agreement among firms in a market about the quantities to produce or the prices to charge. A **cartel** is a group of firms acting in unison.

If Ridge and Vale collude, they produce the monopoly quantity of 24 yards between them, 12 each, at $P = 60 - 24 = 36$. Each plant earns $(36 - 12) \times 12 = 288$, and joint profit is 576.

Nash equilibrium. A situation in which each economic actor, interacting with others, chooses their best strategy given the strategies all the other actors have chosen.

The cartel at 12 and 12 is not a Nash equilibrium, because a plant that deviates to 18 earns 324 against 288. The pair at 16 and 16 is one, since against a rival at 16 every deviation earns less than 256.

What the firms compete over. The same two plants, with the same demand curve and the same marginal cost, produce either the competitive outcome or one well above it. Which one depends only on what the firms choose to compete over, and neither outcome involves an agreement.

When Ridge and Vale choose quantities, they settle at 32 yards and $28, and each earns 256. When they post prices instead, they undercut each other down to $12, where the market takes 48 yards and each earns zero.

Game. A strategic situation described by three lists: the players, the strategies each may choose, and the payoff each receives for every combination of choices.

Mr Red and Mr Blue are questioned in separate rooms, and each may stay silent or confess. The payoffs are years in jail: one year each if both stay silent, two each if both confess, and zero for the one who confesses while the other stays silent and serves three.

Dominant strategy. A strategy that is best for a player no matter what strategy the other players choose.

Two food trucks each choose whether to keep their prices or to discount. If the rival keeps, a truck earns 650 by discounting against 500 by keeping, and if the rival discounts, it earns 400 against 300, so discounting is dominant.

Individual and collective rationality. Mutual silence gives one year each, which is better for Red and better for Blue than the two years each that rational play produces. So there is an outcome that is strictly better for both players at the same time, and rational play does not reach it. Individual rationality does not imply collective rationality.

Two streaming services that each choose a big advertising campaign end up with 80 each, although small campaigns would have given both of them 100.

Repeated game. The same game, played by the same players, round after round, with each round's choices visible before the next.

Suppose Vale plays the grim strategy, mixing 16 forever once Ridge exceeds its quota. Cheating gains Ridge $324 - 288 = 36$ once and costs it $288 - 256 = 32$ in every later round, so if the plants expect to meet even twice more, $36 < 32 + 32$ and cheating is a losing move.

The Sherman and Clayton Acts. The Sherman Antitrust Act of 1890 declares illegal every contract, combination or conspiracy in restraint of trade. It makes it a crime to monopolize or attempt to monopolize, and it lets a private party injured by a violation sue for three times its losses. The Clayton Act of 1914 carries those treble damages forward and adds private injunctive relief, under which courts may order conduct stopped.

If Ridge and Vale agreed on output, that would be naked price fixing, illegal in itself. It would move the market from 32 yards at \$28 toward 24 yards at \$36.

13.2 Worked examples

The valley concrete market.

Q	$P = 60 - Q$	$TR = P \times Q$	$TC = 12Q$	profit $= TR - TC$
0	60	$60 \times 0 = 0$	0	0
6	54	$54 \times 6 = 324$	72	252
12	48	$48 \times 12 = 576$	144	432
18	42	$42 \times 18 = 756$	216	540
24	36	$36 \times 24 = 864$	288	576
30	30	$30 \times 30 = 900$	360	540
36	24	$24 \times 36 = 864$	432	432
42	18	$18 \times 42 = 756$	504	252
48	12	$12 \times 48 = 576$	576	0
54	6	$6 \times 54 = 324$	648	−324
60	0	$0 \times 60 = 0$	720	−720

Only Q is given. The price comes off the demand curve, total revenue is price times quantity, total cost is \$12 a yard, and profit is the difference. Take the row for $Q = 24$ as a check. The price is $P = 60 - 24 = 36$. Total revenue is $TR = 36 \times 24 = 864$. Total cost is $TC = 12 \times 24 = 288$. So profit is $864 - 288 = 576$.

The profit column starts at zero, rises to a peak of 576 at $Q = 24$, falls back to zero at $Q = 48$, and goes negative beyond that.

Ridge's private arithmetic.

Ridge's q	12	14	16	18	20
profit $(36-q)q$	288	308	320	324	320

Ridge's profit in Equation (13.1) peaks at $q = 18$, where it is worth $18 \times 18 = 324$. The neighboring whole numbers confirm it. At 17, profit is $19 \times 17 = 323$, and at 19 it is $17 \times 19 = 323$. So cheating is worth $324 - 288 = 36$ a day.

Where the sliding stops. Now try 16 yards each. Total output is $Q = 32$, the price is $60 - 32 = 28$, and each plant earns $(28 - 12) \times 16 = 256$.

Now test it. If the rival produces 16, a plant's profit is $(60-16-q-12)q$, which simplifies to $(32 - q)q$.

its q	15	16	17
its profit	255	256	255

A move in either direction costs the plant a dollar, so neither plant moves (Figure 13.1).

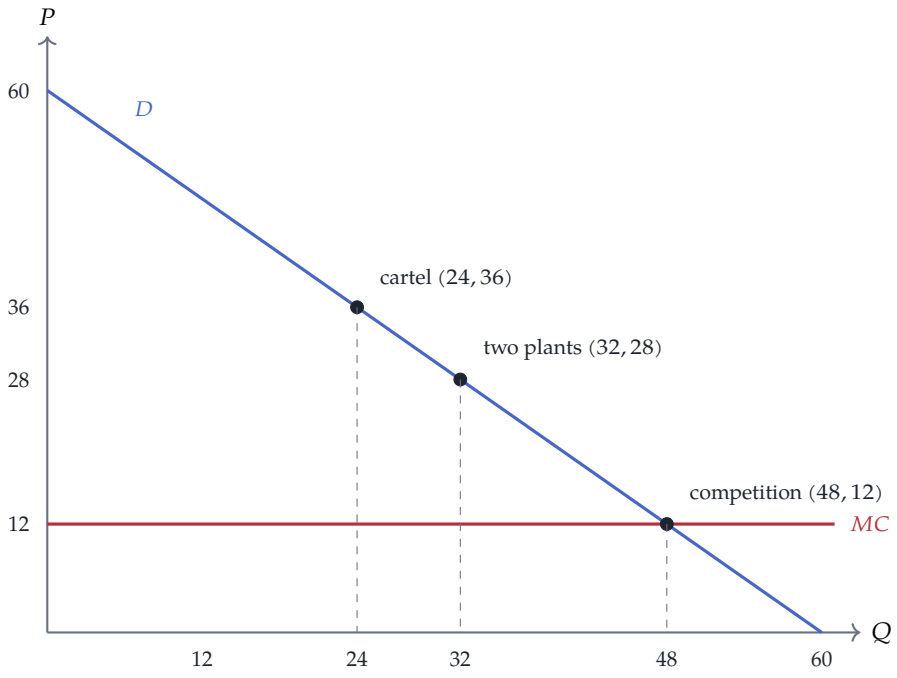


Figure 13.1: Three outcomes from the same demand curve and the same \$12 marginal cost. The duopoly's resting point lies between the cartel and the competitive outcome on both axes, at 32 yards and \$28.

The prisoners' payoff matrix.

	Blue stays silent	Blue confesses
Red stays silent	(1, 1)	(3, 0)
Red confesses	(0, 3)	(2, 2)

Red picks the row and Blue picks the column. In every cell the first number is Red's payoff and the second is Blue's. The payoffs are years in jail, so fewer is better. The same ordering holds for every matrix in this chapter: the row player's payoff comes first.

Reading the four cells in turn: if both stay silent, each serves one year on the minor charge. If Red stays silent and Blue confesses, Red serves three years while Blue walks. The reverse case is the mirror image. And if both confess, each serves two years.

The cartel as a payoff matrix.

	Vale mixes 12	Vale mixes 16
Ridge mixes 12	(288, 288)	(240, 320)
Ridge mixes 16	(320, 240)	(256, 256)

Ridge's profit comes first in each cell, and this time more is better. If Vale mixes 12, Ridge earns 320 by mixing 16, against 288 by staying at 12. If Vale mixes 16, Ridge earns 256 against 240. Mixing 16 wins both comparisons, so it is dominant, and by symmetry so is Vale's.

Both plants mix 16, and the equilibrium is 256 each. Meanwhile 288 each is in plain view, preferred by both, and unreachable.

Four rounds with a known end. Ridge and Vale will interact for exactly four rounds, both of them know it, and Vale plays grim. Cooperating pays 288 a round, cheating against a rival at 12 pays 324, and the rounds after a cheat pay 256.

Cooperating throughout pays $288 \times 4 = 1152$. Cheating in round one pays $324 + 256 \times 3 = 324 + 768 = 1092$. That is worse, so cheating early loses. Cooperating in rounds one to three and then cheating in round four pays $288 \times 3 + 324 = 864 + 324 = 1188$, which is the best of the three.

Cheating in the last round costs nothing, because there is no round left in which to be punished. So a known final round unravels cooperation at the end. What sustains real cartels is the absence of a fixed end.

Monopolistic Competition

14.1 Definitions

Monopolistic competition. A market in which there are many sellers, the products are differentiated, and firms are free to enter and exit.

Consider a coffee roaster in a city of two hundred roasters, each with its own blend, its own shop and its own customers. Its output q is measured in thousands of pounds a year, its price P in dollars a pound, and its total cost is $TC = 144 + 10q + q^2$ in thousands of dollars a year.

What makes the demand curve slope down. The firm's demand curve slopes down because its product is distinguishable from the others, not because it is the only seller. If a firm raises its price above every rival's, it loses some of its customers rather than all of them. The ones who stay are those who preferred this seller by more than the amount of the price rise.

The roaster faces its own demand curve $P = A - 3q$, so to sell one thousand pounds more it has to cut its price by \$3. A price taker would need no price cut at all.

Four steps, in this order. First, set marginal revenue equal to marginal cost and solve for the quantity. Second, take that quantity up to the *demand* curve to find the price. Third, take the same quantity up to average total cost to find the cost per unit. Finally, profit is price minus average total cost, multiplied by the quantity.

With demand at $P = 74 - 3q$, so that $MR = 74 - 6q$, and $MC = 10 + 2q$, the roaster produces $q = 8$ at a price of 50, its average total cost is 36, and its profit is $(50 - 36) \times 8 = 112$. With demand at $P = 42 - 3q$ the same steps give $q = 4$, a price of 30 and average total cost of 50, so it loses $(30 - 50) \times 4 = -80$.

Entry and exit move the firm's own demand curve. Fewer customers at every price means the demand curve has shifted left. Read the other way, the firm must now charge less to sell any given quantity. Losses run the same mechanism backwards. When a roaster closes, its customers still want coffee, and they spread themselves among the shops that remain. So every surviving firm has more customers at any given price, and that pushes the intercept back up. The cost function does not move,

whether firms are entering or leaving.

Entry moves the roaster's intercept from 74 down to 58, a fall of \$16 at every quantity. At six units it could charge $74 - 18 = 56$ before entry and only $58 - 18 = 40$ after it.

Excess capacity. The gap between the output a firm produces and its efficient scale.

In the long run the roaster produces 6 thousand pounds a year, and its efficient scale is 12, so its excess capacity is $12 - 6 = 6$. Its average total cost is 40 where it could be 34.

The markup. Price minus marginal cost.

At its long-run position the roaster charges 40 and its last unit costs 22, so the markup is $40 - 22 = 18$ dollars a pound. A price-taking firm with the same costs charges 34 for a last unit that costs 34, a markup of nothing.

Against perfect competition and monopoly. Monopolistic competition shares free entry with perfect competition, along with its consequence, zero economic profit in the long run. It shares a downward-sloping demand curve with monopoly, along with its consequences: marginal revenue below price, and therefore a price above marginal cost.

With the roaster's costs, perfect competition gives a quantity of 12 at a price of 34, and monopolistic competition gives 6 at a price of 40. Profit is zero in both, which is why the differences between them are not about profit.

Advertising as a signal. The content of an advertisement may say nothing useful, and yet the willingness to spend on it can still be informative. Advertising separates good firms from poor ones when the customers it wins come back. A firm that expects repeat purchases can recoup a cost that a firm selling something poor, which makes just one sale, cannot.

A firm whose customer is worth \$60 over all the repeat purchases will pay \$20 to win one. A firm whose customer is worth \$6 will not, so the campaign itself tells the buyer which kind of firm is selling.

The product-variety externality. When a new firm enters with a product of its own, the consumers who prefer that product gain surplus that the entrant does not collect. The entrant charges one price, so buyers who would have paid far more for this particular blend keep the difference.

A customer who values a new roaster's blend at \$70 pays 40 like everyone else and keeps 30 of surplus. None of that 30 is the entrant's revenue, so it plays no part in the decision to open.

The business-stealing externality. When a new firm enters, it takes customers from firms already in the market, and their loss plays no part in its calculation.

A new roaster spends the fixed 144 to open, and some of its sales are the same customers buying from a different shop. That part of its profit is not new output but a loss to the roasters it has never met.

14.2 Worked examples

Revenue for the roaster.

q	0	1	2	3	4	5	6	7	8
$P = 58 - 3q$	58	55	52	49	46	43	40	37	34
$TR = P \times q$	0	55	104	147	184	215	240	259	272
added by the unit	—	55	49	43	37	31	25	19	13

In the table, only q is given. The price is read off the demand curve, total revenue is price times quantity, and the last row is the difference between one total revenue and the next (Figure 14.1).

Solving the two conditions together. Start from $MR = MC$ and leave A unknown:

$$A - 6q = 10 + 2q \implies A - 10 = 8q \implies A = 8q + 10.$$

Every position in this chapter satisfies this equation: $q = 8$ gives $A = 74$, $q = 6$ gives $A = 58$, and $q = 4$ gives $A = 42$.

Next, from $P = ATC$:

$$A - 3q = \frac{144}{q} + 10 + q \implies A = \frac{144}{q} + 10 + 4q. \tag{14.1}$$

At $q = 4$ this requires $A = 36 + 10 + 16 = 62$, but the loss-making firm had $A = 42$. The shortfall of 20 is exactly the loss per unit that we found there, $ATC(4) - P = 50 - 30 = 20$. So Equation (14.1) does more than agree with that arithmetic. It reproduces it.

Now set the expression for A from $MR = MC$ equal to Equation (14.1). The 10 cancels from both sides, and $4q$ comes off both sides too, leaving

$$4q = \frac{144}{q} \implies 4q^2 = 144 \implies q^2 = 36 \implies q = 6, \tag{14.2}$$

taking the positive root, because a negative quantity means nothing. Substituting back, the first equation gives $A = 8(6) + 10 = 58$, and Equation (14.1) gives $A = 24 + 10 + 24 = 58$.

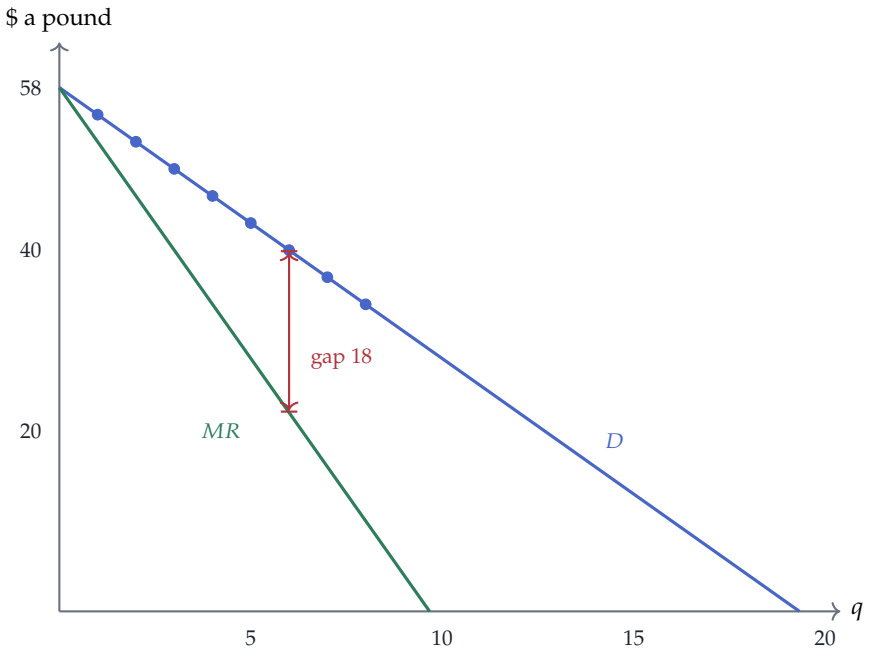


Figure 14.1: The firm's own demand curve and its marginal revenue at $A = 58$. Both start at 58, and marginal revenue reaches zero at a little over 9 while demand runs on to a little over 19, which is twice as far. The marked gap at six units is $3 \times 6 = 18$, the price cut given up on every unit already sold.

The two equations were independent of each other, so the fact that they agree is a check on the answer.

The same result, from the profit column. The table uses $P = 58 - 3q$ and $TC = 144 + 10q + q^2$:

q	3	4	5	6	7	8	9
P	49	46	43	40	37	34	31
TR	147	184	215	240	259	272	279
TC	183	200	219	240	263	288	315
profit	−36	−16	−4	0	−4	−16	−36

The best the firm can do is to make exactly nothing, and it does that at 6. The profit row is symmetric about 6, and it is never positive, because

$$(58 - 3q)q - (144 + 10q + q^2) = -4q^2 + 48q - 144 = -4(q - 6)^2,$$

which is zero at $q = 6$ and negative everywhere else. A profit function whose maximum is exactly zero, reached at one quantity only, is tangency written in numbers rather than drawn as geometry (Figure 14.2).

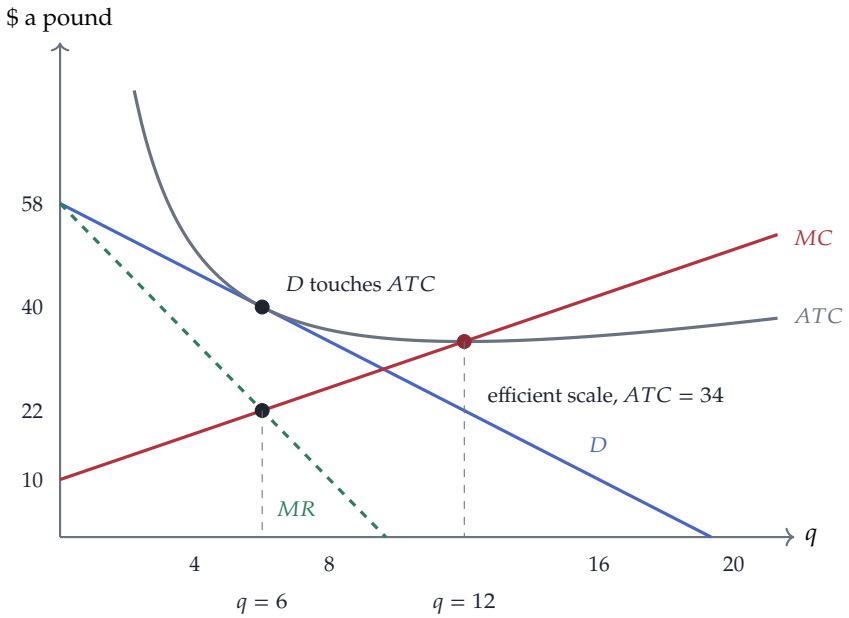


Figure 14.2: The firm in the long run, with the same cost curves as before and only the demand curve moved. The upper black dot at $(6, 40)$ is where the demand curve touches average total cost, so the price equals the cost per unit and the profit rectangle has closed to nothing. The lower black dot is $MR = MC$ at the same quantity. The red dot at $(12, 34)$ is the efficient scale, which the firm does not reach.

The trade that does not happen. In the long-run position, the firm sells 6 at a price of 40, and its marginal cost there is 22.

what a buyer would pay for a seventh unit	$58 - 3(7) = 37$
what a seventh unit costs to make	$TC(7) - TC(6) = 263 - 240 = 23$
gain if it were made and sold	$37 - 23 = 14$

The seventh thousand pounds is worth 38.5 to its buyers in total, since their valuations run from 40 down to 37. It costs 23 to make. So making it would leave the town 15.5 better off, and yet it is not made. The reason is that the firm does not collect that value. It collects marginal revenue, which is the 37 the new unit sells for, less the \$3 it must give up on each of the six existing buyers, or $37 - 18 = 19$, against a cost of 23. That is a loss of 4, and the profit column above shows exactly -4 at $q = 7$.

Consumer Choice

15.1 Definitions

Bundle. A list of quantities, so much of one good and so much of another. When there are two goods, it is just a pair of numbers, (x, y).

The buyer in this chapter has \$120 a week to spend and faces $p_y = \$2$ and $p_x = \$3$. The bundle $(20, 30)$ means 20 units of x and 30 units of y.

The three assumptions. **More is better**: a bundle that has more of both goods is preferred. **Completeness**: any two bundles can be compared with each other. **Consistency**: if A is preferred to B and B is preferred to C, then A is preferred to C.

By more is better, the buyer prefers $(40, 60)$ to $(20, 30)$, because it has more of both goods. Nothing in the list says the buyer is selfish, well informed or sensible.

Utility. A number attached to a bundle, arranged so that a preferred bundle gets a higher number. Only the ordering of those numbers carries meaning.

With $U(x, y) = \sqrt{xy}$, the bundle $(20, 30)$ scores $\sqrt{600} \approx 24.5$ and $(10, 30)$ scores $\sqrt{300} \approx 17.3$. All that tells us is that the buyer prefers the first. It does not say the first is 1.4 times as good.

Indifference curve. The set of bundles among which the buyer is indifferent, with every point on it delivering the same utility. For this function it is the graph of $y = k/x$, which is a rectangular hyperbola. To find the curve through a particular bundle, multiply its two coordinates together.

Through $(20, 30)$ the constant is $k = 20 \times 30 = 600$. So $(10, 60)$ and $(60, 10)$ lie on the same curve, $y = 600/x$, and the buyer is indifferent among all three bundles.

Marginal rate of substitution. The amount of y that the buyer will give up for one more unit of x, with utility held constant. It is the slope of the indifference curve without its minus sign, and we write it as MRS.

Moving along $y = 600/x$ from $(10, 60)$ to $(20, 30)$, the buyer gives up 30 units of y for 10 of x, an average of 3 per unit. From $(20, 30)$ to $(40, 15)$ the rate is only $15/20 = 0.75$, so the rate falls as x rises.

The budget constraint. With two goods, prices p_x and p_y and income m, the bundles that exhaust income are the ones that satisfy $p_x x + p_y y = m$. Every coefficient is a price, and the right-hand side is income. So nothing in the equation refers to taste.

Here the constraint is $3x + 2y = 120$, or $y = 60 - 1.5x$. The bundle $(16, 36)$ exhausts the income exactly, since $48 + 72 = 120$.

The slope is a rate of exchange between goods. The size of the slope, 1.5, is not an amount of money. Three dollars buys one unit of x, and three dollars also buys one and a half units of y. So the price of x, measured in units of y, is 1.5. To get one more unit of x, the market requires the buyer to give up 1.5 units of y.

If x cost \$6 instead, the ratio would be $6/2 = 3$, and each unit of x would cost 3 units of y. The line would be steeper.

The tangency condition. The only bundle from which the buyer wants no move is the one where the buyer's rate and the market's rate coincide:

$$MRS = \frac{p_x}{p_y}, \qquad \text{which here reads} \qquad \frac{y}{x} = \frac{p_x}{p_y}.$$

For a line to be tangent to a curve means that they have equal slopes at a shared point. So the optimum is a tangency.

At $(10, 45)$ the buyer's rate is $45/10 = 4.5$ against the market's 1.5, so at the margin x is worth more than it costs and the buyer buys more of it. At $(20, 30)$ the condition reads $30/20 = 1.5$ against $3/2 = 1.5$, so it holds exactly.

Marginal utility. The extra utility from one more unit of a good, holding the other good fixed. Mathematically, it is a partial derivative. For $U = x^{1/2} y^{1/2}$,

$$MU_x = \frac{\partial U}{\partial x} = \tfrac{1}{2}\sqrt{\frac{y}{x}}, \qquad MU_y = \frac{\partial U}{\partial y} = \tfrac{1}{2}\sqrt{\frac{x}{y}}.$$

With $y = 30$, the marginal utility of x falls from 0.866 at $x = 10$ to 0.612 at $x = 20$, and to 0.433 at $x = 40$.

Equal marginal utility per dollar. At the optimum

$$\frac{MU_x}{p_x} = \mu = \frac{MU_y}{p_y},$$

so the last dollar spent on each good buys the same amount of utility.

At $(20, 30)$, $MU_x/p_x = 0.6124/3 = 0.2041$ and $MU_y/p_y = 0.4082/2 = 0.2041$. Dividing one condition by the other gives $y/x = p_x/p_y$, and putting that into the budget gives the chosen bundle,

$$x^* = \frac{m}{2p_x}, \qquad y^* = \frac{m}{2p_y}. \tag{15.1}$$

A demand function. A rule that maps each price to the quantity a buyer chooses when doing the best that is available at that price. In effect it is a solved maximization problem, with one answer for each price.

With $m = 120$, Equation (15.1) gives $x^* = 60/p_x$. Spending on x is then $p_x x^* = \$60$ whatever the price, so half the budget always goes on each good.

Slutsky compensation. Enough income is given to the buyer for the original bundle to be affordable at the new prices. Costing $A = (20, 30)$ at the new prices gives

$$m' = p_x^{\text{new}} x_A + p_y y_A = 6(20) + 2(30) = 120 + 60 = \$180,$$

which is \$60 more than the buyer had before.

With \$180 and $p_x = \$6$, the buyer chooses $C = (15, 45)$, since $180/12 = 15$ and $180/4 = 45$.

Income and substitution effects. When the price of x rises, the good becomes dearer relative to y, and the buyer also becomes poorer. The substitution effect is the move from A to C, which changes the relative price while purchasing power is held fixed. The income effect is the move from C to B, which takes the compensation away, with prices unchanged.

When p_x doubles from \$3 to \$6, the buyer moves from $A = (20, 30)$ to $B = (10, 30)$. The substitution effect takes x from 20 to 15 and the income effect from 15 to 10, so each is -5 and together they make -10.

Revealed preference. If a bundle is chosen when another bundle was affordable, the chosen bundle is at least as good as the other. No utility function need exist for the principle to apply. It concerns choices and prices alone.

Under the compensated income of \$180, bundle A costs $6(20) + 2(30) = 180$, so it was affordable, and yet the buyer chose C. So C is at least as good as A.

Giffen good. A good whose ordinary demand curve slopes upward. This happens when the income effect is positive and large enough to outweigh the substitution effect.

Such goods are rare. The clearest evidence is the finding of Jensen and Miller (2008) for rice among very poor households in China.

15.2 Worked examples

Seven bundles the buyer cannot choose between.

x	10	12	15	20	30	40	60
$y = 600/x$	60	50	40	30	20	15	10
xy	600	600	600	600	600	600	600

Only x is given in the table. The second row divides 600 by it, and the third row multiplies the two back together. Every column returns 600 (Figure 15.1). So the buyer is indifferent between all seven bundles, and each one delivers $U = \sqrt{600}$.

The curve is symmetric. Both $(10, 60)$ and $(60, 10)$ lie on it, and so do $(12, 50)$ and $(50, 12)$, although the table lists only the first of that pair. Symmetry is a feature of $xy = k$, because swapping the two coordinates leaves the product alone. In economic terms, the table says that this buyer would exchange $(10, 60)$ for $(20, 30)$ without hesitating and without gaining. At those quantities, giving up thirty units of y for ten units of x is an even swap.

Every affordable bundle, evaluated.

x	8	12	16	**20**	24	28	32
$y = 60 - 1.5x$	48	42	36	**30**	24	18	12
xy	384	504	576	**600**	576	504	384

The column climbs until $x = 20$ and then falls away symmetrically. So the best affordable bundle is $(20, 30)$, and we used no calculus to find it.

The symmetry is not an accident of the numbers. Substituting gives $xy = x(60 - 1.5x) = 60x - 1.5x^2$. This is a downward parabola in x, whose roots are $x = 0$ and $x = 40$, so its peak lies halfway between them, at $x = 20$. The table suggests the answer among the bundles

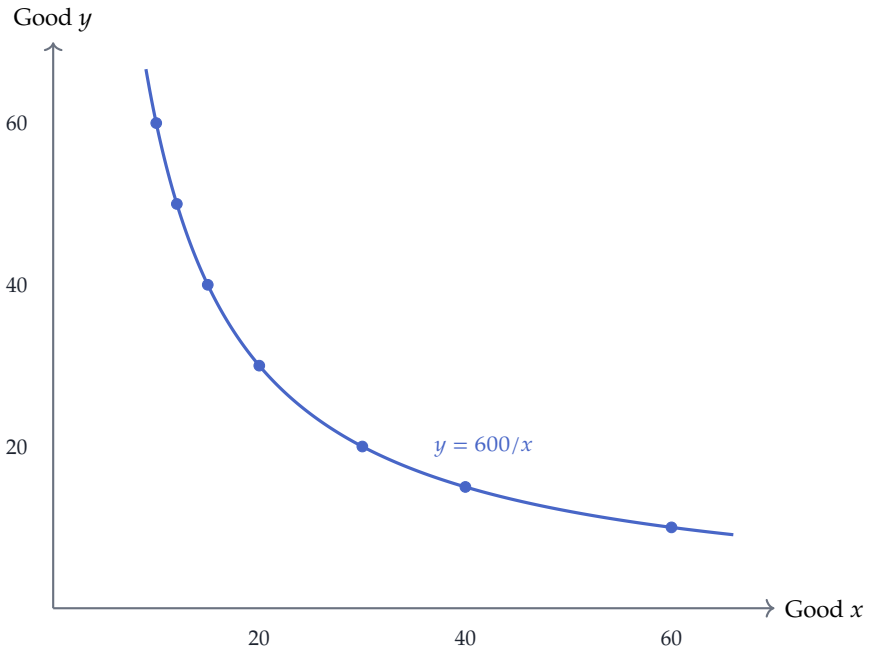


Figure 15.1: The indifference curve through $(20, 30)$. The dots are the seven bundles of the table and the curve is the equation $y = 600/x$, and they coincide because both come from the same constant. The curve falls, because holding utility fixed while taking more x requires giving up y, and it flattens as x rises.

that it tries, and the parabola confirms that answer for every point on the line. What this approach cannot do is answer the same question when p_x changes. For that we need the tangency condition, and the formula in Equation (15.1).

The chosen bundle, and both conditions checked. Putting in the numbers $m = 120, p_x = 3$ and $p_y = 2$,

$$x^* = \frac{120}{2(3)} = 20, \qquad y^* = \frac{120}{2(2)} = 30.$$

check	arithmetic	
budget constraint	$3(20) + 2(30) = 60 + 60 = 120$	holds
tangency	$y/x = 30/20 = 1.5$ and $p_x/p_y = 3/2 = 1.5$	holds

This is the same bundle that the table picked out, so three routes agree. The table computed xy at seven bundles. The check above showed that the bundle is a tangency. And Equation (15.1) returns $(20, 30)$. The first route required no theory, and the third requires no arithmetic.

The quantity chosen at six prices.

p_x	\$1	\$2	\$3	\$4	\$5	\$6
$x^* = 60/p_x$	60	30	20	15	12	10
spent on x, $p_x x^*$	\$60	\$60	\$60	\$60	\$60	\$60

Every entry in the middle row is 60 divided by the price above it. The bottom row never changes, which shows the constant half-share that we proved above.

Two of these columns are already familiar. At $p_x = \$3$ the quantity is 20, which is bundle A. At $p_x = \$6$ it is 10, which is bundle B. The two problems we solved at length above are two entries in the table, and the other four came for free (Figure 15.2). In every column, quantity falls as price rises. Here that is not an assumption imported from elsewhere. It is simply what $60/p_x$ does.

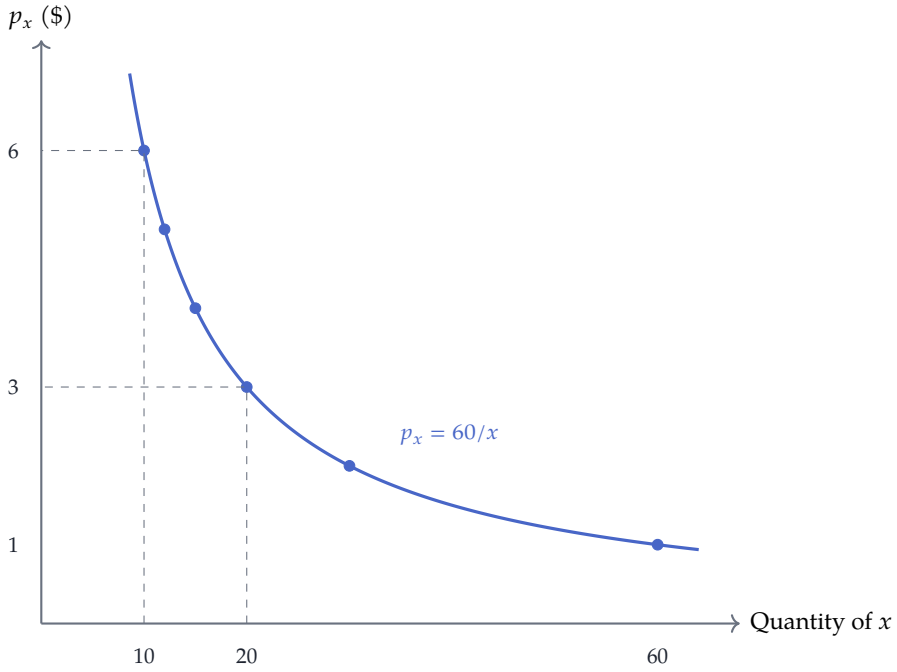


Figure 15.2: The demand curve $p_x = 60/x$. The dots are the six columns of the table and the curve is the equation. The dashed lines mark the two cases solved in full: 20 units at \$3, which is bundle A, and 10 units at \$6, which is bundle B. Both lie on the curve.

Asymmetric Information

16.1 Definitions

Symmetric information. Both parties to a trade know the same things about the object of the trade. This does not mean that everything is known. It means that whatever is known is known by both.

Suppose buyer and seller can both see that a used car is high quality, worth $8,000 to its owner and $10,000 to a buyer. Then it sells at a price such as $9,000, and each side gains $1,000.

Asymmetric information. One party to a trade knows something about the object of the trade that the other party does not.

The seller of a used car has driven it and knows its condition. The buyer has not, and a test drive and an hour with the service history do not settle the question.

Adverse selection, or hidden type. One party knows a **characteristic** of what is being traded, and the other party cannot observe it. The characteristic is fixed before any bargaining takes place.

A used car is either good or bad before anyone bargains over it, and no contract can alter that. So a single price must serve for cars of different worth.

Moral hazard, or hidden action. One party takes an **action** that the other party cannot observe. The action is chosen after the contract has been signed.

A chief executive's effort does not exist until the executive supplies it, after signing. The board sees only whether the outcome is $400 thousand or $200 thousand.

Expected value. Let λ stand for the share of the lot that is high quality, so that $1 - \lambda$ is the share that is low quality. Then the average value of a car drawn from the lot is

$$\lambda\, v_B^H + (1 - \lambda)\, v_B^L,$$

where v_B^H is the value of a high-quality car to a buyer, and v_B^L is what a low-quality car

is worth to a buyer. This is called an expected value. Expected value is just a name for a weighted average, and nothing more.

With sixty good cars in a hundred, $\lambda = 0.6$, and with $v_B^H = 10$ and $v_B^L = 4$ in thousands, the expected value is $0.6(10) + 0.4(4) = 7.6$, or \$7,600. No car on the lot is worth \$7,600.

Reservation value. The least a seller will accept for a good, or the most a buyer will pay for it. Below the seller's number, there is no sale. Above the buyer's number, there is no purchase. A trade can happen between the two.

The owner of a high-quality car will not accept less than \$8,000 for it, and a buyer will not pay more than \$10,000. A price anywhere between the two leaves both better off.

Private information. Something that one party to a trade knows and the other party does not observe.

The seller knows whether the car is one of the sixty good ones or one of the forty bad ones. The buyer knows only the shares, 60 and 40 in a hundred.

The threshold at which the good side withdraws. A high-quality car reaches the market only if the most a buyer will pay reaches the least its owner will accept, $\lambda\, v_B^H + (1-\lambda)\, v_B^L \geq v_S^H$. Solved for the share of good cars, the condition is

$$\lambda \;\geq\; \lambda^* \equiv \frac{v_S^H - v_B^L}{v_B^H - v_B^L}.$$

With the four numbers of this market, $\lambda^* = (8-4)/(10-4) = \frac{2}{3}$. A lot with sixty good cars in a hundred falls short, so its good cars are not offered, and $60 \times \$2{,}000 = \$120{,}000$ of surplus is lost.

Signal. A costly action taken to reveal private information. It is credible because the incentive to take it differs between the types, most often because the action costs more for the party who would take it falsely.

A warranty that pays the buyer W after a major repair is a signal. It costs a high-quality seller $0.10W$ and a low-quality seller $0.60W$, and any warranty between \$10,000 and \$20,000 separates the two.

Screening. The uninformed party offers a set of options, designed so that different types choose differently. It then learns the type from the choice, without observing the type directly.

An insurer offers a high-deductible policy at a low premium and a low-deductible policy at a high premium. An applicant who expects few claims takes the high deductible, and one who expects many claims does not.

The principal–agent problem. One party, the principal, hires another party, the agent, to act on its behalf, and the principal cannot observe what the agent does.

Shareholders and a chief executive are one example, and an employer and an employee are another. In each case the principal can verify hours and presence, but not effort.

Incentive compatibility constraint. The requirement that, given the contract, the action the agent prefers is the action the principal wants.

For the chief executive this reads $0.75w_H + 0.25w_L - 36 \geq 0.45w_H + 0.55w_L$, which simplifies to $w_H - w_L \geq 120$. Only the gap between the two wages matters.

Participation constraint. The requirement that the agent is at least as well off accepting the contract as taking the next-best alternative.

The executive's next-best job is worth \$24 thousand, so the contract must satisfy $0.75w_H + 0.25w_L - 36 \geq 24$. This depends on the level of pay, not the gap.

Limited liability. A wage cannot be negative. The agent can be paid nothing, but the agent cannot be required to pay the principal.

Selling the firm to the executive for a fee of 290 would give a wage of $200 - 290 = -90$ in the bad state. Limited liability rules that out, so the board sets $w_L = 0$ and $w_H = 120$.

Agency cost. The difference between what the principal pays when the action is hidden and what the principal would pay if the action could be observed.

With effort observable the board pays $24+36 = 60$. With effort hidden it pays $0.75\times120 = 90$, so the agency cost is $90 - 60 = 30$, and it goes to the executive.

16.2 Worked examples

The value of the lot, and of one car from it.

	how many	each worth	together
high quality	60	\$10,000	\$600,000
low quality	40	\$4,000	\$160,000
all	100		\$760,000

Multiply each count by its value. The two products are $60 \times 10{,}000 = 600{,}000$ and $40 \times 4{,}000 = 160{,}000$, and adding them gives 760,000. That total is the value to buyers of every car on the lot. It is not a price, and nobody pays it.

Dividing that total by the number of cars gives the value of a single car drawn from the lot:

$$\frac{\$760{,}000}{100} = \$7{,}600.$$

As a check, the answer lies between \$4,000 and \$10,000, and it is closer to \$10,000, because most of the lot is high quality. Any answer outside that range would be wrong.

No car on the lot is worth \$7,600. Every car is worth either \$10,000 or \$4,000. The average is a property of the whole collection, rather than of any member of it.

The four numbers in this market.

	seller will not accept below	buyer will not pay above	gap
high quality	\$8,000	\$10,000	\$2,000
low quality	\$2,000	\$4,000	\$2,000

In symbols, these values are

$$v_S^H = 8, \qquad v_B^H = 10, \qquad v_S^L = 2, \qquad v_B^L = 4,$$

measured in thousands of dollars.

Take a low-quality car first. It is worth \$2,000 to its owner and \$4,000 to a buyer. Suppose it sells at a price of \$3,000. The seller gives up something worth \$2,000 and receives \$3,000, so the seller gains \$1,000. The buyer pays \$3,000 for something worth \$4,000, so the buyer also gains \$1,000. A high-quality car works the same way. It is worth \$8,000 to its owner and \$10,000 to a buyer, so at a price of \$9,000 each side gains \$1,000.

The gap is the same size on both rows, and that is a deliberate feature of the example. If the high-quality gap were smaller, we could say that the market destroys the trades that matter least. Here the two gaps are identical. So the trades that are about to disappear are worth exactly as much as the ones that survive.

Figure 16.1 draws the buyer's ceiling against the good owner's floor for these four numbers, crossing at $\lambda^* = \frac{2}{3}$.

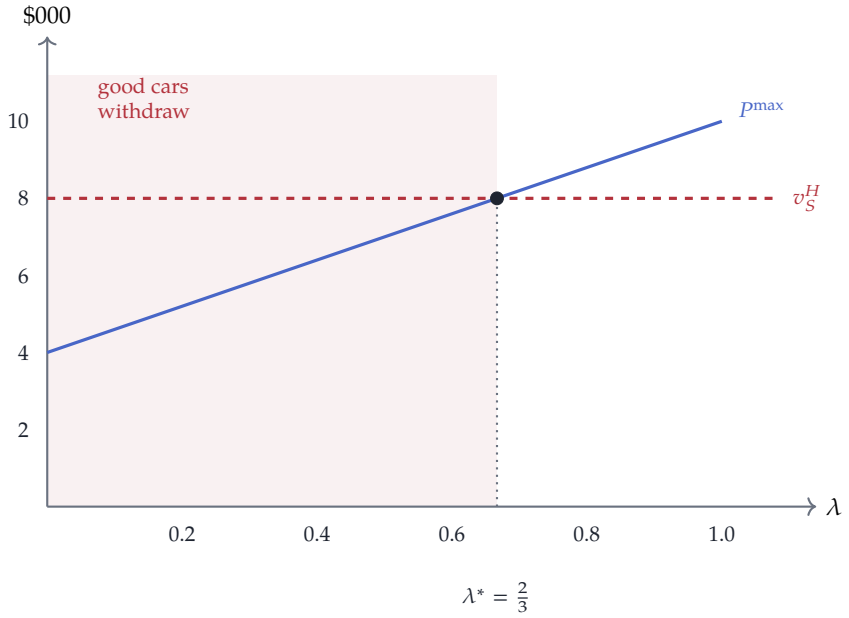


Figure 16.1: The buyer's ceiling and the good owner's floor. The rising line is $P^{\max} = 4 + 6\lambda$, worth \$4,000 when the lot is all low quality and \$10,000 when it is all high quality. The flat dashed line is v_S^H, which does not move with λ because what a good car is worth to its owner has nothing to do with what else is on the market. Left of the crossing the ceiling is below the floor and no price works.

What the same promise costs each type.

	repairs per hundred cars	expected cost of the warranty
high quality	10	$0.10\,W$
low quality	60	$0.60\,W$

A high-quality seller expects to pay out on 10 cars in every 100, so the warranty costs that seller $\frac{10}{100}W = 0.10W$. A low-quality seller expects to pay out on 60 in 100, so for that seller it costs $\frac{60}{100}W = 0.60W$. The ratio of the two is $0.60W/0.10W = 6$. In other words, the identical promise is six times as expensive for the seller of a bad car.

Nothing about the warranty itself distinguishes the two sellers, since the words of the promise are the same. What differs is how much the promise will cost to keep. That cost is determined by the very characteristic that the buyer cannot observe. A seller cannot make a bad car cheap to warrant, and that is precisely why offering the warranty is informative.

Figure 16.2 draws what each seller nets as the warranty grows. The low type's line falls six times as steeply, and the band runs from $W = 10$ to $W = 20$.

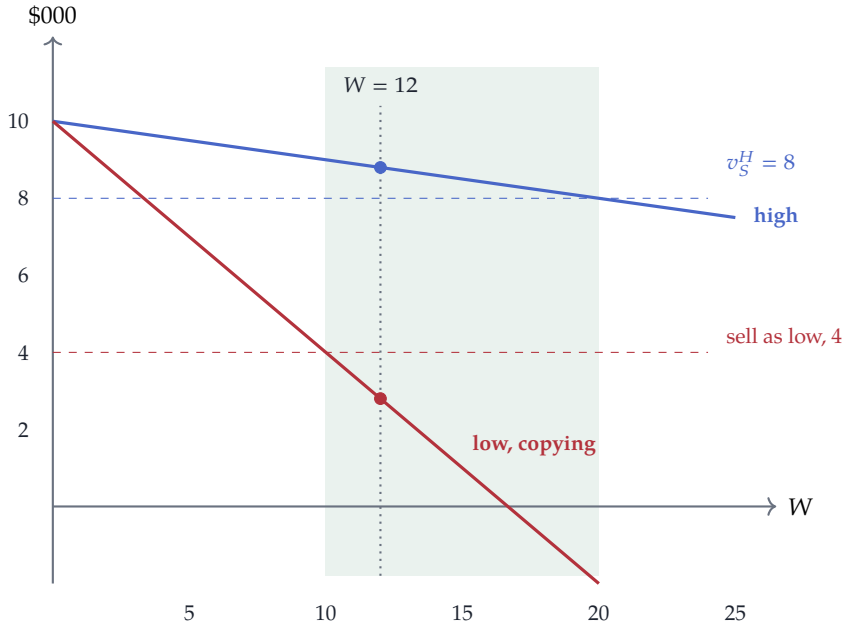


Figure 16.2: What each seller ends up with, meaning the price received less the expected cost of honoring the promise. Both lines start at 10 where the warranty is nothing. The low type's line falls six times as steeply, because 60 low-quality cars in a hundred need a repair against 10 high-quality ones. The shaded band runs from where the red line crosses its own floor of 4, at $W = 10$, to where the blue line crosses its floor of 8, at $W = 20$.

A board hiring a chief executive.

	good outcome \$400	bad outcome \$200	average output
effort supplied	75 in 100	25 in 100	\$350
effort not supplied	45 in 100	55 in 100	\$290

All figures are in thousands. The two averages use the arithmetic from Section 16.1:

$$\text{with effort: } 0.75(400) + 0.25(200) = 300 + 50 = 350,$$

$$\text{without effort: } 0.45(400) + 0.55(200) = 180 + 110 = 290.$$

Supplying effort costs the executive \$36 in forgone leisure and attention. The executive's next-best job is worth \$24.

What makes the problem hard is that the outcome is an imperfect guide to effort. Forty-five in every hundred executives who supply no effort still deliver the good outcome. And twenty-five in every hundred who do supply effort still deliver the bad outcome. So a good year is not proof of effort, and a bad year is not proof that effort was absent.

Modern Data Analysis in Economics

17.1 Definitions

Correlation. Two variables that move together. Here the relationship is negative, because more of one variable goes with less of the other.

Eight students record their weekly hours of tutoring and their exam scores, in invented numbers. The two with no tutoring average 84.5 and the six who were tutored average 70.5, so more tutoring goes with lower scores.

Reverse causality. X and Y move together in the data, but the influence runs from Y to X, rather than from X to Y.

Cities with more police officers per resident can have more crime, because a crime wave leads the city council to put more officers on the street. In the tutoring data, students who expect to do badly seek out tutoring, so the low scores came first.

Omitted variable. A third variable Z, which has been left out of the comparison, and which drives both X and Y.

Among schoolchildren, shoe size and reading ability are strongly correlated, but big feet do not help anyone to read. Age drives both.

Selection. The units that receive a treatment differ from the units that do not, and they differ before the treatment has done anything.

The students who sign up for tutoring are the ones already struggling. So the tutored group was headed for lower scores before the first session took place.

Counterfactual. What would have happened to the same person, firm or town had the treatment not occurred.

Take a tutored student who scored 64. The counterfactual is the score that the same student, in the same week, would have obtained without tutoring.

The fundamental problem of causal inference. The counterfactual is unobservable by definition, because a unit cannot both receive a treatment and not receive it.

A student either took the tutoring or did not, so we observe each one in a single column. The cell the comparison needs, that student's score in the other column, is always the one that cannot be filled.

Control group. Units that do not receive the treatment. Their outcome stands in for what the treated group's outcome would have been.

In January 2025, Fairview starts paying a $500 grant to any landlord who re-lets an empty storefront, and Grantham, one county over, does not. Grantham is then the control group, and its change in vacancies stands in for what Fairview's change would have been without the grant.

Randomized controlled trial. Treatment is assigned by pure chance, so that a coin decides which units are treated and which units are the controls.

The coin takes no account of motivation, health or talent, so with enough subjects each of them splits roughly evenly between the groups. Economists rarely have the option, since a trial of the minimum wage would mean flipping a coin to decide which states raise it.

Experimental and observational data. **Experimental** data comes from a researcher who controlled how the treatment was assigned. **Observational** data comes from the world going about its business, and it is recorded in surveys, tax records, censuses and payrolls.

A count of the vacant storefronts on each Main Street is observational, because nobody assigned anything. Nearly all economic data is of this kind.

Panel data. Observations on many units at many dates. Each unit can then be compared both with other units and with itself at other dates.

Fairview and Grantham, with their vacant storefronts counted every December, make a small panel of two units at three dates. Fairview's counts are 27, 24 and 18 for 2023, 2024 and 2025, and Grantham's are 33, 30 and 27.

Natural experiment. An event, such as a law, a border or a storm, that gives a treatment to one group and not to another, in a way that is unrelated to how the groups were already doing.

On 1 April 1992, New Jersey raised its minimum wage from $4.25 to $5.05 an hour, and Pennsylvania, next door, kept its minimum at $4.25. David Card and Alan Krueger surveyed 410 fast-food restaurants on both sides of the line, before the increase and after it.

Difference-in-differences. The change in the treated group minus the change in the control group. It is an estimate of the treatment's effect, and it is built from four averages.

Applied to Fairview and Grantham, the first subtraction, along time, removes the permanent difference between the two towns. The second, across the towns, removes the year they share, and what survives both is the estimate of the grant's effect.

Parallel trends. Without the treatment, the treated group's outcome would have changed by the same amount as the control group's outcome did.

The levels need not match, because they cancel in the first subtraction. In the year before the grant, Fairview and Grantham each fell by 3, so the assumption held in the one period in which it could be checked.

Controlling for an outcome. A control variable that is itself partly produced by the process being studied. If the effect runs through that variable, holding it fixed removes that part of the effect we are trying to measure.

Blau and Kahn (2017) find that the gender wage gap among full-time workers falls from 20.7 to 8.4 points once education, experience, demographics, industry, occupation and union membership are controlled for. But if hiring or promotion steers women away from high-paying occupations, part of the gap runs through occupation, and holding occupation fixed deletes that part.

Event study. Difference-in-differences stretched over many periods. It tracks the treated and comparison outcomes for years around an event, rather than for one period before and one period after.

Kleven, Landais and Søgaard (2019) line up Danish mothers and fathers by the number of years relative to the first birth. Mothers' earnings drop by about 30 percent around the birth and settle about 20 percent below their pre-birth path, while fathers' earnings are essentially unchanged.

17.2 Worked examples

Tutoring hours and exam scores.

student	A	B	C	D	E	F	G	H
tutoring, hours a week	0	0	1	1	2	3	4	5
exam score	88	81	78	84	72	69	62	58

If we read along the rows, scores tend to fall as the hours of tutoring rise. The two students with no tutoring scored 88 and 81. The student with five hours scored 58. The pattern is consistent, and it is not produced by one unusual student.

Figure 17.1 plots the eight students with the two group averages.

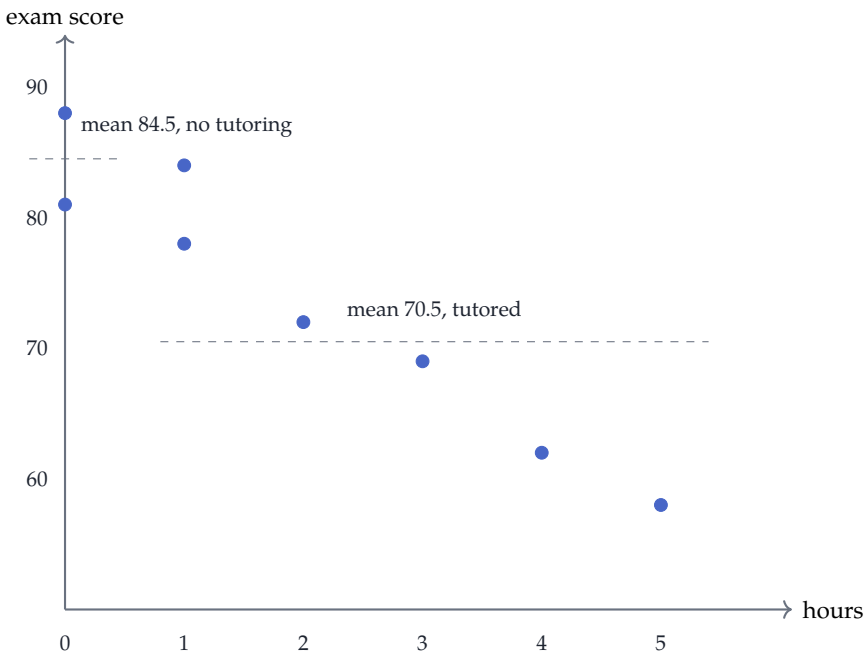


Figure 17.1: The eight students, with the two group averages marked. The relationship is real, consistent and visible, and it is not the work of one outlying point. What the picture cannot record is which way any arrow runs, because the same scatter is consistent with several different causal stories.

Four numbers.

vacant storefronts	Dec 2024, before	Dec 2025, after
Fairview, grant	24	18
Grantham, no grant	30	27

We have two groups at two dates, and this table holds everything the method needs.

The first subtraction runs along time, and gives each town's change:

$$\text{Fairview: } 18 - 24 = -6, \qquad \text{Grantham: } 27 - 30 = -3.$$

The second subtraction runs across the towns and takes the difference between those changes, $(-6) - (-3) = -3$. Figure 17.2 draws the estimate.

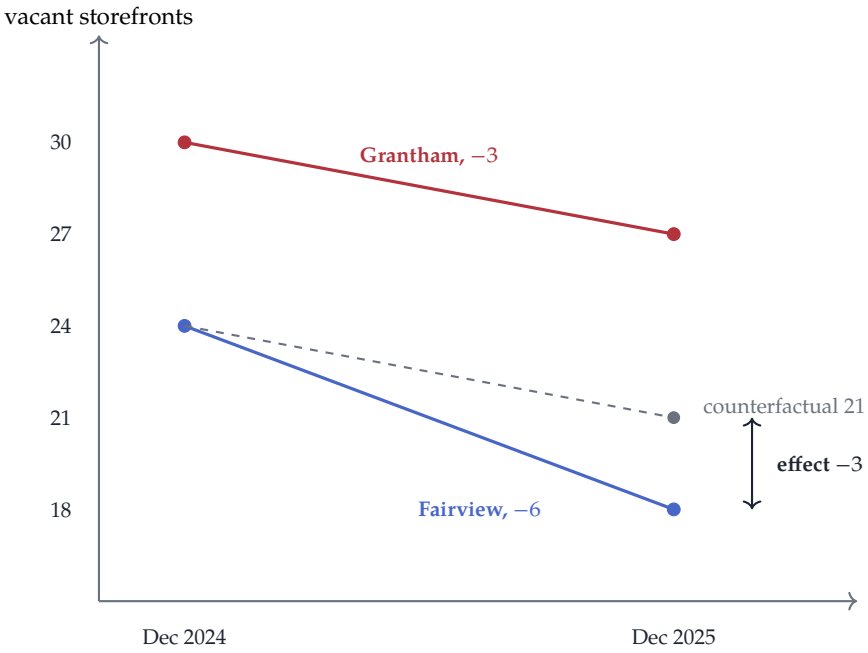


Figure 17.2: Fairview falls by 6 and Grantham by 3. Applying Grantham's change to Fairview's starting point gives the counterfactual of $24 - 3 = 21$, which is what Fairview would have reached without the grant if the two towns would otherwise have moved together. The gap between that counterfactual and the observed 18 is the estimate.

Two other towns, Hazel with the grant and Kirby without it, have the same two-by-two table as Fairview and Grantham, and so the same estimate of -3.

The same table, a different history.

	Dec 2023	Dec 2024	pre-trend	Dec 2025
Hazel, grant	30	24	$24 - 30 = -6$	18
Kirby, none	33	30	$30 - 33 = -3$	27

Before any policy existed, Hazel was already improving twice as fast as Kirby. So the two paths were diverging rather than running in parallel, and the assumption the estimate rests on was false before anyone announced the grant.

Figure 17.3 sets the two histories side by side.

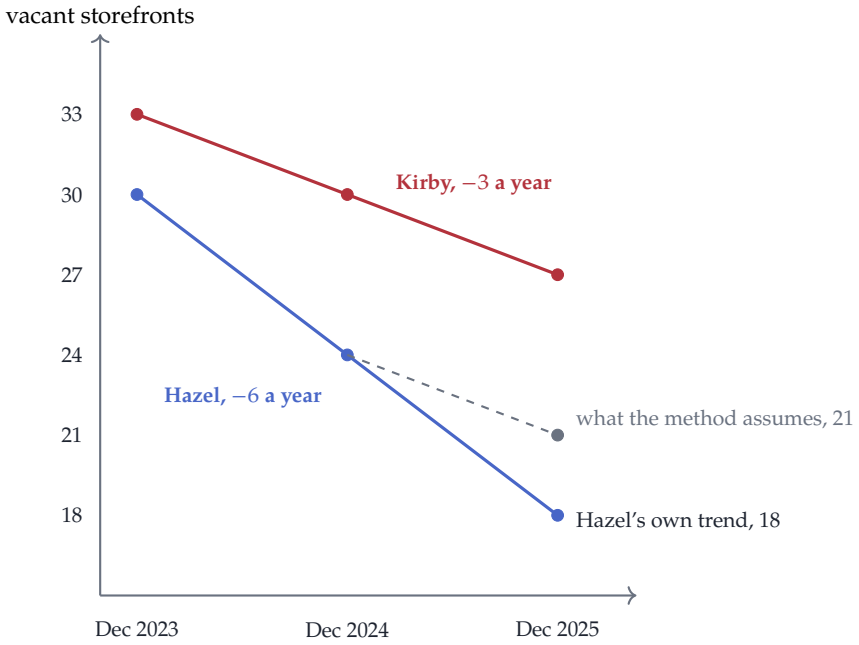


Figure 17.3: Hazel and Kirby have the same two-by-two table as Fairview and Grantham, so the arithmetic returns the same estimate of −3. Their pre-period is different. Hazel was falling twice as fast before the grant, and continuing that path reaches 18 with no grant at all, which is exactly the figure observed.

Full-time-equivalent employment per restaurant.

	before	after	change
New Jersey, raise	20.44	21.03	21.03 − 20.44 = +0.59
Pennsylvania, none	23.33	21.17	21.17 − 23.33 = −2.16

So the difference in differences comes to

$$0.59 - (-2.16) = 0.59 + 2.16 = +2.75.$$

If we subtract the other way, we get the same answer. Before the increase, the gap between the states was $20.44 - 23.33 = -2.89$. After it, the gap was $21.03 - 21.17 = -0.14$. And $(-0.14) - (-2.89) = +2.75$.

These figures are the means published in the paper's own table. Card and Krueger report 2.76, which they computed before rounding.